

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

CS407

[Routing and switching]

[Reading material]

WEEK 7 TO 16

TOPIC [67 TO 168]

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HY KUCH OR KUCH JAGH PY KUCH IMPORTANT SCREEN SHOT MISSING HAIN SIR NY WO SCREEN SHOT READING
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BEST OF LUCK !!...

Week no 7

67. TOPIC

Synopsis/Description

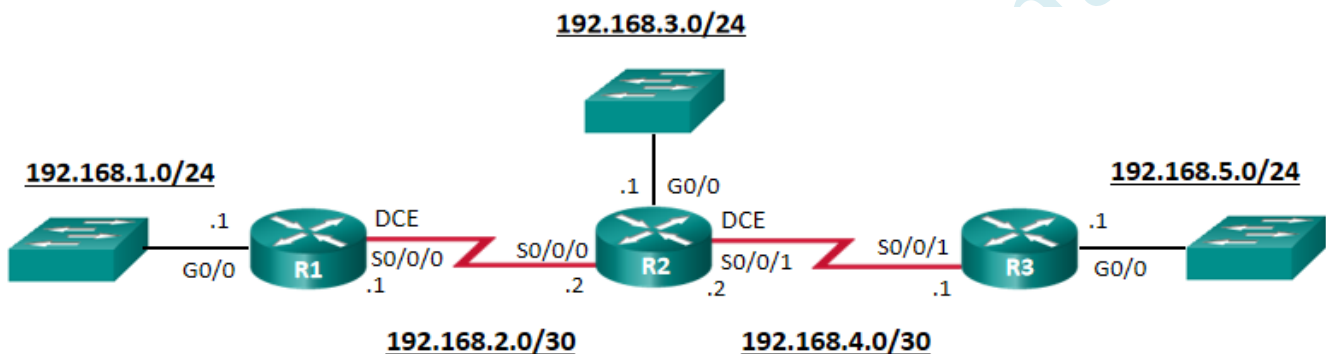
- I. Routing Information Protocol
- II. Configuring RIP
- III. Verifying RIP

Although RIP is rarely used in modern networks, it is useful as a foundation for understanding basic network routing. For this reason, this section provides a brief overview of how to configure basic RIP settings and to verify IPv2.

Please see the slides for more information about RIP configuration and other parameters.

Although RIP is rarely used in modern networks, it is useful as a foundation for understanding basic network routing. For this reason, this section provides a brief overview of how to configure basic RIP settings and to verify RIPv2.

Refer to the reference topology in Figure 1 and the addressing table in Figure 2. In this scenario, all routers have been configured with basic management features and all interfaces identified in the reference topology are configured and enabled. There are no static routes configured and no routing protocols enabled; therefore, remote network access is currently impossible. RIPv2 is used as the dynamic routing protocol. To enable RIP, use the **router rip** command, as shown in Figure 3. This command does not directly start the RIP process. Instead, it provides access to the router configuration mode where the RIP routing settings are configured.



R1(config-router)# ?

Router configuration commands:

address-family	Enter Address Family command mode
auto-summary	Enable automatic network number summarization
default	Set a command to its defaults
default-information	Control distribution of default information
default-metric	Set metric of redistributed routes
distance	Define an administrative distance
distribute-list	Filter networks in routing updates
exit	Exit from routing protocol configuration mode
flash-update-threshold	Specify flash update threshold in second
help	Description of the interactive help system
input-queue	Specify input queue depth
maximum-paths	Forward packets over multiple paths
neighbor	Specify a neighbor router
network	Enable routing on an IP network
no	Negate a command or set its defaults
offset-list	Add or subtract offset from RIP metrics
output-delay	Interpacket delay for RIP updates
passive-interface	Suppress routing updates on an interface
redistribute	Redistribute information from another routing protocol
timers	Adjust routing timers
traffic-share	How to compute traffic share over alternate paths
validate-update-source	Perform sanity checks against source address of routing updates
version	Set routing protocol version

To disable and eliminate RIP, use the **no router rip** global configuration command. This command stops the RIP process and erases all existing RIP configurations.

The **show ip protocols** command displays the IPv4 routing protocol settings currently configured on the router. This output displayed in Figure 1 confirms most RIP parameters including:

1. RIP routing is configured and running on router R1.
2. The values of various timers; for example, the next routing update, is sent by R1 in 16 seconds.
3. The version of RIP configured is currently RIPv1.
4. R1 is currently summarizing at the classful network boundary.
5. The classful networks are advertised by R1. These are the networks that R1 include in its RIP updates.
6. The RIP neighbors are listed including their next-hop IP address, the associated AD that R2 uses for updates sent by this neighbor and when the last update was received from this neighbor

68. TOPIC

Synopsis/Description

- i. Routing Information Protocol v2
- ii. Configuring RIPv2
- iii. Verifying RIPv2

RIP Versions

By default, when a RIP process is configured on a Cisco router, it is running RIPv1. However, even though the router only sends RIPv1 messages, it can interpret both RIPv1 and RIPv2 messages. A RIPv1 router ignores the RIPv2 fields in the route entry.

Use the **version 2** router configuration mode command to enable RIPv2. Please see the slides for information about how to use RIP version 2.

Disabling Auto Summarization

RIPv2 automatically summarizes networks at major network boundaries by default, just like RIPv1.

To modify the default RIPv2 behavior of automatic summarization, use the **no auto-summary** router configuration mode command. This command has no effect when using RIPv1. When automatic summarization has been disabled, RIPv2 no longer summarizes networks to their classful address at boundary

routers. RIPv2 now includes all subnets and their appropriate masks in its routing updates. The **show ip protocols** now states that automatic network summarization is not in effect.

Please see the slides for example of auto-summarization.

RIP Passive Interfaces

By default, RIP updates are forwarded out all RIP enabled interfaces. However, RIP updates really only need to be sent out interfaces connecting to other RIP enabled routers.

Sending out unneeded updates on a LAN impacts the network in three ways:

Wasted Bandwidth - Bandwidth is used to transport unnecessary updates. Because RIP updates are either broadcasted or multicasted; therefore, switches also forward the updates out all ports.

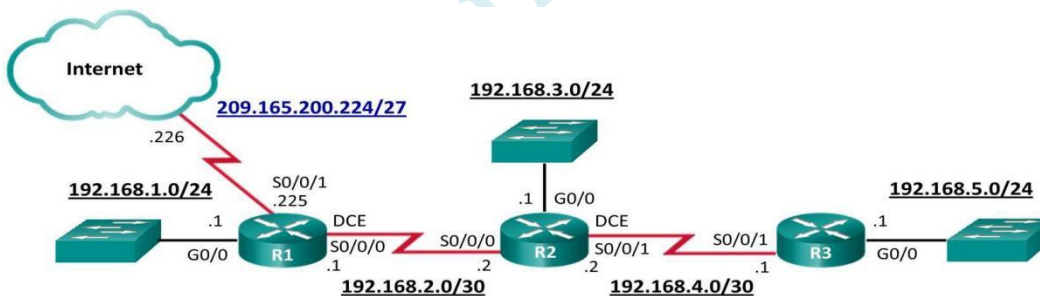
Wasted Resources - All devices on the LAN must process the update up to the transport layers, at which point the devices will discard the update.

Security Risk - Advertising updates on a broadcast network is a security risk. RIP updates can be intercepted with packet sniffing software. Routing updates can be modified and sent back to the router, corrupting the routing table with false metrics that misdirect traffic.

Use the **passive-interface** router configuration command to prevent the transmission of routing updates through a router interface, but still allow that network to be advertised to other routers. The command stops routing updates out the specified interface. However, the network that the specified interface belongs to is still advertised in routing updates that are sent out other interfaces.

Please see the slides for an example of passive interfaces.

Propagating a Default Route



Refer to Figure above. In this scenario, R1 is single-homed to a service provider. Therefore, all that is required for R1 to reach the Internet is a default static route going out of the Serial 0/0/1 interface.

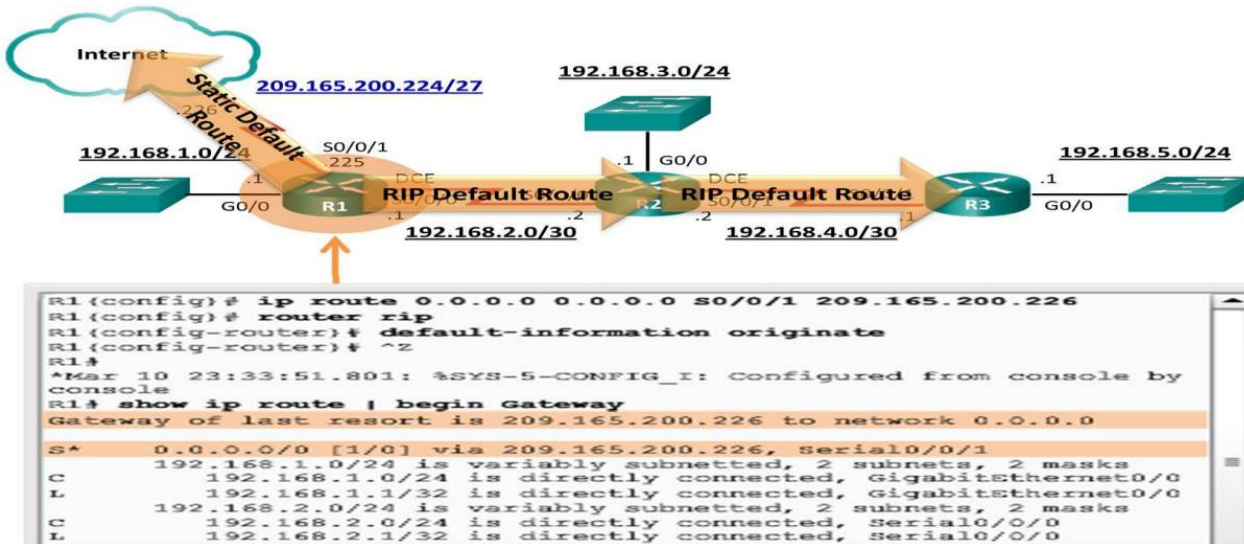
Similar default static routes could be configured on R2 and R3, but it is much more scalable to enter it one time on the edge router R1 and then have R1 propagate it to all other routers using RIP. To provide Internet connectivity to all other networks in the RIP routing domain, the default static route needs to be advertised to all other routers that use the dynamic routing protocol.

To propagate a default route, the edge router must be configured with:

A default static route using the **ip route 0.0.0.0 0.0.0.0 exit-intf next-hop-ip** command.

The **default-information originate** router configuration command. This instructs R1 router to originate default information, by propagating the static default route in RIP updates.

The example in Figure below configures a fully specified default static route to the service provider and then the route is propagated by RIP. Notice that R1 now has a Gateway of Last Resort and default route installed in its routing table.



69. TOPIC

Synopsis/Description

- Link-state routing protocols
- Dijkstra's Shortest Path First Algorithm

Link-state routing protocols are also known as shortest path first protocols and are built around Edsger Dijkstra's shortest path first (SPF) algorithm.

Link-state routing protocols have the reputation of being much more complex than their distance vector counterparts. However, the basic functionality and configuration of link-state routing protocols is equally straight-forward.

Just like RIP and EIGRP, basic OSPF operations can be configured using the:

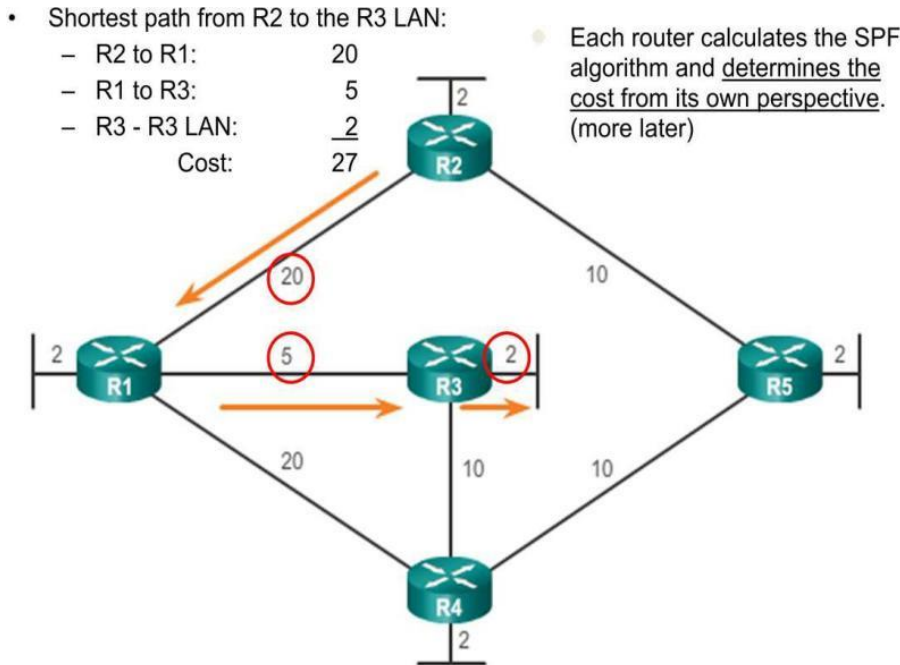
router ospf process-id global configuration command

network command to advertise networks

Dijkstra's Shortest Path First Algorithm

All link-state routing protocols apply Dijkstra's algorithm to calculate the best path route. The algorithm is commonly referred to as the shortest path first (SPF) algorithm. This algorithm uses accumulated costs along each path, from source to destination, to determine the total cost of a route.

In the figure below, each path is labeled with an arbitrary value for cost. The cost of the shortest path for R2 to send packets to the LAN attached to R3 is 27. Each router determines its own cost to each destination in the topology. In other words, each router calculates the SPF algorithm and determines the cost from its own perspective.



70. TOPIC

So exactly how does a link-state routing protocol work? With link-state routing protocols, a link is an interface on a router. Information about the state of those links is known as link-states.

Examine the topology in the figure below. All routers in the topology will complete the following generic link-state routing process to reach a state of convergence:

1. Each router learns about its own links and its own directly connected networks. This is done by detecting that an interface is in the up state.
2. Each router is responsible for meeting its neighbors on directly connected networks. Link state routers do this by exchanging Hello packets with other link-state routers on directly connected networks.
3. Each router builds a Link-State Packet (LSP) containing the state of each directly connected link. This is done by recording all the pertinent information about each neighbor, including neighbor ID, link type, and bandwidth.
4. Each router floods the LSP to all neighbors. Those neighbors store all LSPs received in a database. They then flood the LSPs to their neighbors until all routers in the area have received the LSPs. Each router stores a copy of each LSP received from its neighbors in a local database.

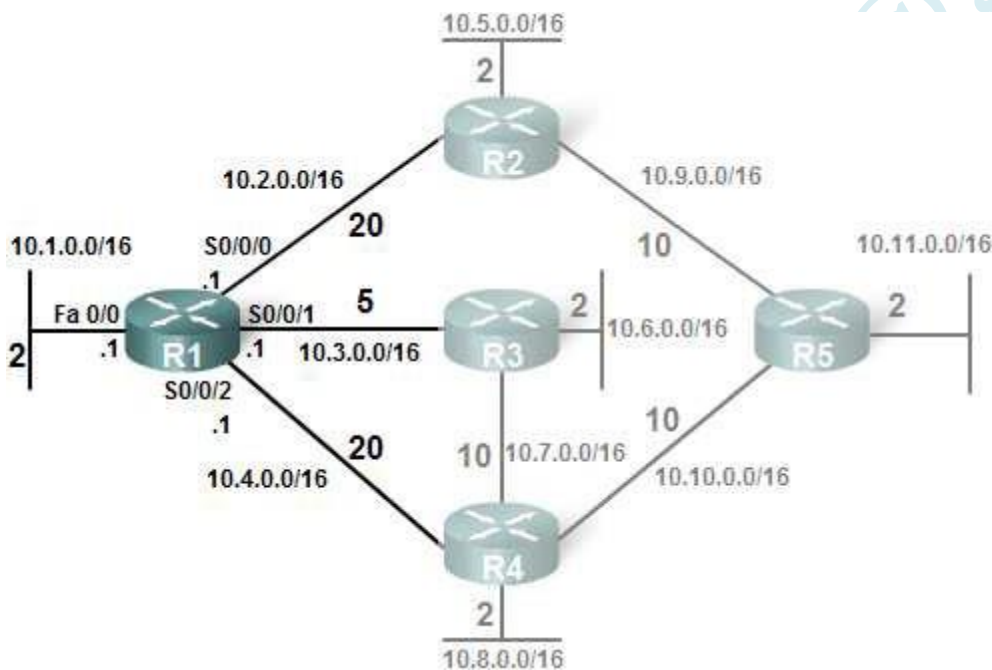
- Each router uses the database to construct a complete map of the topology and computes the best path to each destination network. Like having a road map, the router now has a complete map of all destinations in the topology and the routes to reach them. The SPF algorithm is used to construct the map of the topology and to determine the best path to each network.

Note: This process is the same for both OSPF for IPv4 and OSPF for IPv6. The examples in this section refer to OSPF for IPv4.

Step 1: Learning About Directly Connected Networks

The first step in the link-state routing process is that each router learns about its own links, its own directly connected networks. When a router interface is configured with an IP address and subnet mask, the interface becomes part of that network.

As with distance vector protocols and static routes, the interface must be properly configured with an IPv4 address and subnet mask, and the link must be in the up state before the link-state routing protocol can learn about a link. Also, like distance vector protocols, the interface must be included in one of the network router configuration statements before it can participate in the link-state routing process.



Step 2: Sending Hello Packets to Neighbors

The second step in the link-state routing process is that each router is responsible for meeting its neighbors on directly connected networks.

Routers with link-state routing protocols use a Hello protocol to discover any neighbors on its links. A neighbor is any other router that is enabled with the same link-state routing protocol.

When two link-state routers learn that they are neighbors, they form an adjacency. These small Hello packets continue to be exchanged between two adjacent neighbors and serves as a keepalive function to monitor the state of the neighbor. If a router stops receiving Hello packets from a neighbor, that neighbor is considered unreachable and the adjacency is broken.

Please see the slides for working example of Link state routing protocol.

71. TOPIC

Synopsis/Description

- i. Working of link-state routing protocol
- ii. Link-State Packets
- iii. Sending Hello Packets to Neighbors

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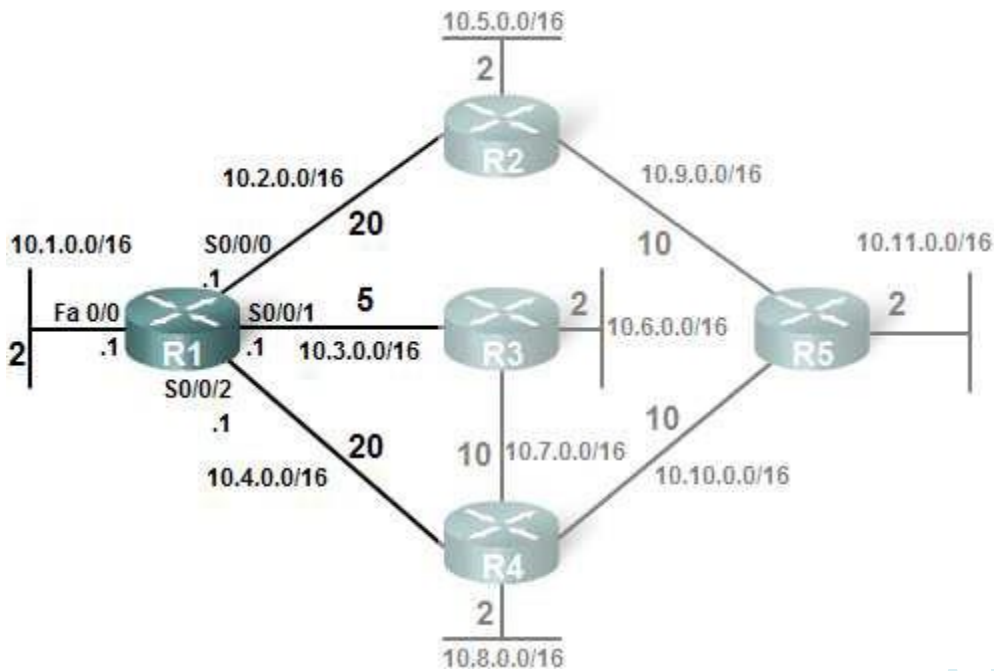
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Advantages of Link State Routing Protocols

There are several advantages of link-state routing protocols compared to distance vector routing protocols.

Builds a Topological Map - Link-state routing protocols create a topological map, or SPF tree of the network topology. Because link-state routing protocols exchange link-states, the SPF algorithm can build an SPF tree of the network. Using the SPF tree, each router can independently determine the shortest path to every network.

Fast Convergence - When receiving an LSP, link-state routing protocols immediately flood the LSP out all interfaces except for the interface from which the LSP was received. In contrast, RIP needs to process each routing update and update its routing table before flooding them out other interfaces.

Event-driven Updates - After the initial flooding of LSPs, link-state routing protocols only send out an LSP when there is a change in the topology. The LSP contains only the information regarding the affected link. Unlike some distance vector routing protocols, link-state routing protocols do not send periodic updates.

Hierarchical Design - Link-state routing protocols use the concept of areas. Multiple areas create a hierarchical design to networks, allowing for better route aggregation (summarization) and the isolation of routing issues within an area.

Link-state protocols also have a few disadvantages compared to distance vector routing protocols:

Memory Requirements - Link-state protocols require additional memory to create and maintain the link-state database and SPF tree.

Processing Requirements - Link-state protocols can also require more CPU processing than distance vector routing protocols. The SPF algorithm requires more CPU time than distance vector algorithms such as Bellman-Ford, because link-state protocols build a complete map of the topology.

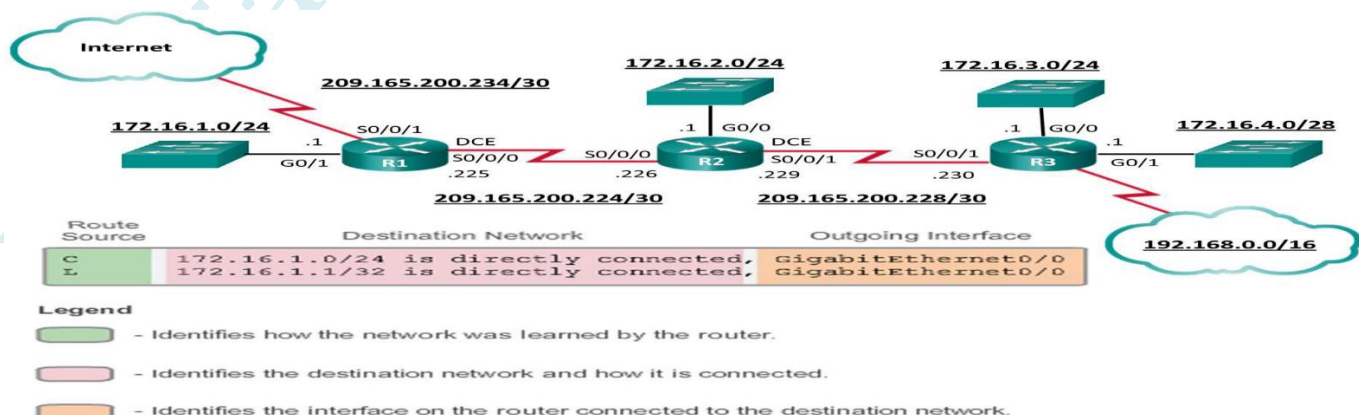
Bandwidth Requirements - The flooding of link-state packets can adversely affect the available bandwidth on a network. This should only occur during initial startup of routers, but can also be an issue on unstable networks.

Addressing Disadvantages

Modern link-state routing protocols are designed to minimize the effects on memory, CPU, and bandwidth. The use and configuration of multiple areas can reduce the size of the link-state databases. Multiple areas can also limit the amount of link-state information flooding in a routing domain and send LSPs only to those routers that need them. When there is a change in the topology, only those routers in the affected area receive the LSP and run the SPF algorithm. This can help isolate an unstable link to a specific area in the routing domain.

73. TOPIC

Reference Topology – Connected and Local



The topology displayed in Figure above is used as the reference topology. Notice that in the topology:

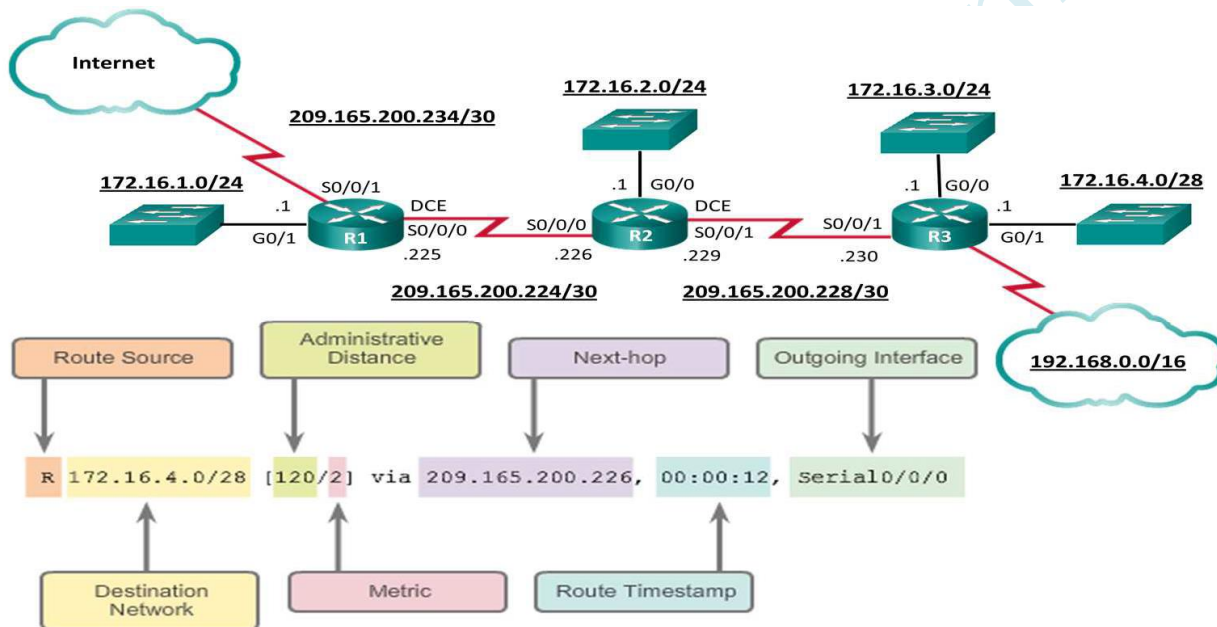
R1 is the edge router that connects to the Internet. Therefore, it is propagating a default static route to R2 and R3.

R1, R2, and R3 contain discontinuous networks separated by another classful network. R3 is also introducing a 192.168.0.0/16 supernet route.

Figure above displays the IPv4 routing table of R1 with directly connected, static, and dynamic routes.

Note: The routing table hierarchy in Cisco IOS was originally implemented with the classful routing scheme. Although the routing table incorporates both classful and classless addressing, the overall structure is still built around this classful scheme.

Remote Networks



The figure above displays an IPv4 routing table entry on R1 for the route to remote network 172.16.4.0 on R3. The entry identifies the following information:

- Route source - Identifies how the route was learned.
- Destination network - Identifies the address of the remote network.
- Administrative distance - Identifies the trustworthiness of the route source.
- Metric - Identifies the value assigned to reach the remote network. Lower values indicate preferred routes.
- Next hop - Identifies the IPv4 address of the next router to forward the packet to.
- Route timestamp - Identifies from when the route was last heard.
- Outgoing interface - Identifies the exit interface to use to forward a packet toward the final destination.

IPv4 Routing Table – Ultimate Route

A dynamically built routing table provides a great deal of information. Therefore, it is crucial to understand the output generated by the routing table. Special terms are applied when discussing the contents of a routing table.

The Cisco IP routing table is not a flat database. The routing table is actually a hierarchical structure that is used to speed up the lookup process when locating routes and forwarding packets. Within this structure, the hierarchy includes several levels.

Routes are discussed in terms of:

- Ultimate route
- Level 1 route
- Level 1 parent route
- Level 2 child routes

IPv4 Routing Table – Level 1 Route

A level 1 route is a route with a subnet mask equal to or less than the classful mask of the network address. Therefore, a level 1 route can be a:

Network route - A network route that has a subnet mask equal to that of the classful mask.

Supernet route - A supernet route is a network address with a mask less than the classful mask, for example, a summary address.

Default route - A default route is a static route with the address 0.0.0.0/0.

The source of the level 1 route can be a directly connected network, static route, or a dynamic routing protocol.

IPv4 Routing Table – Level 2 Child Route

A level 2 child route is a route that is a subnet of a classful network address. A level 1 parent route is a level 1 network route that is subnetted. A level 1 parent routes contain level 2 child routes.

Like a level 1 route, the source of a level 2 route can be a directly connected network, a static route, or a dynamically learned route. Level 2 child routes are also ultimate routes.

Please see the slides for an example of it.

74. TOPIC

Synopsis/ Description

- Route Summarization
- Configuration of IPv4 Route Summarization
- Verifying Connectivity

Topology

Addressing Table

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
R1	G0/0	172.31.157.1	255.255.255.192	N/A
	S0/0/0	172.31.157.97	255.255.255.252	N/A
R2	G0/0	172.31.157.65	255.255.255.224	N/A
	S0/0/0	172.31.157.98	255.255.255.252	N/A
	S0/0/1	172.22.1.129	255.255.255.224	N/A
R3	G0/0	172.22.1.161	255.255.255.224	N/A
	S0/0/1	172.22.1.158	255.255.255.224	N/A
PC1	NIC	172.31.157.62	255.255.255.192	172.31.157.1
PC2	NIC	172.31.157.94	255.255.255.224	172.31.157.65
PC3	NIC	172.22.1.190	255.255.255.224	172.22.1.161

Part 1: Calculate Summary Routes

Step 1: Calculate a summary route on R1 to reach LANs on R3.

- a. List the 172.22.1.128/27 and 172.22.1.160/27 networks in binary format.

172.22.1.128: 10101100.00010110.00000001.10000000

172.22.1.160: 10101100.00010110.00000001.10100000

- b. Count the left-most matching bits to determine the mask for the summary route. They have 26 left-most bits in common.

172.22.1.128: 10101100.00010110.00000001.10000000

172.22.1.160: 10101100.00010110.00000001.10100000

- c. Copy the matching bits and fill in the remaining bits with zeros to determine the summarized network address.

10101100.00010110.00000001.10000000

- d. What is the summarized network address and subnet mask? 172.22.1.128 255.255.255.192

Step 2: Calculate a summary route on R3 to reach LANs on R1 and R2.

a. Calculate the summary route for the 172.31.157.0/26, 172.31.157.64/27, and 172.31.157.96/30 networks. List the networks in binary format. Then, count the left-most matching bits to determine the mask for the summary route.

10101100.00011111.10011101.00000000

10101100.00011111.10011101.01000000

10101100.00011111.10011101.01100000

b. What is the summarized network address and subnet mask? **172.31.157.0 255.255.255.128**

Part 2: Configure Summary Routes

Step 1: Configure a summary route for R1.

Configure the recursive summary route that you calculated in Part 1, Step 1.

R1(config)# ip route 172.22.1.128 255.255.255.192 172.31.157.98

Step 2: Configure a summary route for R3.

Configure the directly attached summary route that you calculated in Part 1, Step 2.

R3(config)# ip route 172.31.157.0 255.255.255.128 serial 0/0/1

Part 3: Verify Connectivity

Verify that all PC hosts and routers can ping other PC hosts and routers in the topology. If not, troubleshoot and correct the issues.

75. TOPIC

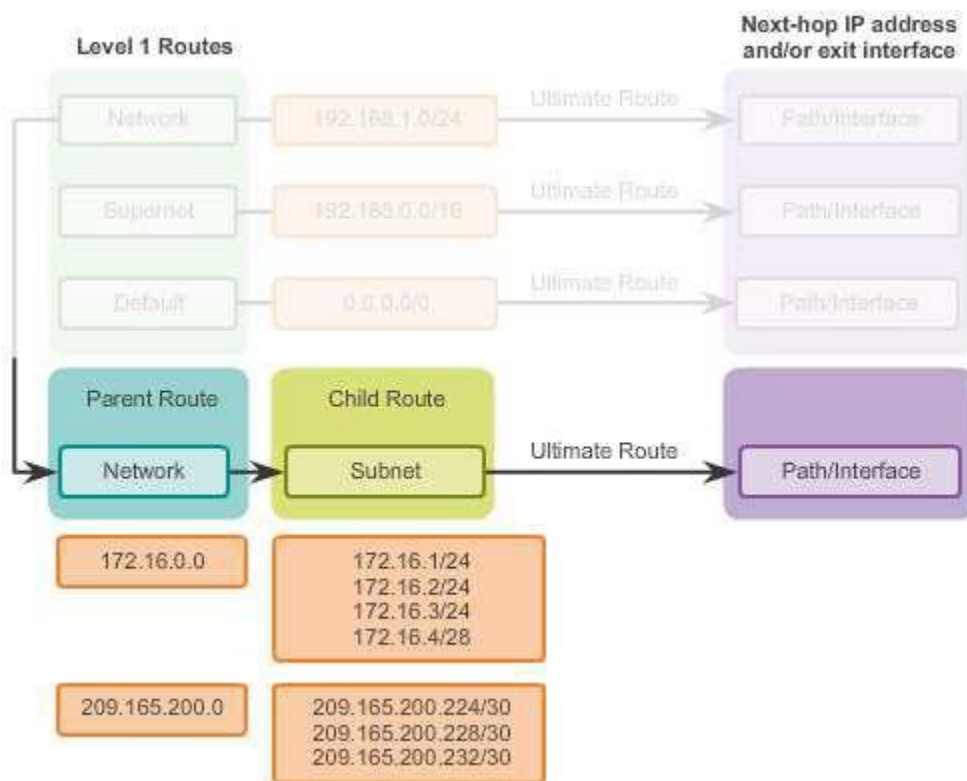
Only practical demonstration in the topic

76. TOPIC

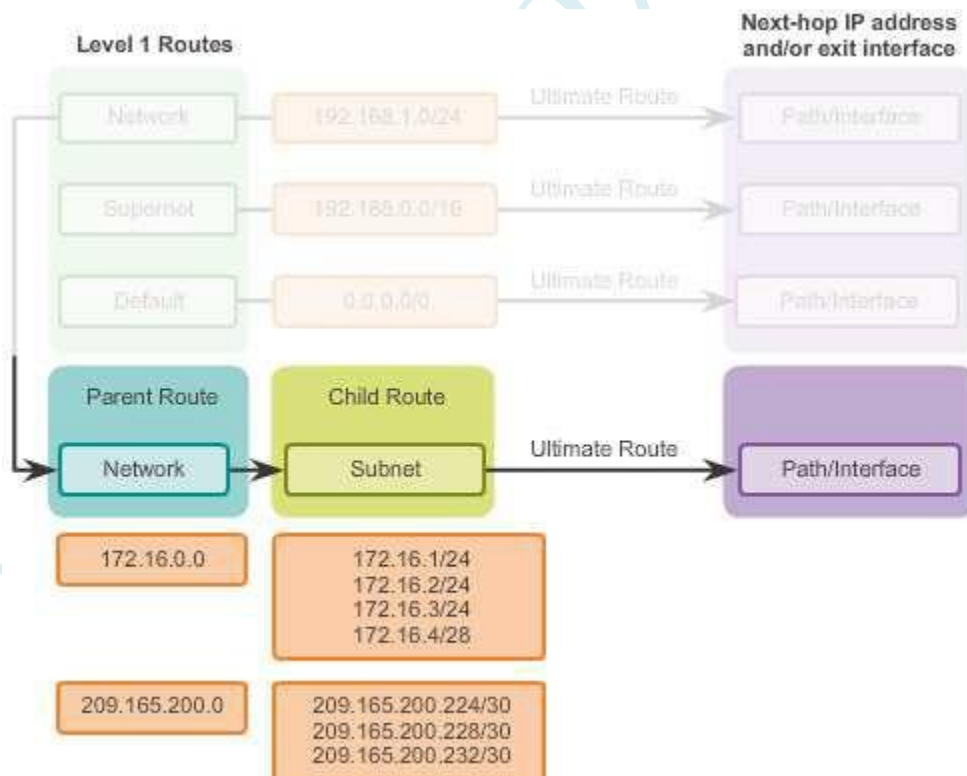
When a packet arrives on a router interface, the router examines the IPv4 header, identifies the destination IPv4 address, and proceeds through the router lookup process.

In Figure below, the router examines level 1 network routes for the best match with the destination address of the IPv4 packet.

1. If the best match is a level 1 ultimate route, then this route is used to forward the packet.



2. If the best match is a level 1 parent route, proceed to the next step.



In Figure above, the router examines child routes (the subnet routes) of the parent route for a best match.

3. If there is a match with a level 2 child route, that subnet is used to forward the packet.



If there is not a match with any of the level 2 child routes, proceed to the next step.

In Figure above, the router continues searching level 1 supernet routes in the routing table for a match, including the default route, if there is one.

5. If there is now a lesser match with a level 1 supernet or default routes, the router uses that route to forward the packet.
6. If there is not a match with any route in the routing table, the router drops the packet.

Note: A route referencing only a next-hop IP address and not an exit interface must be resolved to a route with an exit interface. A recursive lookup is performed on the next-hop IP address until the route is resolved to an exit interface.

Best Route = Longest Match

What is meant by the router must find the best match in the routing table? Best match is equal to the longest match.

For there to be a match between the destination IPv4 address of a packet and a route in the routing table, a minimum number of far left bits must match between the IPv4 address of the packet and the route in the routing table. The subnet mask of the route in the routing table is used to determine the minimum number of far left bits that must match. Remember that an IPv4 packet only contains the IPv4 address and not the subnet mask.

The best match is the route in the routing table that has the most number of far left matching bits with the destination IPv4 address of the packet. The route with the greatest number of equivalent far left bits, or the longest match, is always the preferred route.

In the figure below, a packet is destined for 172.16.0.10. The router has three possible routes that match this packet: 172.16.0.0/12, 172.16.0.0/18, and 172.16.0.0/26. Of the three routes, 172.16.0.0/26 has the longest match and is therefore chosen to forward the packet. Remember, for any of these routes to be considered a match there must be at least the number of matching bits indicated by the subnet mask of the route.

IP Packet Destination	172.16.0.10	10101100.00010000.00000000.00001010
Route 1	172.16.0.0/12	10101100.00010000.00000000.00000000
Route 2	172.16.0.0/18	10101100.00010000.00000000.00000000
Route 3	172.16.0.0/26	10101100.00010000.00000000.00000000

Longest Match to IP Packet Destination

77. TOPIC

	Interior Gateway Protocols		Exterior Gateway Protocols	
	Distance Vector Routing Protocols		Link State Routing Protocols	Path Vector
Classful	RIP	IGRP		EGP
Classless	RIPv2	EIGRP	OSPFv2	BGPv4
IPv6	RIPng	EIGRP for IPv6	OSPFv3	BGPv4 for IPv6

As shown in Figure above, OSPF version 2 (OSPFv2) is available for IPv4 while OSPF version 3 (OSPFv3) is available for IPv6.

The initial development of OSPF began in 1987 by the Internet Engineering Task Force (IETF) OSPF Working Group. At that time, the Internet was largely an academic and research network funded by the U.S. government.

In 1989, the specification for OSPFv1 was published in RFC 1131. Two implementations were written. One implementation was developed to run on routers and the other to run on UNIX workstations. The

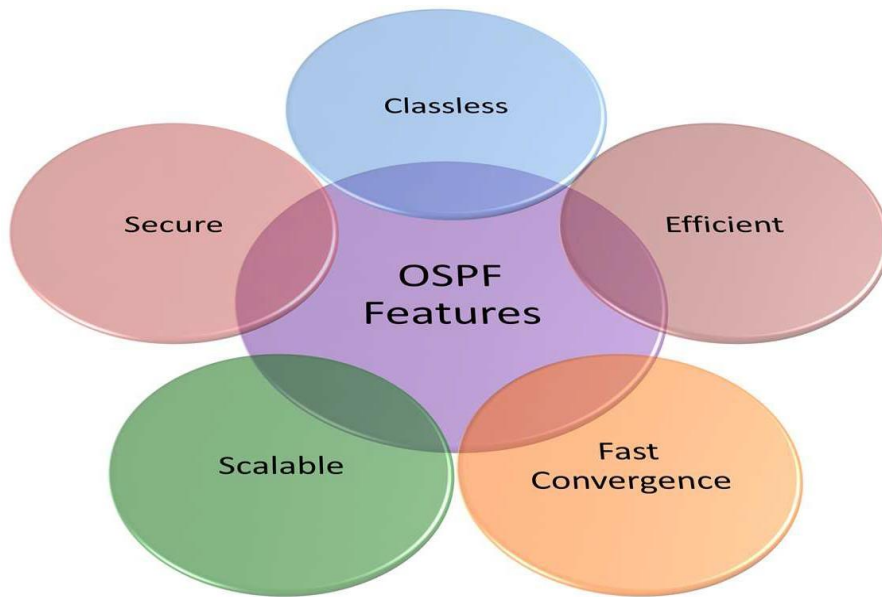
latter implementation became a widespread UNIX process known as GATED. OSPFv1 was an experimental routing protocol and was never deployed.

In 1991, OSPFv2 was introduced in RFC 1247 by John Moy. OSPFv2 offered significant technical improvements over OSPFv1. It is classless by design; therefore, it supports VLSM and CIDR.

At the same time the OSPF was introduced, ISO was working on a link-state routing protocol of their own, Intermediate System-to-Intermediate System (IS-IS). IETF chose OSPF as their recommended Interior Gateway Protocol (IGP).

In 1998, the OSPFv2 specification was updated in RFC 2328, which remains the current RFC for OSPF. In 2008, OSPFv3 was updated in RFC 5340 as OSPF for IPv6.

Features of OSPF



OSPF features, as shown in Figure above, include:

Classless - It is classless by design; therefore, it supports VLSM and CIDR.

Efficient - Routing changes trigger routing updates (no periodic updates). It uses the SPF algorithm to choose the best path.

Fast convergence - It quickly propagates network changes.

Scalable - It works well in small and large network sizes. Routers can be grouped into areas to support a hierarchical system.

Secure - It supports Message Digest 5 (MD5) authentication. When enabled, OSPF routers only accept encrypted routing updates from peers with the same pre-shared password.

Administrative distance (AD) is the trustworthiness (or preference) of the route source. OSPF has a default administrative distance of 110.

Components of OSPF

All routing protocols share similar components. They all use routing protocol messages to exchange route information. The messages help build data structures, which are then processed using a routing algorithm.

The three main components of the OSPF routing protocol include:

Data Structures

OSPF creates and maintains three databases:

- Adjacency database - Creates the neighbor table
- Link-state database (LSDB) - Creates the topology table
- Forwarding database - Creates the routing table

These tables contain a list of neighboring routers to exchange routing information with and are kept and maintained in RAM.

Routing Protocol Messages

OSPF exchanges messages to convey routing information using five types of packets.

- Hello packet
- Database description packet
- Link-state request packet
- Link-state update packet
- Link-state acknowledgment packet

These packets are used to discover neighboring routers and also to exchange routing information to maintain accurate information about the network.

Algorithm

The CPU processes the neighbor and topology tables using Dijkstra's SPF algorithm. The SPF algorithm is based on the cumulative cost to reach a destination.

The SPF algorithm creates an SPF tree by placing each router at the root of the tree and calculating the shortest path to each node. The SPF tree is then used to calculate the best routes. OSPF places the best routes into the forwarding database, which is used to make the routing table.

Link State Operation

To maintain routing information, OSPF routers complete the following generic link-state routing process to reach a state of convergence:

1. Establish Neighbor Adjacencies - OSPF-enabled routers must recognize each other on the network before they can share information. An OSPF-enabled router sends Hello packets out all OSPF-enabled interfaces to determine if neighbors are present on those links. If a neighbor is present, the OSPF-enabled router attempts to establish a neighbor adjacency with that neighbor.
2. Exchange Link-State Advertisements - After adjacencies are established, routers then exchange link-state advertisements (LSAs). LSAs contain the state and cost of each directly connected link. Routers flood their LSAs to adjacent neighbors. Adjacent neighbors receiving the LSA immediately flood the LSA to other directly connected neighbors, until all routers in the area have all LSAs.

3. Build the Topology Table - After LSAs are received, OSPF-enabled routers build the topology table (LSDB) based on the received LSAs. This database eventually holds all the information about the topology of the network.
4. Execute the SPF Algorithm - Routers then execute the SPF algorithm. The SPF algorithm creates the SPF tree.

From the SPF tree, the best paths are inserted into the routing table. Routing decisions are made based on the entries in the routing table.

Single Area and Multiarea OSPF

To make OSPF more efficient and scalable, OSPF supports hierarchical routing using areas. An OSPF area is a group of routers that share the same link-state information in their LSDBs.

OSPF can be implemented in one of two ways:

- **Single-Area OSPF** - All routers are in one area called the backbone area (area 0).
- **Multiarea OSPF** - OSPF is implemented using multiple areas, in a hierarchical fashion. All areas must connect to the backbone area (area 0). Routers interconnecting the areas are referred to as Area Border Routers (ABR).

With multiarea OSPF, OSPF can divide one large autonomous system (AS) into smaller areas, to support hierarchical routing. With hierarchical routing, routing still occurs between the areas (interarea routing), while many of the processor intensive routing operations, such as recalculating the database, are kept within an area.

For instance, any time a router receives new information about a topology change within the area (including the addition, deletion, or modification of a link) the router must rerun the SPF algorithm, create a new SPF tree, and update the routing table. The SPF algorithm is CPU-intensive and the time it takes for calculation depends on the size of the area.

Note: Topology changes are distributed to routers in other areas in a distance vector format. In other words, these routers only update their routing tables and do not need to rerun the SPF algorithm.

Too many routers in one area would make the LSDBs very large and increase the load on the CPU. Therefore, arranging routers into areas effectively partitions a potentially large database into smaller and more manageable databases.

The hierarchical-topology possibilities of multiarea OSPF have these advantages:

- **Smaller routing tables** - Fewer routing table entries because network addresses can be summarized between areas. Route summarization is not enabled by default.
- **Reduced link-state update overhead** - Minimizes processing and memory requirements.
- **Reduced frequency of SPF calculations** - Localizes the impact of a topology change within an area. For instance, it minimizes routing update impact because LSA flooding stops at the area boundary.

78. TOPIC

OSPF Message Encapsulation

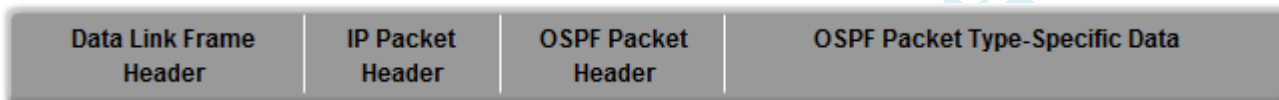
OSPF messages transmitted over an Ethernet link contain the following information:

Data Link Ethernet Frame Header - Identifies the destination multicast MAC addresses 01-00-5E-00-00-05 or 01-00-5E-00-00-06.

IP Packet Header - Identifies the IPv4 protocol field 89 which indicates that this is an OSPF packet. It also identifies one of two OSPF multicast addresses, 224.0.0.5 or 224.0.0.6.

OSPF Packet Header - Identifies the OSPF packet type, the router ID and the area ID.

OSPF Packet Type Specific Data - Contains the OSPF packet type information. The content differs depending on the packet type. In this case, it is an IPv4 Header.



OSPF Hello Packet

Hello Packet

The OSPF Type 1 packet is the Hello packet. Hello packets are used to:

Discover OSPF neighbors and establish neighbor adjacencies.

Advertise parameters on which two routers must agree to become neighbors.

Elect the Designated Router (DR) and Backup Designated Router (BDR) on multiaccess networks like Ethernet and Frame Relay. Point-to-point links do not require DR or BDR.

Fields contained in the Type 1 Hello packet:

Type - Identifies the type of packet. A one (1) indicates a Hello packet. A value 2 identifies a DBD packet, 3 an LSR packet, 4 an LSU packet, and 5 an LSAck packet.

Router ID - A 32-bit value expressed in dotted decimal notation (an IPv4 address) used to uniquely identifying the originating router.

Area ID - Area from which the packet originated.

Network Mask - Subnet mask associated with the sending interface.

Hello Interval - Specifies the frequency, in seconds, at which a router sends Hello packets. The default Hello interval on multiaccess networks is 10 seconds. This timer must be the same on neighboring routers; otherwise, an adjacency is not established.

Router Priority - Used in a DR/BDR election. The default priority for all OSPF routers is 1, but can be manually altered from 0 to 255. The higher the value, the more likely the router becomes the DR on the link.

Dead Interval - Is the time in seconds that a router waits to hear from a neighbor before declaring the neighboring router out of service. By default, the router Dead Interval is four times the Hello interval. This timer must be the same on neighboring routers; otherwise, an adjacency is not established.

Designated Router (DR) - Router ID of the DR.

Backup Designated Router (BDR) - Router ID of the BDR.

List of Neighbors - List that identifies the router IDs of all adjacent routers.

OSPF Hello Interval

OSPF Hello packets are transmitted to multicast address 224.0.0.5 in IPv4 and FF02::5 in IPv6 (all OSPF routers) every:

- 10 seconds (default on multiaccess and point-to-point networks)
- 30 seconds (default on nonbroadcast multiaccess [NBMA] networks; for example, Frame Relay)

The Dead interval is the period that the router waits to receive a Hello packet before declaring the neighbor down. If the Dead interval expires before the routers receive a Hello packet, OSPF removes that neighbor from its LSDB. The router floods the LSDB with information about the down neighbor out all OSPF-enabled interfaces.

Cisco uses a default of 4 times the Hello interval:

- 40 seconds (default on multiaccess and point-to-point networks)
- 120 seconds (default on NBMA networks; for example, Frame Relay)

OSPF Link State Updates

Routers initially exchange Type 2 DBD packets, which is an abbreviated list of the sending router's LSDB and is used by receiving routers to check against the local LSDB.

A Type 3 LSR packet is used by the receiving routers to request more information about an entry in the DBD.

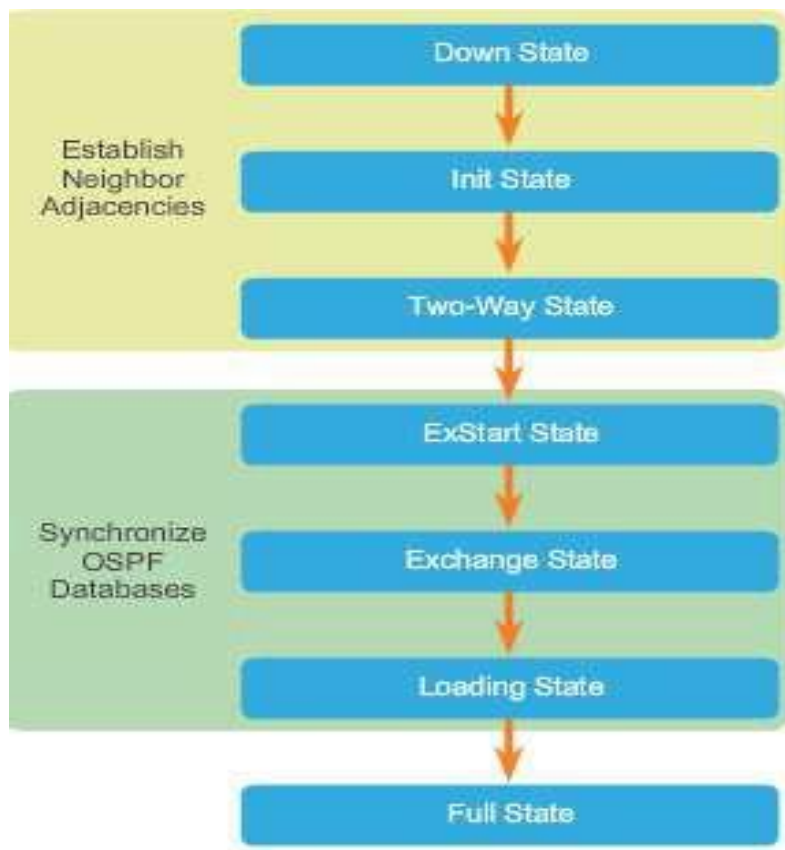
The Type 4 LSU packet is used to reply to an LSR packet.

LSUs are also used to forward OSPF routing updates, such as link changes. Specifically, an LSU packet can contain 11 different types of OSPFv2 LSAs. OSPFv3 renamed several of these LSAs and also contains two additional LSAs.

Note: The difference between the LSU and LSA terms can sometimes be confusing because these terms are often used interchangeably. However, an LSU contains one or more LSAs.

79. TOPIC

OSPF Operational States



Establish Neighbor Adjacencies

When OSPF is enabled on an interface, the router must determine if there is another OSPF neighbor on the link. To accomplish this, the router forwards a Hello packet that contains its router ID out all OSPF-enabled interfaces. The OSPF router ID is used by the OSPF process to uniquely identify each router in the OSPF area. A router ID is an IP address assigned to identify a specific router among OSPF peers.

When a neighboring OSPF-enabled router receives a Hello packet with a router ID that is not within its neighbor list, the receiving router attempts to establish an adjacency with the initiating router.

The action performed in Two-Way state depends on the type of inter-connection between the adjacent routers:

If the two adjacent neighbors are interconnected over a point-to-point link, then they immediately transition from the Two-Way state to the database synchronization phase.

If the routers are interconnected over a common Ethernet network, then a designated router DR and a BDR must be elected.

Hello packets are continually exchanged to maintain router information. OSPF
DR and BDR

Why is a DR and BDR election necessary?

Multiaccess networks can create two challenges for OSPF regarding the flooding of LSAs:

Creation of multiple adjacencies - Ethernet networks could potentially interconnect many OSPF routers over a common link. Creating adjacencies with every router is unnecessary and undesirable. It would lead to an excessive number of LSAs exchanged between routers on the same network.

Extensive flooding of LSAs - Link-state routers flood their LSAs any time OSPF is initialized, or when there is a change in the topology. This flooding can become excessive.

The solution to managing the number of adjacencies and the flooding of LSAs on a multiaccess network is the DR. On multiaccess networks, OSPF elects a DR to be the collection and distribution point for LSAs sent and received. A BDR is also elected in case the DR fails. All other routers become DROTHERs. A DROTHER is a router that is neither the DR nor the BDR.



Synchronizing OSPF Databases

After the Two-Way state, routers transition to database synchronization states. While the Hello packet was used to establish neighbor adjacencies, the other four types of OSPF packets are used during the process of exchanging and synchronizing LSDBs.

In the ExStart state, a master and slave relationship is created between each router and its adjacent DR and BDR. The router with the higher router ID acts as the master for the Exchange state. In Figure above, R2 becomes the master.

In the Exchange state, the master and slave routers exchange one or more DBD packets. A DBD packet includes information about the LSA entry header that appears in the router's LSDB. The entries can be about a link or about a network. Each LSA entry header includes information about the link-state type, the address of the advertising router, the link's cost, and the sequence number. The router uses the sequence number to determine the newness of the received link-state information.

In Figure below, R2 sends a DBD packet to R1. When R1 receives the DBD, it performs the following actions:

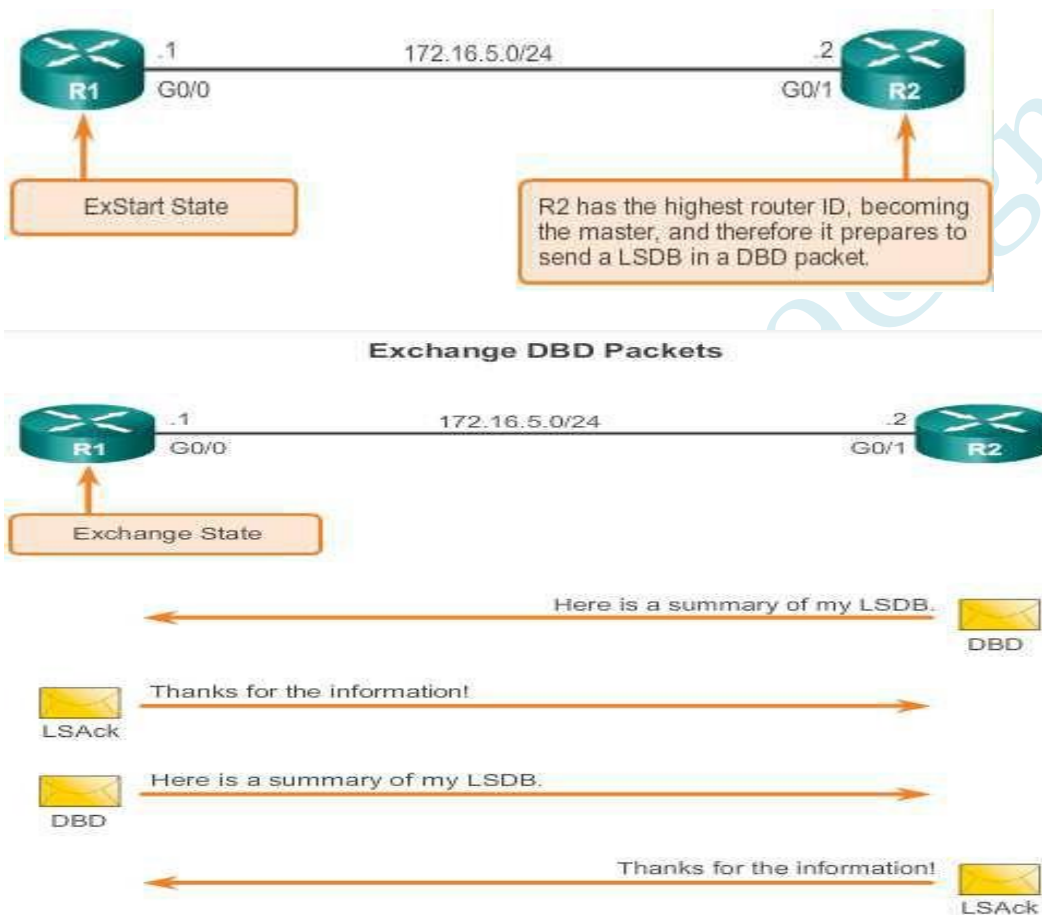
1. It acknowledges the receipt of the DBD using the LSack packet.
2. R1 then sends DBD packets to R2.
3. R2 acknowledges R1.

R1 compares the information received with the information it has in its own LSDB. If the DBD packet has a more current link-state entry, the router transitions to the Loading state.

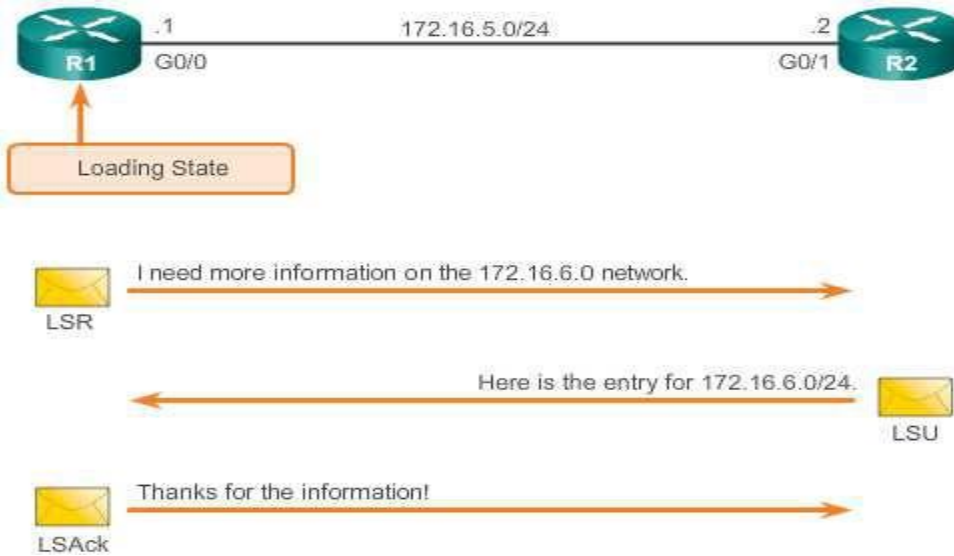
After all LSRs have been satisfied for a given router, the adjacent routers are considered synchronized and in a full state.

As long as the neighboring routers continue receiving Hello packets, the network in the transmitted LSAs remain in the topology database. After the topological databases are synchronized, updates (LSUs) are sent only to neighbors when:

- A change is perceived (incremental updates)
- Every 30 minutes



Getting Additional Route Information



80. TOPIC

Introduced in 1991, OSPFv2 is a link-state routing protocol for IPv4. OSPF was designed as an alternative to another IPv4 routing protocol, RIP.

Every router requires a router ID to participate in an OSPF domain. The router ID can be defined by an administrator or automatically assigned by the router. The router ID is used by the OSPF-enabled router to:

Uniquely identify the router - The router ID is used by other routers to uniquely identify each router within the OSPF domain and all packets that originate from them.

Participate in the election of the DR - In a multiaccess LAN environment, the election of the DR occurs during initial establishment of the OSPF network. When OSPF links become active, the routing device configured with the highest priority is elected the DR. Assuming there is no priority configured, or there is a tie, then the router with the highest router ID is elected the DR. The routing device with the second highest router ID is elected the BDR.

But how does the router determine the router ID? Cisco routers derive the router ID based on one of three criteria, in the following preferential order:

The router ID is explicitly configured using the OSPF router-id `router configuration mode` command. The `router-id` value is any 32-bit value expressed as an IPv4 address. This is the recommended method to assign a router ID.

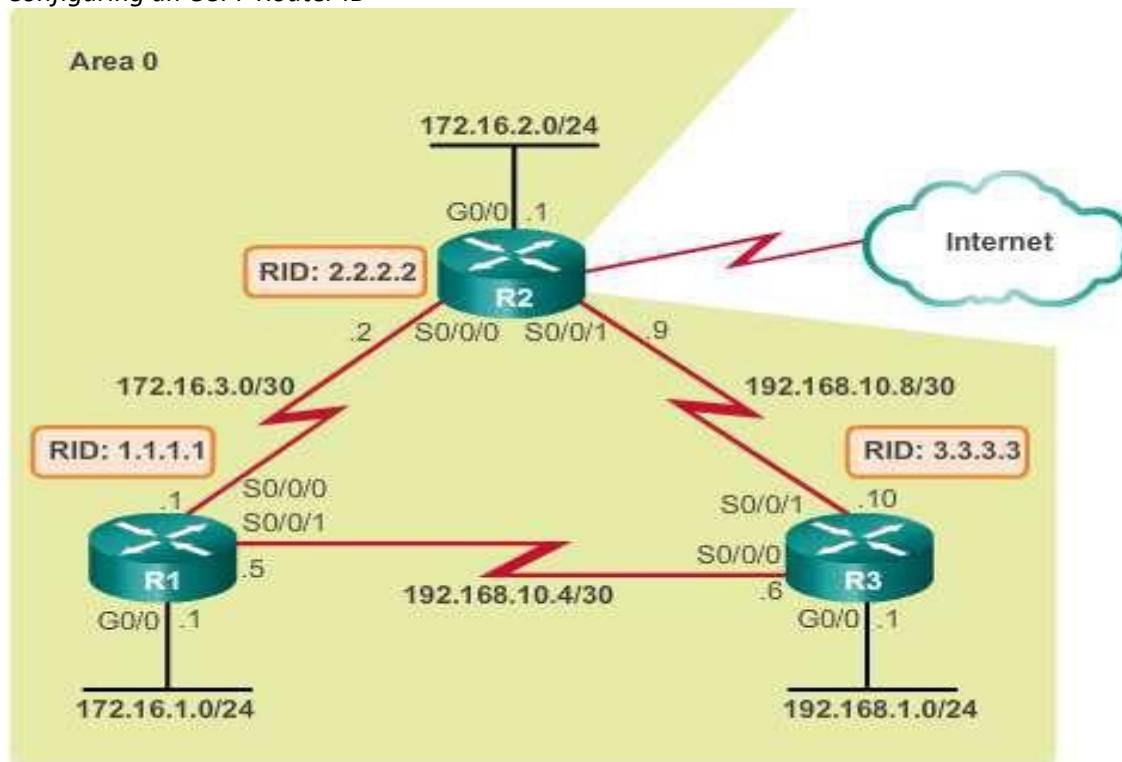
If the router ID is not explicitly configured, the router chooses the highest IPv4 address of any of configured loopback interfaces. This is the next best alternative to assigning a router ID.

If no loopback interfaces are configured, then the router chooses the highest active IPv4 address of any of its physical interfaces. This is the least recommended method because it makes it more difficult for administrators to distinguish between specific routers.

If the router uses the highest IPv4 address for the router ID, the interface does not need to be OSPF-enabled. This means that the interface address does not need to be included in one of the OSPF network commands for the router to use that IP address as the router ID. The only requirement is that the interface is active and in the up state.

Note: The router ID looks like an IP address, but it is not routable and, therefore, is not included in the routing table, unless the OSPF routing process chooses an interface (physical or loopback) that is appropriately defined by a network command.

Configuring an OSPF Router ID



Use the router-id rid router configuration mode command to manually assign a 32-bit value expressed as an IPv4 address to a router. An OSPF router identifies itself to other routers using this router ID.

As shown in Figure above, R1 is configured with a router ID of 1.1.1.1, R2 with 2.2.2.2, and R3 with 3.3.3.3.

Use the show ip protocols command to verify the router ID.

Sometimes a router ID needs to be changed, for example, when a network administrator establishes a new router ID scheme for the network. However, after a router selects a router ID, an active OSPF router does not allow the router ID to be changed until the router is reloaded or the OSPF process cleared.

Clearing the OSPF process is the preferred method to reset the router ID.

The OSPF routing process is cleared using the clear ip ospf process privileged EXEC mode command. This forces OSPF on R1 to transition to the Down and Init states. The show ip protocols command verifies that the router ID has changed.

Using a Loopback Interface as the Router ID

A router ID can also be assigned using a loopback interface.

The IPv4 address of the loopback interface should be configured using a 32-bit subnet mask (255.255.255.255). This effectively creates a host route. A 32-bit host route does not get advertised as a route to other OSPF routers.

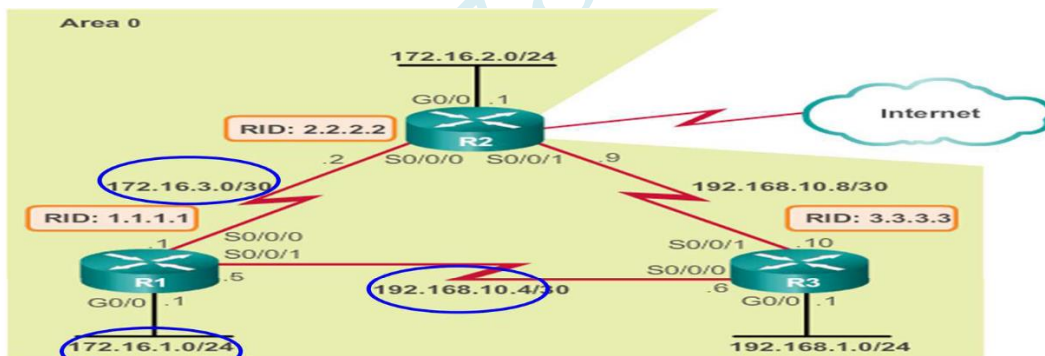
Enabling OSPF on Interfaces

The network command determines which interfaces participate in the routing process for an OSPF area. Any interfaces on a router that match the network address in the network command are enabled to send and receive OSPF packets. As a result, the network (or subnet) address for the interface is included in OSPF routing updates.

The basic command syntax is `network network-address wildcard-mask area area-id`.

The area area-id syntax refers to the OSPF area. When configuring single-area OSPF, the network command must be configured with the same area-id value on all routers. Although any area ID can be used, it is good practice to use an area ID of 0 with single-area OSPF. This convention makes it easier if the network is later altered to support multiarea OSPF.

The figure below displays the reference topology.



```
R1(config)# router ospf 10
R1(config-router)# route-id 1.1.1.1
R1(config-router)# network 172.16.1.0 0.0.0.255 area 0
R1(config-router)# network 172.16.3.0 0.0.0.3 area 0
R1(config-router)# network 192.168.10.4 0.0.0.3 area 0
R1(config-router)# end
```

As an alternative, OSPFv2 can be enabled using the `network intf-ip-address 0.0.0.0 area area-id router configuration mode` command.

The advantage of specifying the interface is that the wildcard mask calculation is not necessary. OSPFv2 uses the interface address and subnet mask to determine the network to advertise.

81. TOPIC

Synopsis/Description

- Network Convergence
- Routing Protocols and Network Convergence

Convergence is the process of routers agreeing on optimal routes for forwarding packets and thereby completing the updating of their routing tables. When this occurs, each router independently runs a routing algorithm to recalculate metrics and build a new routing table based on this information.

The network has converged when all routers have complete and accurate information about the entire network. Convergence time is the time it takes routers to share information, calculate best paths, and update their routing tables. A network is not completely operable until the network has converged; therefore, most networks require short convergence times.

Convergence is both collaborative and independent. The routers share information with each other, but must independently calculate the impacts of the topology change on their own routes. Because they develop an agreement with the new topology independently, they are said to converge on this consensus.

Convergence properties include the speed of propagation of routing information and the calculation of optimal paths. The speed of propagation refers to the amount of time it takes for routers within the network to forward routing information.

Routing protocols can be rated based on the speed to convergence; the faster the convergence, the better the routing protocol. Generally, older protocols, such as RIP, are slow to converge, whereas modern protocols, such as EIGRP and OSPF, converge more quickly.

82. TOPIC

Synopsis/Description

- i. Build the Network
- ii. Configure Device Settings
- iii. Configure RIPv2
- iv. Verify RIPv2

Background/Scenario

In this activity, you will configure static and default routes. A static route is a route that is entered manually by the network administrator to create a route that is reliable and safe. There are four different static routes that are used in this activity: a recursive static route, a directly connected static route, a fully specified static route, and a default route.

83. TOPIC

Synopsis/Description

- I. Routing Information Protocol

- II. Extended Interior Gateway Routing Protocol
- III. RIP vs EIGRP

SR.NO	RIP	EIGRP
1.	RIP stands for Routing Information Protocol.	EIGRP stands for Enhanced Interior Gateway Routing Protocol.
2.	RIP works on Bellman Ford algorithm.	EIGRP works on DUAL(Diffusing Update Algorithm) Algorithm.
3.	It is a industry standard dynamic routing protocol.	It is a Cisco standard routing protocol.
4.	It is basically use for smaller size organization.	It is basically use for lager size organization as compared to RIP.
5.	RIP is a distance vector protocol.	EIGRP is derived from Integrated Gateway Routing Protocol.
6.	It allow maximum hop count upto 15.	It allow maximum hop count upto 255.
7.	It's administrative distance is 120.	It's administrative distance is 90.
8.	It is not a more intelligent dynamic routing protocol.	It is a more intelligent routing protocol than RIP.
9.	It calculates the metric in terms of Hop Count.	It calculates the metric in terms of bandwidth and delay.
10.	RIP routing protocol creates two table in the router: Routing Table, and Topology Table.	EIGRP routing protocol creates three table In the router: Neighbor Table, Topology Table, and Routing Tables

84. TOPIC

- **Metric** - The RFCs for both OSPFv2 and OSPFv3 define the metric as the cost of sending packets out the interface. OSPFv2 and OSPFv3 can be modified using the **auto-cost reference bandwidth ref-bw** router configuration mode command. The command only influences the OSPF metric where it was configured. For example, if this command was entered for OSPFv3, it does not affect the OSPFv2 routing metrics.
- **Areas** - The concept of multiple areas in OSPFv3 is the same as in OSPFv2. Multiareas that minimize link-state flooding and provide better stability with the OSPF domain.
- **OSPF packet types** - OSPFv3 uses the same five basic packet types as OSPFv2 (Hello, DBD, LSR, LSU, and LSAck).
- **Neighbor discovery mechanism** - The neighbor state machine, including the list of OSPF neighbor states and events, remains unchanged. OSPFv2 and OSPFv3 use the Hello mechanism to learn about neighboring routers and form adjacencies. However, in OSPFv3, there is no requirement for matching subnets to form neighbor adjacencies. This is because neighbor adjacencies are formed using link-local addresses, not global unicast addresses.
- **DR/BDR election process** - The DR/BDR election process remains unchanged in OSPFv3.
- **Router ID** - Both OSPFv2 and OSPFv3 use a 32-bit number for the router ID represented in dotted-decimal notation. Typically this is an IPv4 address. The OSPF **router-id** command must be used to

configure the router ID. The process in determining the 32-bit Router ID is the same in both protocols. Use an explicitly-configured router ID; otherwise, the highest loopback IPv4 address becomes the router ID.

OSPFv2 vs. OSPFv3

The differences between OSPFv2 and OSPFv3:

- **Advertises** - OSPFv2 advertises IPv4 routes, whereas OSPFv3 advertises routes for IPv6.
- **Source address** - OSPFv2 messages are sourced from the IPv4 address of the exit interface. In OSPFv3, OSPF messages are sourced using the link-local address of the exit interface.
- **All OSPF router multicast addresses**- OSPFv2 uses 224.0.0.5; whereas, OSPFv3 uses FF02::5.
- **DR/BDR multicast address** - OSPFv2 uses 224.0.0.6; whereas, OSPFv3 uses FF02::6.
- **Advertise networks** - OSPFv2 advertises networks using the **network** router configuration command; whereas, OSPFv3 uses the **ipv6 ospf process-id area area-id** interface configuration command.
- **IP unicast routing** - Enabled, by default, in IPv4; whereas, the **ipv6 unicast-routing** global configuration command must be configured.

Authentication - OSPFv2 uses either plaintext authentication or MD5 authentication. OSPFv3 uses IPv6 authentication.

85. TOPIC

Verify OSPFv2:

Figure below shows the reference topology.

Use the **show ip ospf neighbor** command to verify that the router has formed an adjacency with its neighboring routers. If the router ID of the neighboring router is not displayed, or if it does not show as being in a state of FULL, the two routers have not formed an OSPF adjacency.

If two routers do not establish adjacency, link-state information is not exchanged. Incomplete LSDBs can cause inaccurate SPF trees and routing tables. Routes to destination networks may not exist, or may not be the most optimum path.

CS407

Routing and switching
Reading Material

Made by Mah jabeen
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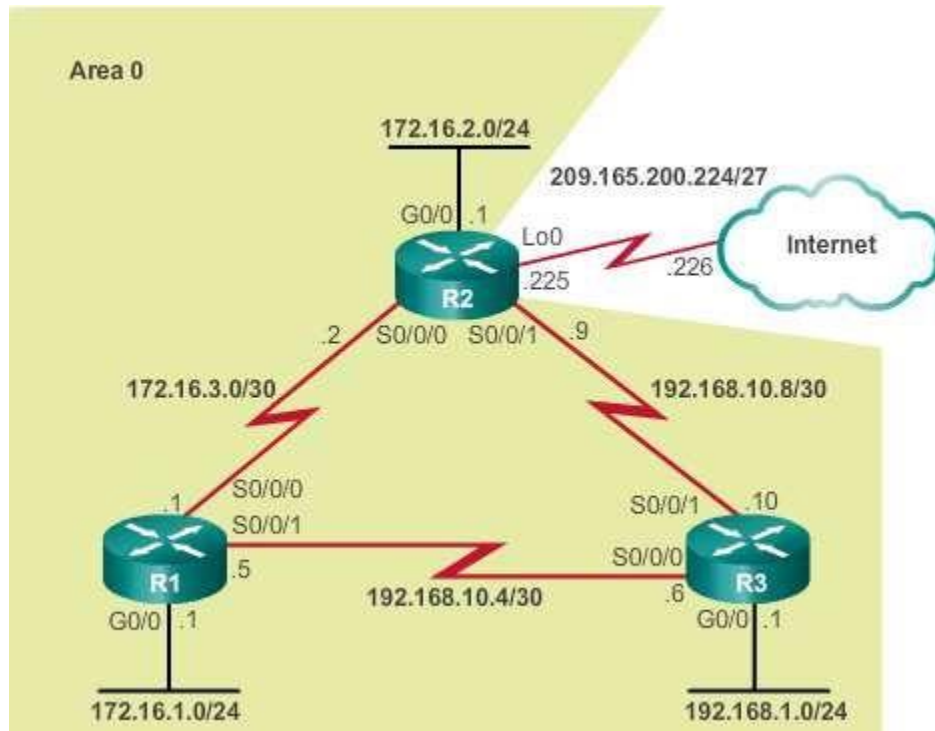


Figure below displays the neighbor adjacency of R1. For each neighbor, this command displays the following output:

Neighbor ID - The router ID of the neighboring router.

Pri - The OSPF priority of the interface. This value is used in the DR and BDR election.

State - The OSPF state of the interface. FULL state means that the router and its neighbor have identical OSPF LSDBs. On multi-access networks, such as Ethernet, two routers that are adjacent may have their states displayed as 2WAY. The dash indicates that no DR or BDR is required because of the network type.

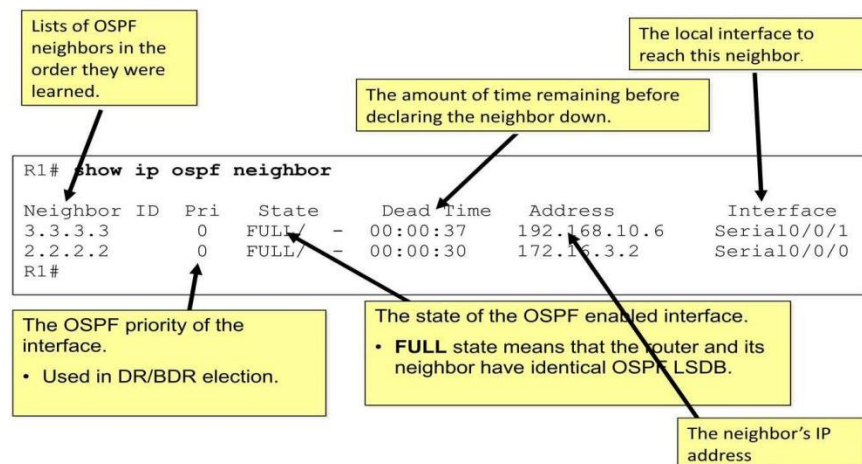
Dead Time - The amount of time remaining that the router waits to receive an OSPF Hello packet from the neighbor before declaring the neighbor down. This value is reset when the interface receives a Hello packet.

Address - The IPv4 address of the neighbor's interface to which this router is directly connected.

Interface - The interface on which this router has formed adjacency with the neighbor.

Two routers may not form an OSPF adjacency if:

- The subnet masks do not match, causing the routers to be on separate networks.
- OSPF Hello or Dead Timers do not match.
- OSPF Network Types do not match.
- There is a missing or incorrect OSPF **network** command.



Verify OSPF Interface Settings

The quickest way to verify OSPF interface settings is to use the **show ip ospf interface** command. This command provides a detailed list for every OSPF-enabled interface. The command is useful to determine whether the **network** statements were correctly composed.

To get a summary of OSPF-enabled interfaces, use the **show ip ospf interface brief**.

```
R1# show ip ospf interface brief
```

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Se0/0/1	10	0	192.168.10.5/30	15625	P2P	1/1	
Se0/0/0	10	0	172.16.3.1/30	647	P2P	1/1	
Gi0/0	10	0	172.16.1.1/24	1	DR	0/0	

R1#

Clearing Routing Table:

To clear all routes from the IP routing table, use:

- Router# clear ip route *

To clear a specific route from the IP routing table, use:

- Router# clear ip route A.B.C.D

OSPF Commands and Description:

Show ip protocols: The **show ip protocol** is used to verify the type of protocols running on the router and the network. It displays the OSPF process ID, OSPF router ID, type of OSPF area, networks and administrative distance for OSPF neighboring router's ID, etc.

show ip ospf neighbor: The **show ip ospf neighbor** displays the summary of ospf information regarding the **neighbor relationship**. It is also used for **the production network**.

show ip route: The “show ip route” command is used to display the routing table.

show ip ospf interface: The “show ip ospf interface” command is used to display hello interval and dead interval.

show ip ospf: The show ip ospf command is used to display the OSPF information like process ID, router ID, area information, Shortest Path First (STF) statistics, and the Last time SPF algorithm calculated.

86. TOPIC

OSPFv3 vs. OSPFv2:

- OSPFv3 is the OSPFv2 equivalent for exchanging IPv6 prefixes. Recall that in IPv6, the network address is referred to as the prefix and the subnet mask is called the prefix-length.
- Similar to its IPv4 counterpart, OSPFv3 exchanges routing information to populate the IPv6 routing table with remote prefixes.
- OSPFv2 runs over the IPv4 network layer, communicating with other OSPF IPv4 peers, and advertising only IPv4 routes.
- OSPFv3 has the same functionality as OSPFv2, but uses IPv6 as the network layer transport, communicating with OSPFv3 peers and advertising IPv6 routes. OSPFv3 also uses the SPF algorithm as the computation engine to determine the best paths throughout the routing domain.
- As with all IPv6 routing protocols, OSPFv3 has separate processes from its IPv4 counterpart. The processes and operations are basically the same as in the IPv4 routing protocol, but run independently. OSPFv2 and OSPFv3 each have separate adjacency tables, OSPF topology tables, and IP routing tables.
- The OSPFv3 configuration and verification commands are similar those used in OSPFv2.

OSPFv2 and OSPFv3 Similarities

- **Link-state** - OSPFv2 and OSPFv3 are both classless link-state routing protocols.
- **Routing algorithm** - OSPFv2 and OSPFv3 use the SPF algorithm to make routing decisions.
- **Metric** - The RFCs for both OSPFv2 and OSPFv3 define the metric as the cost of sending packets out the interface. OSPFv2 and OSPFv3 can be modified using the **auto-cost reference bandwidth ref-bw** router configuration mode command. The command only influences the OSPF metric where it was configured. For example, if this command was entered for OSPFv3, it does not affect the OSPFv2 routing metrics.
- **Areas** - The concept of multiple areas in OSPFv3 is the same as in OSPFv2. Multiareas that minimize link-state flooding and provide better stability with the OSPF domain.
- **OSPF packet types** - OSPFv3 uses the same five basic packet types as OSPFv2 (Hello, DBD, LSR, LSU, and LSAck).
- **Neighbor discovery mechanism** - The neighbor state machine, including the list of OSPF neighbor states and events, remains unchanged. OSPFv2 and OSPFv3 use the Hello mechanism to learn about neighboring routers and form adjacencies. However, in OSPFv3, there is no requirement for matching subnets to form neighbor adjacencies. This is because neighbor adjacencies are formed using link-local addresses, not global unicast addresses.
- **DR/BDR election process** - The DR/BDR election process remains unchanged in OSPFv3.
- **Router ID** - Both OSPFv2 and OSPFv3 use a 32-bit number for the router ID represented in dotted-decimal notation. Typically, this is an IPv4 address. The OSPF **router-id** command must be used to

configure the router ID. The process in determining the 32-bit Router ID is the same in both protocols. Use an explicitly-configured router ID; otherwise, the highest loopback IPv4 address becomes the router ID.

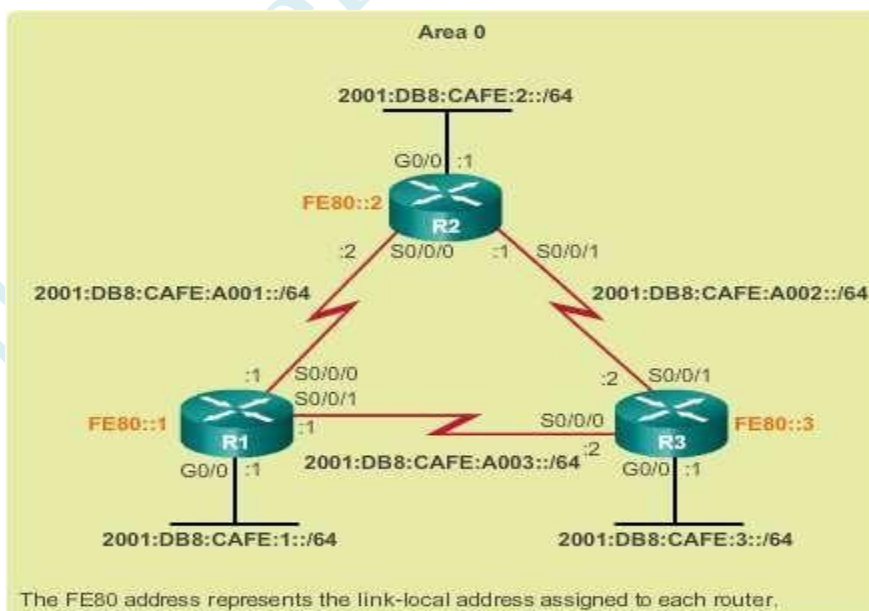
OSPFv2 vs. OSPFv3 Differences

- **Advertises** - OSPFv2 advertises IPv4 routes, whereas OSPFv3 advertises routes for IPv6.
- **Source address** - OSPFv2 messages are sourced from the IPv4 address of the exit interface. In OSPFv3, OSPF messages are sourced using the link-local address of the exit interface.
- **All OSPF router multicast addresses**- OSPFv2 uses 224.0.0.5; whereas, OSPFv3 uses FF02::5.
- **DR/BDR multicast address** - OSPFv2 uses 224.0.0.6; whereas, OSPFv3 uses FF02::6.
- **Advertise networks** - OSPFv2 advertises networks using the **network** router configuration command; whereas, OSPFv3 uses the **ipv6 ospf process-id area area-id** interface configuration command.
- **IP unicast routing** - Enabled, by default, in IPv4; whereas, the **ipv6 unicast-routing** global configuration command must be configured.
- **Authentication** - OSPFv2 uses either plaintext authentication or MD5 authentication. OSPFv3 uses IPv6 authentication.

87. TOPIC

OSPFv3 Network Topology

Figure below displays the network topology that is used to configure OSPFv3.



In this topology, none of the routers have IPv4 addresses configured. A network with router interfaces configured with IPv4 and IPv6 addresses is referred to as dual-stacked. A dual-stacked network can have OSPFv2 and OSPFv3 simultaneously-enabled.

The steps to configure basic OSPFv3 in a single area.

Link Local Addresses

The output of the **show ipv6 interface brief** command confirms that the correct global IPv6 addresses have been successfully configured and that the interfaces are enabled.

Link-local addresses are automatically created when an IPv6 global unicast address is assigned to the interface. Global unicast addresses are not required on an interface; however, IPv6 link-local addresses are.

Unless configured manually, Cisco routers create the link-local address using FE80::/10 prefix and the EUI-64 process. EUI-64 involves using the 48-bit Ethernet MAC address, inserting FFFE in the middle and flipping the seventh bit. For serial interfaces, Cisco uses the MAC address of an Ethernet interface.

Configuring Link Local Addresses on R1

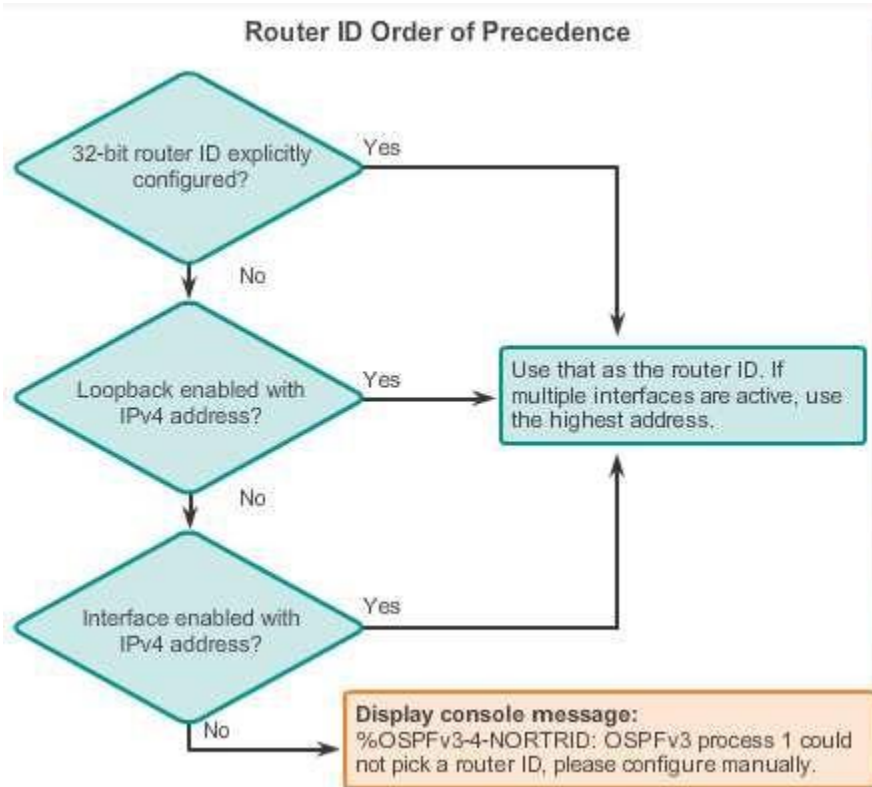
Link-local addresses created using the EUI-64 format or in some cases, random interface IDs, make it difficult to recognize and remember those addresses. Because IPv6 routing protocols use IPv6 link-local addresses for unicast addressing and next-hop address information in the routing table, it is common practice to make it an easily recognizable address.

Configuring the link-local address manually provides the ability to create an address that is recognizable and easier to remember. As well, a router with several interfaces can assign the same link-local address to each IPv6 interface. This is because the link-local address is only required for local communications.

Link-local addresses can be configured manually using the same interface command used to create IPv6 global unicast addresses, but appending the **link-local** keyword to the **ipv6 address** command.

A link-local address has a prefix within the range FE80 to FEBF. When an address begins with this hexet (16-bit segment) the **link-local** keyword must follow the address.

Configuring OSPFv3 Router ID



88. TOPIC

Verify OSPFv3 Neighbors & Protocol Settings

Use the **show ipv6 ospf neighbor** command to verify that the router has formed an adjacency with its neighboring routers. If the router ID of the neighboring router is not displayed, or if it does not show as being in a state of FULL, the two routers have not formed an OSPF adjacency.

```

R1# show ipv6 ospf neighbor

OSPFv3 Router with ID (1.1.1.1) (Process ID 10)

Neighbor ID  Pri  State           Dead Time   Interface ID  Interface
3.3.3.3      0  FULL/-         00:00:39    6             Serial0/0/1
2.2.2.2      0  FULL/-         00:00:36    6             Serial0/0/0
R1#
  
```

If two routers do not establish a neighbor adjacency, link-state information is not exchanged. Incomplete LSDBs can cause inaccurate SPF trees and routing tables. Routes to destination networks may not exist or may not be the most optimum path.

Figure above displays the neighbor adjacency of R1. For each neighbor, this command displays the following output:

- **Neighbor ID** - The router ID of the neighboring router.
- **Pri** - The OSPF priority of the interface. Value is used in the DR and BDR election.

- **State** - The OSPF state of the interface. FULL state means that the router and its neighbor have identical OSPF LSDBs. On multi access networks such as Ethernet, two routers that are adjacent may have their states displayed as 2WAY. The dash indicates that no DR or BDR is required because of the network type.
- **Dead Time** - The amount of time remaining that the router waits to receive an OSPF Hello packet from the neighbor before declaring the neighbor down. This value is reset when the interface receives a Hello packet.
- **Interface ID** - The interface ID or link ID.
- **Interface** - The interface on which this router has formed adjacency with the neighbor.

```
R1# show ipv6 protocols
IPv6 Routing Protocol is "connected"
IPv6 Routing Protocol is "ND"
IPv6 Routing Protocol is "ospf 10"
Router ID 1.1.1.1
Number of areas: 1 normal, 0 stub, 0 nssa
Interfaces (Area 0):
  Serial0/0/1
  Serial0/0/0
  GigabitEthernet0/0
Redistribution:
  None
R1#
```

As shown in Figure above, the **show ipv6 protocols** command is a quick way to verify vital OSPFv3 configuration information, including the OSPF process ID, the router ID, and the interfaces enabled for OSPFv3.

Week no 9

89. TOPIC

Multiarea OSPF

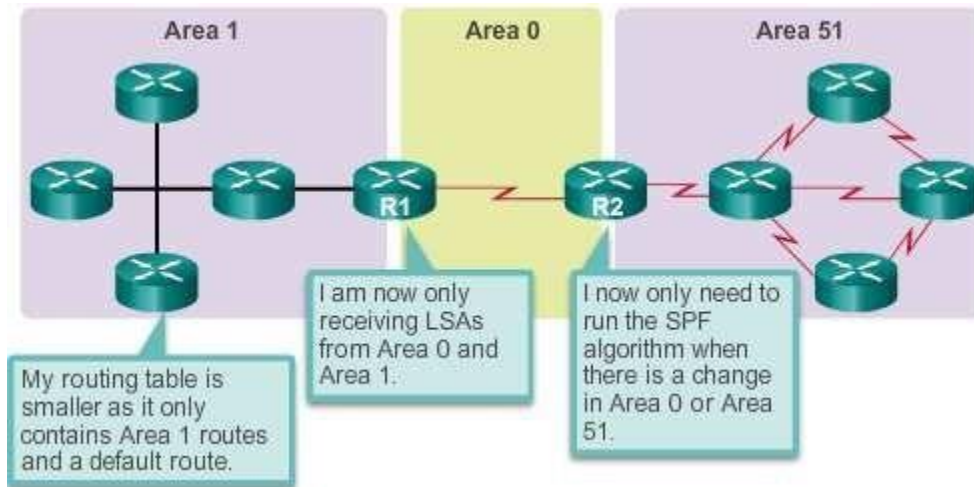
When a large OSPF area is divided into smaller areas, this is called multiarea OSPF. Multiarea OSPF is useful in larger network deployments to reduce processing and memory overhead.

For instance, any time a router receives new information about the topology, as with additions, deletions, or modifications of a link, the router must rerun the SPF algorithm, create a new SPF tree, and update the routing table. The SPF algorithm is CPU-intensive and the time it takes for calculation depends on the size of the area. Too many routers in one area make the LSDB larger and increase the load on the CPU. Therefore, arranging routers into areas effectively partitions one potentially large database into smaller and more manageable databases.

Multiarea OSPF requires a hierarchical network design. The main area is called the backbone area (area 0) and all other areas must connect to the backbone area. With hierarchical routing, routing still occurs between the areas (interarea routing); while many of the tedious routing operations, such as recalculating the database, are kept within an area.

The hierarchical-topology possibilities of multiarea OSPF have these advantages:

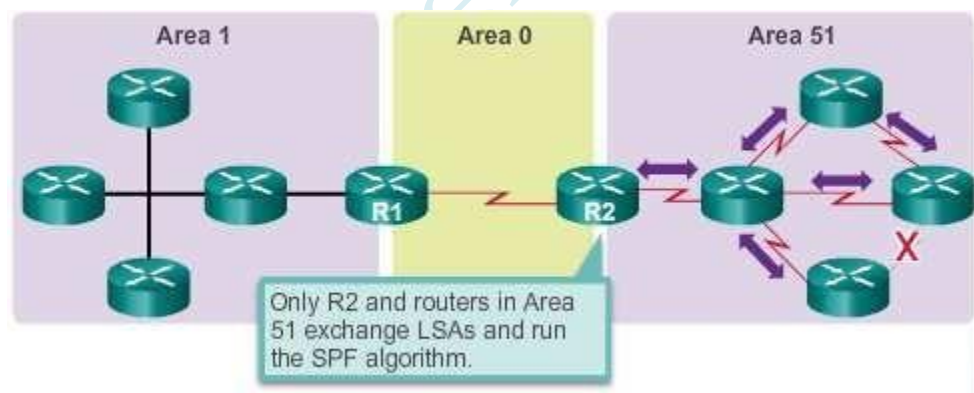
Smaller routing tables - There are fewer routing table entries as network addresses can be summarized between areas. For example in figure below, R1 summarizes the routes from area 1 to area 0 and R2 summarizes the routes from area 51 to area 0. R1 and R2 also propagate a default static route to area 1 and area 51.



Reduced link-state update overhead- Minimizes processing and memory requirements, because there are fewer routers exchanging LSAs.

Reduced frequency of SPF calculations - Localizes impact of a topology change within an area. For instance, it minimizes routing update impact, because LSA flooding stops at the area boundary.

In Figure below, assume a link fails between two internal routers in area 51. Only the routers in area 51 exchange LSAs and rerun the SPF algorithm for this event. R1 does not receive LSAs from area 51 and does not recalculate the SPF algorithm.



OSPF Two-Layer Area Hierarchy

Multiarea OSPF is implemented in a two-layer area hierarchy:

Backbone (Transit) area - An OSPF area whose primary function is the fast and efficient movement of IP packets. Backbone areas interconnect with other OSPF area types. Generally, end users are not found within a backbone area. The backbone area is also called OSPF area 0. Hierarchical networking defines area 0 as the core to which all other areas directly connect.

Regular (Non -backbone) area -Connects users and resources. Regular areas are usually set up along functional or geographical groupings. By default, a regular area does not allow traffic from another area to use its links to reach other areas. All traffic from other areas must cross a transit area.

OSPF enforces this rigid two-layer area hierarchy. The underlying physical connectivity of the network must map to the two-layer area structure, with all non-backbone areas attaching directly to area 0. All traffic moving from one area to another area must traverse the backbone area. This traffic is referred to as interarea traffic.

The optimal number of routers per area varies based on factors such as network stability, but Cisco recommends the following guidelines:

- An area should have no more than 50 routers.
- A router should not be in more than three areas.
- Any single router should not have more than 60 neighbors.

Types of OSPF Routers

OSPF routers of different types control the traffic that goes in and out of areas. The OSPF routers are categorized based on the function they perform in the routing domain.

There are four different types of OSPF routers:

- **Internal router** This is a router that has all of its interfaces in the same area. All internal routers in an area have identical LSDBs.
- **Backbone router** This is a router in the backbone area. Generally, the backbone area is set to area 0.
- **Area Border Router (ABR)** This is a router that has interfaces attached to multiple areas. It must maintain separate LSDBs for each area it is connected to, and can route between areas. ABRs are exit points for the area, which means that routing information destined for another area can get there only via the ABR of the local area. ABRs can be configured to summarize the routing information from the LSDBs of their attached areas. ABRs distribute the routing information into the backbone. The backbone routers then forward the information to the other ABRs. In a multiarea network, an area can have one or more ABRs.
- **Autonomous System Boundary Router (ASBR)** This is a router that has at least one interface attached to an external internetwork (another autonomous system), such as a non-OSPF network. An ASBR can import non-OSPF network information to the OSPF network, and vice versa, using a process called route redistribution.

Redistribution in multiarea OSPF occurs when an ASBR connects different routing domains (e.g., EIGRP and OSPF) and configures them to exchange and advertise routing information between those routing domains.

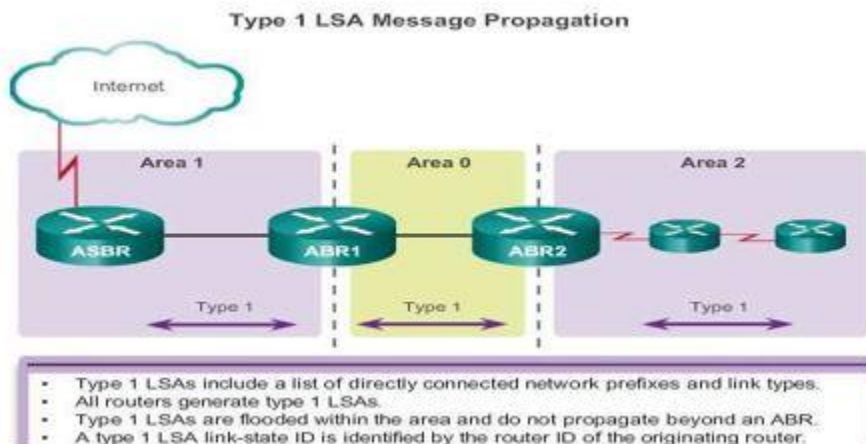
A router can be classified as more than one router type. For example, if a router connects to area 0 and area 1, and in addition, maintains routing information for another, non-OSPF network, it falls under three different classifications: a backbone router, an ABR, and an ASBR.

LSA Type	Description
1	Router LSA
2	Network LSA
3 and 4	Summary LSAs
5	AS External LSA
6	Multicast OSPF LSA
7	Defined for NSSAs
8	External Attributes LSA for Border Gateway Protocol (BGP)
9, 10, or 11	Opaque LSAs

Most Common LSA Types

LSA Type	Description
1	Router LSA
2	Network LSA
3 and 4	Summary LSAs
5	AS External LSA

OSPF LSA Type 1

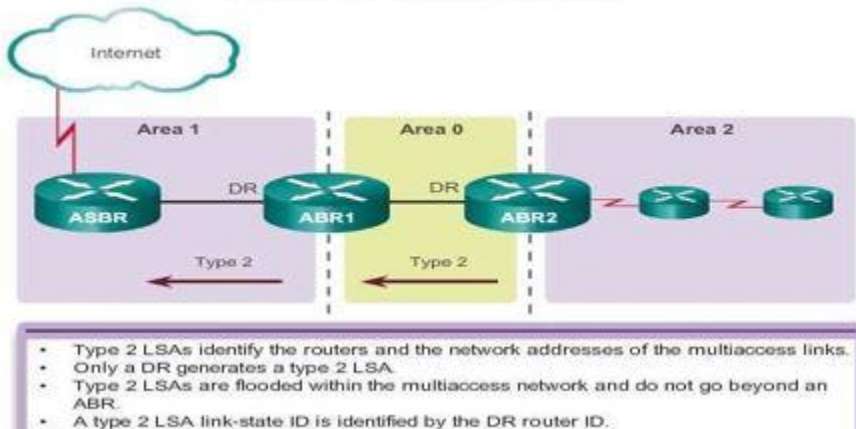


As shown in the figure above, all routers advertise their directly connected OSPF-enabled links in a type 1 LSA and forward their network information to OSPF neighbors. The LSA contains a list of the directly connected interfaces, link types, and link states. Type 1 LSAs are also referred to the router link entries.

- Type 1 LSAs are flooded only within the area in which they originated. ABRs subsequently advertise the networks learned from the type 1 LSAs to other areas as type 3 LSAs.
- The type 1 LSA link ID is identified by the router ID of the originating router.

OSPF LSA Type 2

Type 2 LSA Message Propagation



A type 2 LSA only exists for multiaccess and non-broadcast multiaccess (NBMA) networks where there is a DR elected and at least two routers on the multiaccess segment. The type 2 LSA contains the router ID and IP address of the DR, along with the router ID of all other routers on the multiaccess segment. A type 2 LSA is created for every multiaccess network in the area.

The purpose of an a type 2 LSA is to give other routers information about multiaccess networks within the same area.

The DR floods type 2 LSAs only within the area in which they originated. Type 2 LSAs are not forwarded outside of an area.

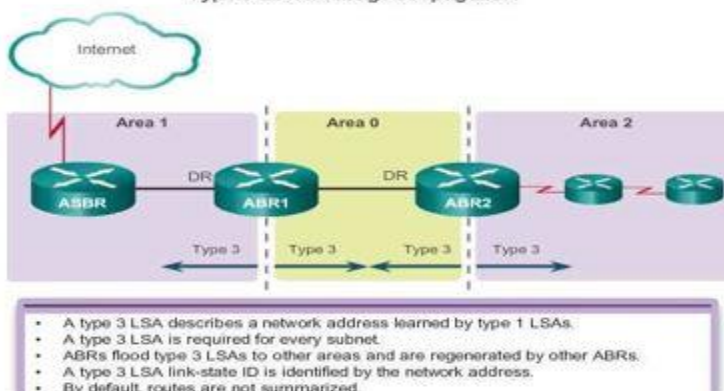
Type 2 LSAs are also referred to the network link entries.

As shown in the figure above, ABR1 is the DR for the Ethernet network in area 1. It generates the type 2 LSA and forwards it into area 1. ABR2 is the DR for the multiaccess network in area 0. There are no multiaccess networks in area 2 and therefore, no type 2 LSAs are ever propagated in that area.

The link-state ID for a network LSA is the IP interface address of the DR that advertises it.

OSPF LSA Type 3

Type 3 LSA Message Propagation



Type 3 LSAs are used by ABRs to advertise networks from other areas. ABRs collect type 1 LSAs in the LSDB. After an OSPF area has converged, the ABR creates a type 3 LSA for each of its learned OSPF networks. Therefore, an ABR with many OSPF routes must create type 3 LSAs for each network.

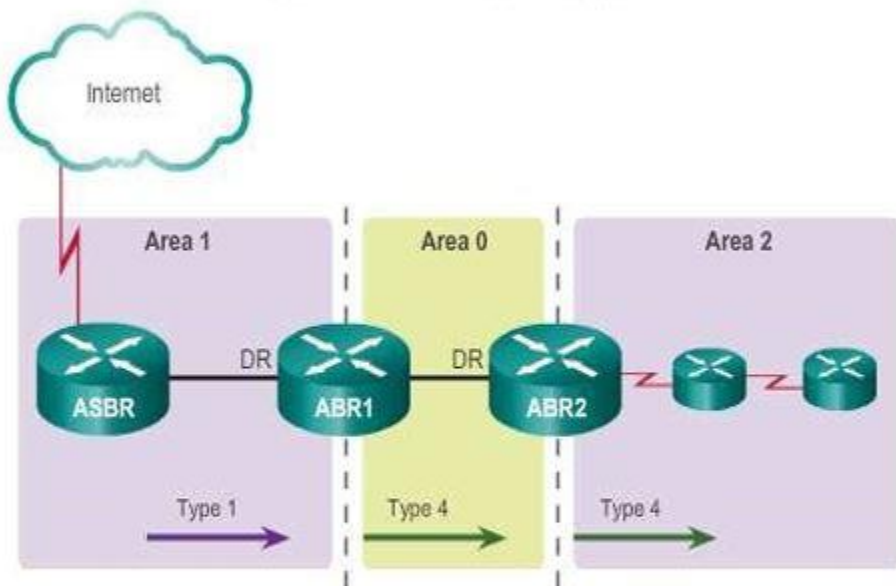
As shown in the figure above, ABR1 and ABR2 flood type 3 LSAs from one area to other areas. The ABRs propagate the type 3 LSAs into other areas. In a large OSPF deployment with many networks, propagating type 3 LSAs can cause significant flooding problems. For this reason, it is strongly recommended that manual route summarization be configured on the ABR.

The link-state ID is set to the network number and the mask is also advertised.

Receiving a type 3 LSA into its area does not cause a router to run the SPF algorithm. The routes being advertised in the type 3 LSAs are appropriately added to or deleted from the router's routing table, but a full SPF calculation is not necessary.

OSPF LSA Type 4

Type 4 LSA Message Propagation



- Type 4 LSAs are used to advertise an ASBR to other areas and provide a route to the ASBR.
- ABRs generate type 4 LSAs.
- A type 4 LSA is generated by the originating ABR and regenerated by other ABRs.
- A type 4 LSA link-state ID is identified by the router ID of the ASBR.

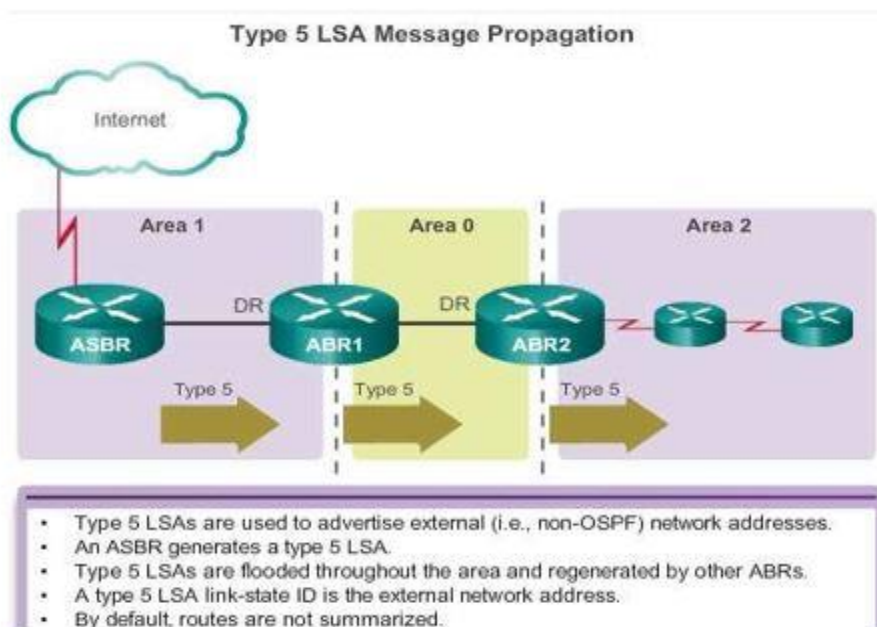
Type 4 and type 5 LSAs are used collectively to identify an ASBR and advertise external networks into an OSPF routing domain.

A type 4 summary LSA is generated by an ABR only when an ASBR exists within an area. A type 4 LSA identifies the ASBR and provides a route to it. All traffic destined to an external autonomous system requires routing table knowledge of the ASBR that originated the external routes.

As shown in the figure above, the ASBR sends a type 1 LSA, identifying itself as an ASBR. The LSA includes a special bit known as the external bit (e bit) that is used to identify the router as an ASBR. When ABR1 receives the type 1 LSA, it notices the e bit, it builds a type 4 LSA, and then floods the type 4 LSA to the backbone (area 0). Subsequent ABRs flood the type 4 LSA into other areas.

The link-state ID is set to the ASBR router ID.

OSPF LSA Type 5



Type 5 external LSAs describe routes to networks outside the OSPF autonomous system. Type 5 LSAs are originated by the ASBR and are flooded to the entire autonomous system.

Type 5 LSAs are also referred to as autonomous system external LSA entries.

In the figure above, the ASBR generates type 5 LSAs for each of its external routes and floods it into the area. Subsequent ABRs also flood the type 5 LSA into other areas. Routers in other areas use the information from the type 4 LSA to reach the external routes.

In a large OSPF deployment with many networks, propagating multiple type 5 LSAs can cause significant flooding problems. For this reason, it is strongly recommended that manual route summarization be configured on the ASBR.

The link-state ID is the external network number.

91. TOPIC

In this activity, the IP addressing is already configured. You are responsible for configuring the three router topology with basic single area OSPFv2 and then verifying connectivity between end devices.

Part 1: Configure OSPFv2 Routing

Step 1: Configure OSPF on the R1, R2 and R3.

Use the following requirements to configure OSPF routing on all three routers:

- Process ID 10
- Router ID for each router: R1 = 1.1.1.1; R2 = 2.2.2.2; R3 = 3.3.3.3
- Network address for each interface
- LAN interface set to passive (do not use the **default** keyword)

Step 2: Verify OSPF routing is operational.

On each router, the routing table should now have a route to every network in the topology.

Part 2: Verify the Configurations

Each PC should be able to ping the other two PCs. If not, check your configurations.

```
!-----
```

```
!R1
```

```
!-----
```

```
ena
```

```
conf t
```

```
!
```

```
router ospf 10
```

```
router-id 1.1.1.1
```

```
network 172.16.1.0 0.0.0.255 area 0
```

```
network 172.16.3.0 0.0.0.3 area 0
```

```
network 192.168.10.4 0.0.0.3 area 0
```

```
passive-interface GigabitEthernet0/0
```

!

End

!-----

!R2

!-----

ena

conf t

!

router ospf 10

router-id 2.2.2.2

network 172.16.2.0 0.0.0.255 area 0

network 172.16.3.0 0.0.0.3 area 0

network 192.168.10.8 0.0.0.3 area 0

passive-interface GigabitEthernet0/0

!

end

!-----

!R3

!-----

ena

conf t

!

router ospf 10

router-id 3.3.3.3

network 192.168.1.0 0.0.0.255 area 0

```
network 192.168.10.4 0.0.0.3 area 0
```

```
network 192.168.10.8 0.0.0.3 area 0
```

```
passive-interface GigabitEthernet0/0
```

```
!
```

```
End
```

Link for the site where same practical is being performed:

<https://examans.com/8-2-2-7-packet-tracer-configuring-ospfv2-in-a-single-area/>

92. TOPIC

In this activity, the IPv6 addressing is already configured. You are responsible for configuring the three router topology with basic single area OSPFv3 and then verifying connectivity between end devices.

Part 1: Configure OSPFv3 Routing

Step 1: Configure OSPFv3 on R1, R2 and R3.

Use the following requirements to configure OSPF routing on all three routers:

Enable IPv6 routing

Process ID 10

Router ID for each router: R1 = 1.1.1.1; R2 = 2.2.2.2; R3 = 3.3.3.3

Enable OSPFv3 on each interface

Adjust the default reference bandwidth to support gigabit links using the auto-cost reference-bandwidth command.

Prevent the LAN interfaces from sending out OSPF routing messages.

Step 2: Verify OSPF routing is operational.

Verify each router has established adjacency with the other two routers. Verify the routing table has a route to every network in the topology.

Part 2: Verify Connectivity

Each PC should be able to ping the other two PCs. If not, check your configurations.

!R1

!-----

ena

conf t

!

ipv6 unicast-routing

!

ipv6 router ospf 10

router-id 1.1.1.1

interface GigabitEthernet 0/0

ipv6 ospf 10 area 0

!

interface Serial0/0/0

ipv6 ospf 10 area 0

!

interface Serial0/0/1

ipv6 ospf 10 area 0

!

end

!-----

!R2

!-----

ena

conf t

!

ipv6 unicast-routing

!

ipv6 router ospf 10

router-id 2.2.2.2

interface GigabitEthernet 0/0

ipv6 ospf 10 area 0

!

interface Serial0/0/0

ipv6 ospf 10 area 0

!

interface Serial0/0/1

ipv6 ospf 10 area 0

!

end

!-----

!R3

!-----

ena

conf t

!

ipv6 unicast-routing

!

ipv6 router ospf 10

router-id 3.3.3.3

```
interface GigabitEthernet 0/0
```

```
ipv6 ospf 10 area 0
```

```
!
```

```
interface Serial0/0/0
```

```
ipv6 ospf 10 area 0
```

```
!
```

```
interface Serial0/0/1
```

```
ipv6 ospf 10 area 0
```

```
!
```

```
end
```

Link: <https://examans.com/8-3-3-5-packet-tracer-configuring-basic-ospfv3-in-a-single-area/>

93. TOPIC

Packet Tracer – Configuring IPv4 Route Summarization - 2:

This topic is in continuation of the previous topic.

Just checking connectivity of routers, there is nothing for reading mode and post assessment in this lecture.

94. TOPIC

OSPF Routing Table and Types of Routes:

OSPF Routing Table Entries

```

R1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 192.168.10.2 to network 0.0.0.0

O*E2 0.0.0.0/0 [110/1] via 192.168.10.2, 00:00:19, Serial0/0/0
    10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C    10.1.1.0/24 is directly connected, GigabitEthernet0/0
L    10.1.1.1/32 is directly connected, GigabitEthernet0/0
C    10.1.2.0/24 is directly connected, GigabitEthernet0/1
L    10.1.2.1/32 is directly connected, GigabitEthernet0/1
O    10.2.1.0/24 [110/648] via 192.168.10.2, 00:04:34, Serial0/0/0
O IA 192.168.1.0/24 [110/1295] via 192.168.10.2, 00:01:48, Serial0/0/0
O IA 192.168.2.0/24 [110/1295] via 192.168.10.2, 00:01:48, Serial0/0/0
    192.168.10.0/24 is variably subnetted, 3 subnets, 2 masks
C    192.168.10.0/30 is directly connected, Serial0/0/0
L    192.168.10.1/32 is directly connected, Serial0/0/0
O    192.168.10.4/30 [110/1294] via 192.168.10.2, 00:01:55, Serial0/0/0
R1#

```

Figure above provides a sample routing table for a multiarea OSPF topology with a link to an external non-OSPF network. OSPF routes in an IPv4 routing table are identified using the following descriptors:

- Router (type 1) and network (type 2) LSAs describe the details within an area. The routing table reflects this link-state information with a designation of **O**, meaning that the route is intra-area.
- **IA** - When an ABR receives summary LSAs, it adds them to its LSDB and regenerates them into the local area. When an ABR receives external LSAs, it adds them to its LSDB and floods them into the area. The internal routers then assimilate the information into their databases. Summary LSAs appear in the routing table as IA (interarea routes).
- **E1** or **O E2** - External LSAs appear in the routing table marked as external type 1 (E1) or external type 2 (E2) routes.

OSPF Route Calculation

Each router uses the SPF algorithm against the LSDB to build the SPF tree. The SPF tree is used to determine the best paths.

The order in which the best paths are calculated is as follows:

1. All routers calculate the best paths to destinations within their area (intra-area) and add these entries to the routing table. These are the type 1 and type 2 LSAs, which are noted in the routing table with a routing designator of **O**.
2. All routers calculate the best paths to the other areas within the internetwork. These best paths are the interarea route entries, or type 3 and type 4 LSAs, and are noted with a routing designator of **O IA**.

3. All routers (except those that are in a form of stub area) calculate the best paths to the external autonomous system (type 5) destinations. These are noted with either an O E1 or an O E2 route designator, depending on the configuration.

When converged, a router can communicate with any network within or outside the OSPF autonomous system.

95. TOPIC

Implementing Multiarea OSPF

OSPF can be implemented as single-area or multiarea. The type of OSPF implementation chosen depends on the specific requirements and existing topology.

There are 4 steps to implementing multiarea OSPF.

Steps 1 and 2 are part of the planning process.

Step 1. Gather the network requirements and parameters - This includes determining the number of host and network devices, the IP addressing scheme (if already implemented), the size of the routing domain, the size of the routing tables, the risk of topology changes, and other network characteristics.

Step 2. Define the OSPF parameters -Based on information gathered during Step 1, the network administrator must determine if single-area or multiarea OSPF is the preferred implementation. If multiarea OSPF is selected, there are several considerations the network administrator must take into account while determining the OSPF parameters, to include:

- **IP addressing plan** - This governs how OSPF can be deployed and how well the OSPF deployment might scale. A detailed IP addressing plan, along with the IP subnetting information, must be created. A good IP addressing plan should enable the usage of OSPF multiarea design and summarization. This plan more easily scales the network, as well as optimizes OSPF behavior and the propagation of LSA.
- **OSPF areas** - Dividing an OSPF network into areas decreases the LSDB size and limits the propagation of link-state updates when the topology changes. The routers that are to be ABRs and ASBRs must be identified, as are those that are to perform any summarization or redistribution.
- **Network topology** - This consists of links that connect the network equipment and belong to different OSPF areas in a multiarea OSPF design. Network topology is important to determine primary and backup links. Primary and backup links are defined by the changing OSPF cost on interfaces. A detailed network topology plan should also be used to determine the different OSPF areas, ABR, and ASBR as well as summarization and redistribution points, if multiarea OSPF is used.

Step 3. Configure the multiarea OSPF implementation based on the parameters.

Step 4. Verify the multiarea OSPF implementation based on the parameters.

Configuring Multiarea OSPF

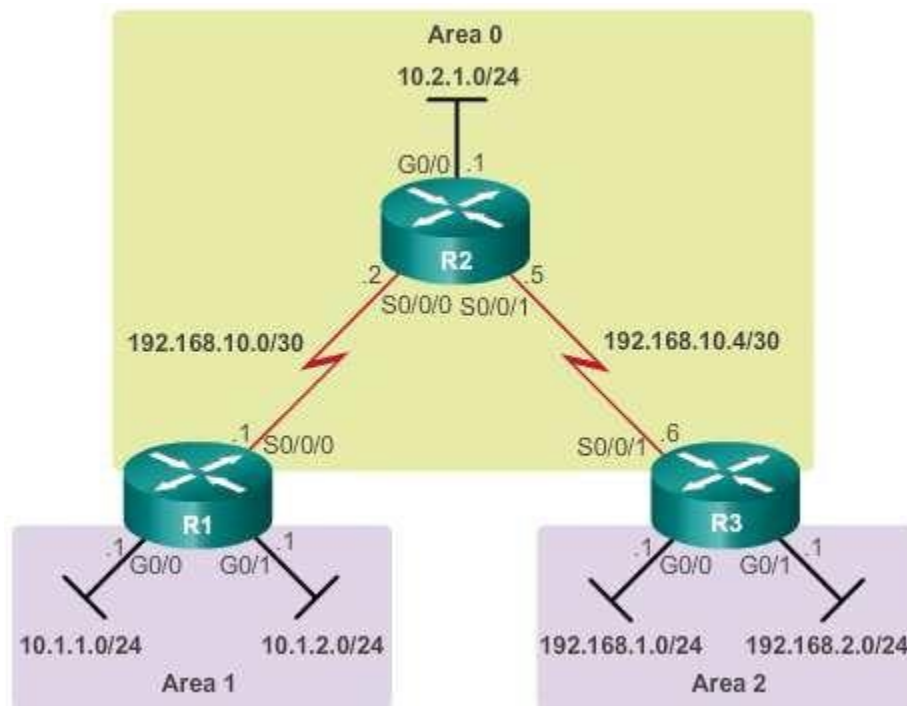


Figure above displays the reference multiarea OSPF topology. In this example:

- R1 is an ABR because it has interfaces in area 1 and an interface in area 0.
- R2 is an internal backbone router because all of its interfaces are in area 0.
- R3 is an ABR because it has interfaces in area 2 and an interface in area 0.

There are no special commands required to implement this multiarea OSPF network. A router simply becomes an ABR when it has two **network** statements in different areas.

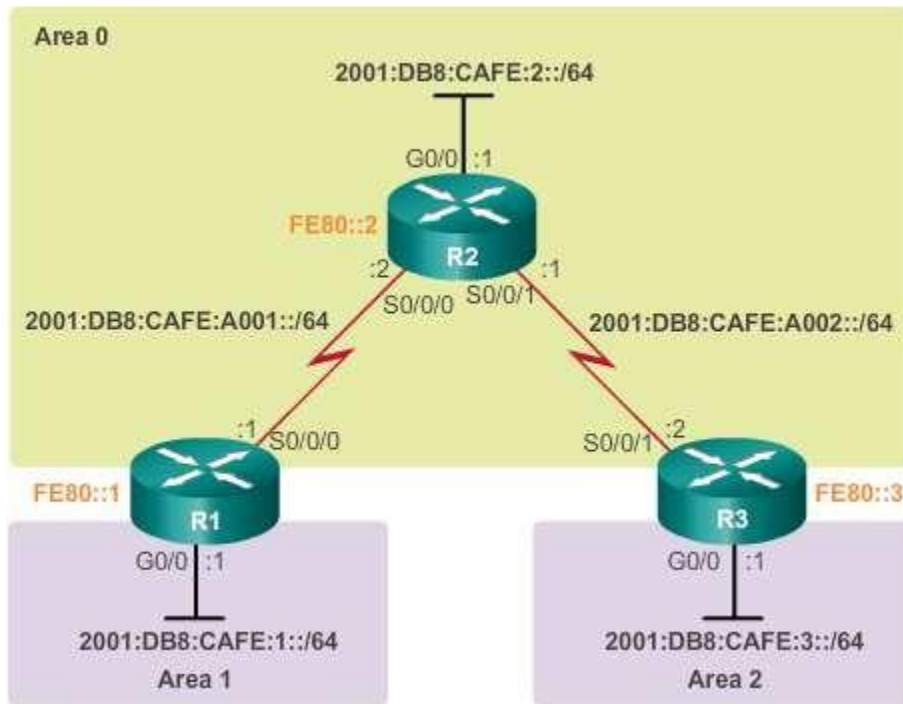
R1 is assigned the router ID 1.1.1.1. This example enables OSPF on the two LAN interfaces in area 1. The serial interface is configured as part of OSPF area 0. Because R2 has interfaces connected to two different areas, it is an ABR.

Upon completion of the R2 and R3 configuration, notice the informational messages informing of the adjacencies with R1 (1.1.1.1).

Upon completion of the R3 configuration, notice the informational messages informing of an adjacency with R1 (1.1.1.1) and R2 (2.2.2.2).

```
R1(config)# router ospf 10
R1(config-router)# router-id 1.1.1.1
R1(config-router)# network 10.1.1.1 0.0.0.0 area 1
R1(config-router)# network 10.1.2.1 0.0.0.0 area 1
R1(config-router)# network 192.168.10.1 0.0.0.0 area 0
R1(config-router)# end
R1#
```

Configuring Multiarea OSPFv3



Like OSPFv2, implementing the multiarea OSPFv3 topology in Figure above is simple. There are no special commands required. A router simply becomes an ABR when it has two interfaces in different areas.

R1 is assigned the router ID 1.1.1.1. The example also enables OSPF on the two LAN interfaces in area 1 and the serial interface in area 0. Because R1 has interfaces connected to two different areas, it becomes an ABR.

Upon completion of the R2 configuration, notice the message that there is an adjacency with R1

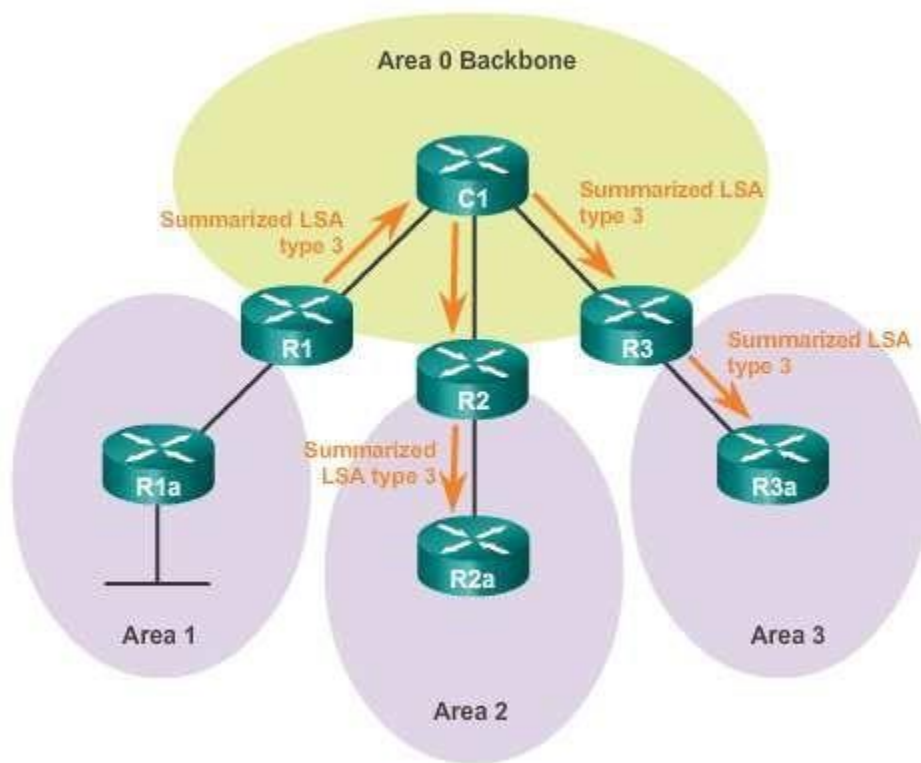
(1.1.1.1).

Upon completion of the R3 configuration, notice the message that there is an adjacency with R2

(2.2.2.2).

```
R1(config)# ipv6 router ospf 10
R1(config-rtr)# router-id 1.1.1.1
R1(config-rtr)# exit
R1(config)#
R1(config)# interface GigabitEthernet 0/0
R1(config-if)# ipv6 ospf 10 area 1
R1(config-if)#
R1(config-if)# interface Serial0/0/0
R1(config-if)# ipv6 ospf 10 area 0
R1(config-if)# end
R1#
```

Summarization helps keep routing tables small. It involves consolidating multiple routes into a single advertisement, which can then be propagated into the backbone area.

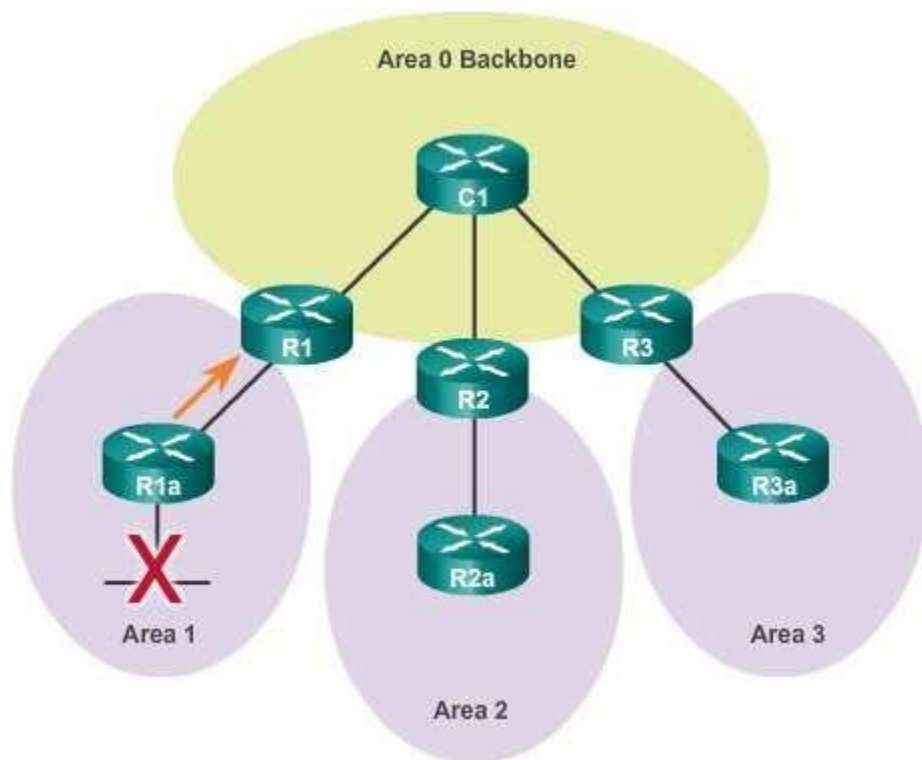


Normally, type 1 and type 2 LSAs are generated inside each area, translated into type 3 LSAs, and sent to other areas. If area 1 had 30 networks to advertise, then 30 type 3 LSAs would be forwarded into the backbone. With route summarization, the ABR consolidates the 30 networks into one of two advertisements.

In Figure above, R1 consolidates all of the network advertisements into one summary LSA. Instead of forwarding individual LSAs for each route in area 1, R1 forwards a summary LSA to the core router C1. C1 in turn, forwards the summary LSA to R2 and R3. R2 and R3 then forward it to their respective internal routers.

Summarization also helps increase the network's stability, because it reduces unnecessary LSA flooding. This directly affects the amount of bandwidth, CPU, and memory resources consumed by the OSPF routing process. Without route summarization, every specific-link LSA is propagated into the OSPF backbone and beyond, causing unnecessary network traffic and router overhead.

In Figure below, a network link on R1a fails. R1a sends an LSA to R1. However, R1 does not propagate the update, because it has a summary route configured. Specific-link LSA flooding outside the area does not occur.



Summarizing Interarea Routes on ABRs

In OSPF, summarization can only be configured on ABRs or ASBRs. Instead of advertising many specific networks, the ABR routers and ASBR routers advertise a summary route. ABR routers summarize type 3 LSAs and ASBR routers summarize type 5 LSAs.

By default, summary LSAs (type 3 LSAs) and external LSAs (type 5 LSAs) do not contain summarized (aggregated) routes; that is, by default, summary LSAs are not summarized.

Route summarization can be configured as follows:

Interarea route summarization - Interarea route summarization occurs on ABRs and applies to routes from within each area. It does not apply to external routes injected into OSPF via redistribution. To perform effective interarea route summarization, network addresses within areas should be assigned contiguously so that these addresses can be summarized into a minimal number of summary addresses.

External route summarization - External route summarization is specific to external routes that are injected into OSPF via route redistribution. Again, it is important to ensure the contiguity of the external address ranges that are being summarized. Generally, only ASBRs summarize external routes.

Note: External route summarization is configured on ASBRs using the **summary-address address mask** router configuration mode command.

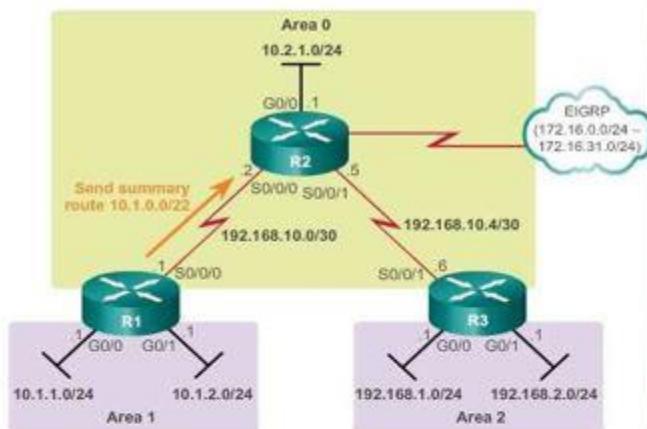
OSPF does not perform auto-summarization. Interarea summarization must be manually configured on ABRs.

Summarization of internal routes can only be done by ABRs. When summarization is enabled on an ABR, it injects into the backbone a single type 3 LSA describing the summary route. Multiple routes inside the area are summarized by the one LSA.

A summary route is generated if at least one subnet within the area falls in the summary address range. The summarized route metric is equal to the lowest cost of all subnets within the summary address range.

Note: An ABR can only summarize routes that are within the areas connected to the ABR.

Figure below shows a multiarea OSPF topology. The routing tables of R1 and R3 are examined to see the effect of the summarization.



```
R1# show ip route ospf | begin Gateway
Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O   10.2.1.0/24 [110/648] via 192.168.10.2, 00:00:49,
    Serial0/0/0
O IA 192.168.1.0/24 [110/1295] via 192.168.10.2, 00:00:49,
    Serial0/0/0
O IA 192.168.2.0/24 [110/1295] via 192.168.10.2, 00:00:49,
    Serial0/0/0
  192.168.10.0/24 is variably subnetted, 3 subnets, 2
    masks
O   192.168.10.4/30 [110/1294] via 192.168.10.2,
    00:00:49, Serial0/0/0
R1#

R3# show ip route ospf | begin Gateway
Gateway of last resort is not set

  10.0.0.0/24 is subnetted, 3 subnets
O IA 10.1.1.0 [110/1295] via 192.168.10.5, 00:27:14, Serial0/0/1
O IA 10.1.2.0 [110/1295] via 192.168.10.5, 00:27:14, Serial0/0/1
O   10.2.1.0 [110/648] via 192.168.10.5, 00:27:57, Serial0/0/1
  192.168.10.0/24 is variably subnetted, 3 subnets, 2 masks
O   192.168.10.0/30 [110/1294] via 192.168.10.5, 00:27:57,
    Serial0/0/1
R3#
```

CS407

Routing and switching

Reading Material

Made by Mahjabeen

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Calculating the Summary Route

Step 1	Step 2	
10.1.1.0	00001010.00000001.00000001.00000000	01.00000000
10.1.2.0	00001010.00000001.00000001.00000000	10.00000000

First 22 Bits Match

Some Bits Are Different

Summarized Route: 10.1.0.0/22 or 10.1.0.0 255.255.252.0

The figure above illustrates that summarizing networks into a single address and mask can be done in three steps:

Step 1. List the networks in binary format. In the example the two area 1 networks 10.1.1.0/24 and 10.1.2.0/24 are listed binary format.

Step 2. Count the number of far left matching bits to determine the mask for the summary route. As highlighted, the first 22 far left matching bits match. This results in the prefix /22 or subnet mask 255.255.252.0.

Step 3. Copy the matching bits and then add zero bits to determine the summarized network address. In this example, the matching bits with zeros at the end result in a network address of 10.1.0.0/22. This summary address summarizes four networks: 10.1.0.0/24, 10.1.1.0/24, 10.1.2.0/24, and 10.1.3.0/24.

In the example the summary address matches four networks although only two networks exist.

Summarizing Area 1 Routes on R1

```
R1(config)# router ospf 10
R1(config-router)# area 1 range 10.1.0.0 255.255.252.0
R1(config-router)#
```

Verify the R1 Routing Table After Summarization

```
R1# show ip route ospf | begin Gateway
Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
O 10.1.0.0/22 is a summary, 00:00:09, Null0
O 10.2.1.0/24 [110/648] via 192.168.10.2, 00:00:09, Serial0/0/0
O IA 192.168.1.0/24 [110/1295] via 192.168.10.2, 00:00:09, Serial0/0/0
O IA 192.168.2.0/24 [110/1295] via 192.168.10.2, 00:00:09, Serial0/0/0
192.168.10.0/24 is variably subnetted, 3 subnets, 2 masks
O 192.168.10.4/30 [110/1294] via 192.168.10.2, 00:00:09, Serial0/0/0
R1#
```

97. TOPIC

Verifying Multiarea OSPF:

The same verification commands used to verify single-area OSPF also can be used to verify the multiarea OSPF topology:

show ip ospf neighbor

show ip ospf

show ip ospf interface

Commands that verify specific multiarea information include:

show ip protocols

show ip ospf interface brief

show ip route ospf

show ip ospf database

Note: For the equivalent OSPFv3 command, simply substitute **ip** with **ipv6**.

Use the **show ip protocols** command to verify the OSPF status. The output of the command reveals which routing protocols are configured on a router. It also includes routing protocol specifics such as the router ID, number of areas in the router, and networks included within the routing protocol configuration.

Use the **show ip ospf interface brief** command to display concise OSPF-related information of OSPF-enabled interfaces. This command reveals useful information, such as the OSPF process ID that the interface is assigned to, the area that the interfaces are in, and the cost of the interface.

Verify the OSPF Routes

The most common command used to verify a multiarea OSPF configuration is the **show ip route** command. Add the **ospf** parameter to display only OSPF-related information.

Use the **show ip ospf database** command to verify the contents of the LSDB.

There are many command options available with the **show ip ospf database** command.

Verify Multiarea OSPFv3

Like OSPFv2, OSPFv3 provides similar OSPFv3 verification commands.

98. TOPIC

Access Control Lists (ACLs):

An ACL is a series of IOS commands that control whether a router forwards or drops packets based on information found in the packet header. ACLs are among the most commonly used features of Cisco IOS software.

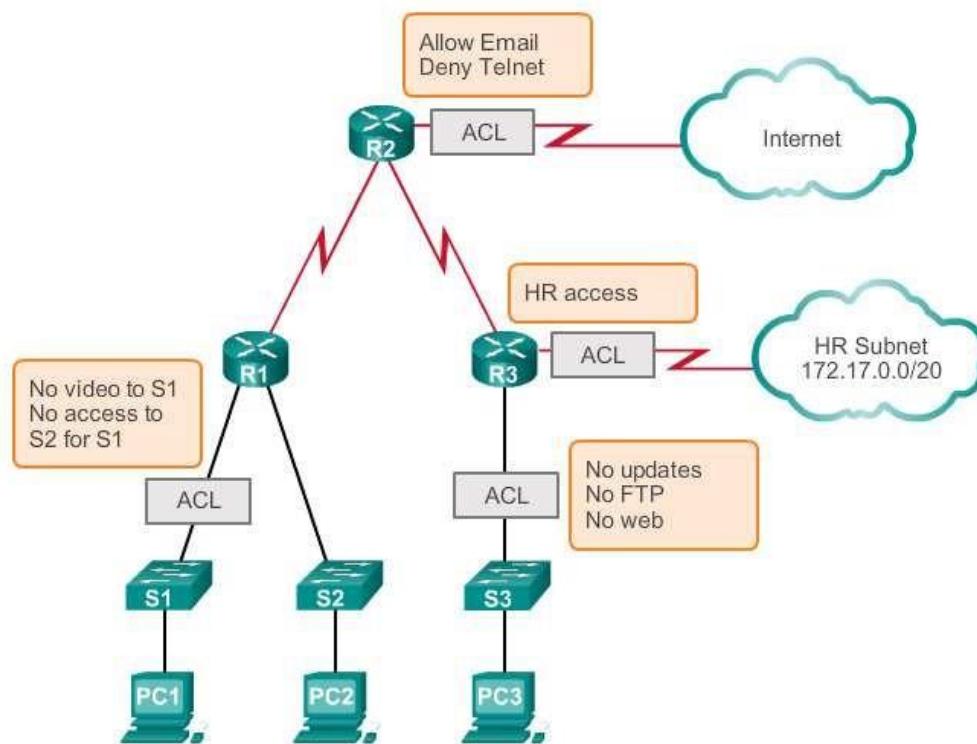
When configured, ACLs perform the following tasks:

- Limit network traffic to increase network performance. For example, if corporate policy does not allow video traffic on the network, ACLs that block video traffic could be configured and applied. This would greatly reduce the network load and increase network performance.
- Provide traffic flow control. ACLs can restrict the delivery of routing updates. If updates are not required because of network conditions, bandwidth is preserved.
- Provide a basic level of security for network access. ACLs can allow one host to access a part of the network and prevent another host from accessing the same area. For example, access to the Human Resources network can be restricted to authorized users.
- Filter traffic based on traffic type. For example, an ACL can permit email traffic, but block all Telnet traffic.
- Screen hosts to permit or deny access to network services. ACLs can permit or deny a user to access file types, such as FTP or HTTP.

By default, a router does not have ACLs configured; therefore, by default a router does not filter traffic. Traffic that enters the router is routed solely based on information within the routing table. However, when an ACL is applied to an interface, the router performs the additional task of evaluating all network packets as they pass through the interface to determine if the packet can be forwarded.

In addition to either permitting or denying traffic, ACLs can be used for selecting types of traffic to be analyzed, forwarded, or processed in other ways. For example, ACLs can be used to classify traffic to enable priority processing. This capability is similar to having a VIP pass at a concert or sporting event. The VIP pass gives selected guests privileges not offered to general admission ticket holders, such as priority entry or being able to enter a restricted area.

The figure below shows a sample topology with ACLs applied.



Packet Filtering

So how does an ACL use the information passed during a TCP/IP conversation to filter traffic?

Packet filtering, sometimes called static packet filtering, controls access to a network by analyzing the incoming and outgoing packets and passing or dropping them based on given criteria, such as the source IP address, destination IP addresses, and the protocol carried within the packet.

A router acts as a packet filter when it forwards or denies packets according to filtering rules. When a packet arrives at the packet-filtering router, the router extracts certain information from the packet header. Using this information, the router makes decisions, based on configured filter rules, as to whether the packet can pass through or be discarded.

A packet-filtering router uses rules to determine whether to permit or deny traffic. A router can also perform packet filtering at Layer 4, the transport layer. The router can filter packets based on the source port and destination port of the TCP or UDP segment. These rules are defined using ACLs.

An ACL is a sequential list of permit or deny statements, known as access control entries (ACEs). ACEs are also commonly called ACL statements. ACEs can be created to filter traffic based on certain criteria such as: the source address, destination address, the protocol, and port numbers. When network traffic passes through an interface configured with an ACL, the router compares the information within the packet against each ACE, in sequential order, to determine if the packet matches one of the statements. If a match is found, the packet is processed accordingly. In this way, ACLs can be configured to control access to a network or subnet.

To evaluate network traffic, the ACL extracts the following information from the Layer 3 packet header:

- Source IP address
- Destination IP address

- ICMP message type

The ACL can also extract upper layer information from the Layer 4 header, including:

- TCP/UDP source port
- TCP/UDP destination port
- ACL Operation

ACLs define the set of rules that give added control for packets that enter inbound interfaces, packets that relay through the router, and packets that exit outbound interfaces of the router. ACLs do not act on packets that originate from the router itself.

ACLs are configured to apply to inbound traffic or to apply to outbound traffic.

Inbound ACLs - Incoming packets are processed before they are routed to the outbound interface. An inbound ACL is efficient because it saves the overhead of routing lookups if the packet is discarded. If the packet is permitted by the tests, it is then processed for routing. Inbound ACLs are best used to filter packets when the network attached to an inbound interface is the only source of the packets needed to be examined.

Outbound ACLs - Incoming packets are routed to the outbound interface, and then they are processed through the outbound ACL. Outbound ACLs are best used when the same filter will be applied to packets coming from multiple inbound interfaces before exiting the same outbound interface.

The last statement of an ACL is always an implicit deny. This statement is automatically inserted at the end of each ACL even though it is not physically present. The implicit deny blocks all traffic. Because of this implicit deny, an ACL that does not have at least one permit statement will block all traffic.

99. TOPIC

Standard vs. Extended IPv4 ACL:

The two types of Cisco IPv4 ACLs are standard and extended.

Standard ACLs

Standard ACLs can be used to permit or deny traffic only from source IPv4 addresses. The destination of the packet and the ports involved are not evaluated.

Extended ACLs

Extended ACLs filter IPv4 packets based on several attributes:

- Protocol type
- Source IPv4 address
- Destination IPv4 address
- Source TCP or UDP ports
- Destination TCP or UDP ports
- Optional protocol type information for finer control

Numbered and Named ACLs

Standard and extended ACLs can be created using either a number or a name to identify the ACL and its list of statements.

Using numbered ACLs is an effective method for determining the ACL type on smaller networks with more homogeneously defined traffic. However, a number does not provide information about the purpose of the ACL.

Wildcard Masking

IPv4 ACEs include the use of wildcard masks. A wildcard mask is a string of 32 binary digits used by the router to determine which bits of the address to examine for a match.

As with subnet masks, the numbers 1 and 0 in the wildcard mask identify how to treat the corresponding IP address bits. However, in a wildcard mask, these bits are used for different purposes and follow different rules.

- Subnet masks use binary 1s and 0s to identify the network, subnet, and host portion of an IP address. Wildcard masks use binary 1s and 0s to filter individual IP addresses or groups of IP addresses to permit or deny access to resources.
- Wildcard masks and subnet masks differ in the way they match binary 1s and 0s. Wildcard masks use the following rules to match binary 1s and 0s:
- Wildcard mask bit 0 - Match the corresponding bit value in the address. • Wildcard mask bit 1 - Ignore the corresponding bit value in the address.

Wildcard Bit Mask Keywords

Working with decimal representations of binary wildcard mask bits can be tedious. To simplify this task, the keywords **host** and **any** help identify the most common uses of wildcard masking. These keywords eliminate entering wildcard masks when identifying a specific host or an entire network. These keywords also make it easier to read an ACL by providing visual clues as to the source or destination of the criteria.

The **host** keyword substitutes for the 0.0.0.0 mask. This mask states that all IPv4 address bits must match or only one host is matched.

The **any** option substitutes for the IP address and 255.255.255.255 mask. This mask says to ignore the entire IPv4 address or to accept any addresses.

Week no 10

100. TOPIC

ACL Creation and Placement:

Writing ACLs can be a complex task. For every interface there may be multiple policies needed to manage the type of traffic allowed to enter or exit that interface.

Here are some guidelines for using ACLs:

- Use ACLs in firewall routers positioned between your internal network and an external network such as the Internet.
- Use ACLs on a router positioned between two parts of your network to control traffic entering or exiting a specific part of your internal network.
- Configure ACLs on border routers, that is, routers situated at the edges of your networks. This provides a very basic buffer from the outside network, or between a less controlled area of your own network and a more sensitive area of your network.
- Configure ACLs for each network protocol configured on the border router interfaces.

The Three Ps

A general rule for applying ACLs on a router can be recalled by remembering the three Ps. You can configure one ACL per protocol, per direction, per interface:

One ACL per protocol -To control traffic flow on an interface, an ACL must be defined for each protocol enabled on the interface.

One ACL per direction -ACLs control traffic in one direction at a time on an interface. Two separate ACLs must be created to control inbound and outbound traffic.

One ACL per interface - ACLs control traffic for an interface, for example, GigabitEthernet 0/0.

Where to Place ACLs

The proper placement of an ACL can make the network operate more efficiently. An ACL can be placed to reduce unnecessary traffic. For example, traffic that will be denied at a remote destination should not be forwarded using network resources along the route to that destination.

Every ACL should be placed where it has the greatest impact on efficiency. The basic rules are:

Extended ACLs - Locate extended ACLs as close as possible to the source of the traffic to be filtered. This way, undesirable traffic is denied close to the source network without crossing the network infrastructure.

Standard ACLs - Because standard ACLs do not specify destination addresses, place them as close to the destination as possible. Placing a standard ACL at the source of the traffic will effectively prevent that traffic from reaching any other networks through the interface where the ACL is applied.

Placement of the ACL and therefore the type of ACL used may also depend on:

The extent of the network administrator's control - Placement of the ACL can depend on whether or not the network administrator has control of both the source and destination networks.

Bandwidth of the networks involved- Filtering unwanted traffic at the source prevents transmission of the traffic before it consumes bandwidth on the path to a destination. This is especially important in low bandwidth networks.

Ease of configuration - If a network administrator wants to deny traffic coming from several networks, one option is to use a single standard ACL on the router closest to the destination. The disadvantage is that traffic from these networks will use bandwidth unnecessarily. An extended ACL could be used on each router where the traffic originated. This will save bandwidth by filtering the traffic at the source but requires creating extended ACLs on multiple routers.

Standard ACL Placement

A standard ACL can only filter traffic based on a source address. The basic rule for placement of a standard ACL is to place the ACL as close as possible to the destination network. This allows the traffic to reach all other networks except the network where the packets will be filtered.

Extended ACL Placement

Like a standard ACL, an extended ACL can filter traffic based on the source address. However, an extended ACL can also filter traffic based on the destination address, protocol, and port number. This allows network administrators more flexibility in the type of traffic that can be filtered and where to place the ACL. The basic rule for placing an extended ACL is to place it as close to the source as possible. This prevents unwanted traffic from being sent across multiple networks only to be denied when it reaches its destination. Network administrators can only place ACLs on devices that they control. Therefore, placement must be determined in the context of where the control of the network administrator extends.

101. TOPIC

Configuring Multiarea OSPFv2:

In this activity, you will configure multiarea OSPFv2. The network is already connected and interfaces are configured with IPv4 addressing. Your job is to enable multiarea OSPFv2.

Part 1: Configure OSPFv2

Step 1: Configure OSPFv2 on R1.

Configure OSPFv2 on R1 with a process ID of 1 and a router ID of 1.1.1.1.

```
R1(config)# router ospf 1
```

```
R1(config-router)# router-id 1.1.1.1
```

Step 2: Advertise each directly connected network in OSPFv2 on R1.

Configure each network in OSPFv2 assigning areas according to the **Addressing Table**.

```
R1(config-router)# network 10.1.1.0 0.0.0.255 area 1
```

```
R1(config-router)# network 10.1.2.0 0.0.0.255 area 1
```

```
R1(config-router)# network 192.168.10.0 0.0.0.3 area 0
```

Step 3: Configure OSPFv2 on R2 and R3.

Repeat the steps above for **R2** and **R3** using a router ID of 2.2.2.2 and 3.3.3.3, respectively.

```
R2(config)# router ospf 1
```

```
R2(config-router)# router-id 2.2.2.2
```

```
R2(config-router)# network 10.2.1.0 0.0.0.255 area 0
```

```
R2(config-router)# network 192.168.10.0 0.0.0.3 area 0
```

```
R2(config-router)# network 192.168.10.4 0.0.0.3 area 0
```

```
R3(config)# router ospf 1
```

```
R3(config-router)# router-id 3.3.3.3
```

```
R3(config-router)# network 192.168.2.0 0.0.0.255 area 2
```

```
R3(config-router)# network 192.168.1.0 0.0.0.255 area 2
```

```
R3(config-router)# network 192.168.10.4 0.0.0.3 area 0
```

Link: <https://examans.com/9-2-2-6-packet-tracer-configuring-multiarea-ospfv2/>

102. TOPIC

Part 2: Verify and Examine Multiarea OSPFv2

Step 1: Verify connectivity to each of the OSPFv2 areas.

From R1, ping each of the following remote devices in area 0 and area 2: 192.168.1.2, 192.168.2.2, and 10.2.1.2.

Step 2: Use show commands to examine the current OSPFv2 operations.

Use the following commands to gather information about your OSPFv2 multiarea implementation.

show ip protocols

show ip route

show ip ospf database

show ip ospf interface

show ip ospf neighbor

103. TOPIC

Background

In this activity, you will configure multiarea OSPFv3. The network is already connected and interfaces are configured with IPv6 addressing. Your job is to enable multiarea OSPFv3, verify connectivity and examine the operation of multiarea OSPFv3.

Part 1: Configure OSPFv3

Step 1: Enable IPv6 routing and configure OSPFv3 on RA.

a. Enable IPv6 routing.

RA(config)# **ipv6 unicast-routing**

b. Configure OSPFv3 on RA with a process ID of 1 and a router ID of 1.1.1.1.

RA(config)# **ipv6 router ospf 1**

RA(config-rtr)# **router-id 1.1.1.1**

Step 2: Advertise each directly connected network in OSPFv3 on RA.

Configure each active IPv6 interface with OSPFv3 assigning each to the area listed in the **Addressing Table**.

RA(config)# **interface GigabitEthernet 0/0**

RA(config-if)# **ipv6 ospf 1 area 1**

RA(config-if)# **interface GigabitEthernet 0/1**

RA(config-if)# **ipv6 ospf 1 area 1**

RA(config-if)# **interface Serial 0/0/0**

RA(config-if)# **ipv6 ospf 1 area 0**

Step 3: Configure OSPFv3 on RB and RC

Repeat the Steps 1 and 2 for **RB** and **RC**, changing the router ID to 2.2.2.2 and 3.3.3.3 respectively.

RB(config)# **ipv6 unicast-routing**

RB(config)# **ipv6 router ospf 1**

RB(config-rtr)# **router-id 2.2.2.2**

RB(config-rtr)# **interface GigabitEthernet0/0**

RB(config-if)# **ipv6 ospf 1 area 0**

RB(config-if)# **interface Serial0/0/0**

RB(config-if)# **ipv6 ospf 1 area 0**

RB(config-if)# **interface Serial0/0/1**

RB(config-if)# **ipv6 ospf 1 area 0**

RC(config)# **ipv6 unicast-routing**

RC(config)# **ipv6 router ospf 1**

RC(config-rtr)# **router-id 3.3.3.3**

RC(config-rtr)# **interface GigabitEthernet 0/0**

RC(config-if)# **ipv6 ospf 1 area 2**

RC(config-if)# **interface GigabitEthernet 0/1**

RC(config-if)# **ipv6 ospf 1 area 2**

RC(config-if)# **interface Serial 0/0/1**

RC(config-if)# **ipv6 ospf 1 area 0**

Part 2: Verify Multiarea OSPFv3 Operations

Step 1: Verify connectivity to each of the OSPFv3 areas.

From RA, ping each of the following remote devices in area 0 and area 2: 2001:DB8:1:B1::2, 2001:DB8:1:A1::2, 2001:DB8:1:A2::2, 2001:DB8:1:C1::2, and 2001:DB8:1:C2::2.

Step 2: Use show commands to examine the current OSPFv3 operations.

Use the following commands to gather information about your OSPFv3 multiarea implementation.

show ipv6 ospf

show ipv6 route

show ipv6 ospf database

show ipv6 ospf interface

show ipv6 ospf neighbor

104. TOPIC

Configure Standard IPv4 ACLs:

[How are ACLs Created?](#)

In Two Steps:

- 1. Create an ACL definition.**
 - Enter global configuration mode.
 - Define statements of what to filter.
- 2. Apply the ACL to an interface.**
 - Enter interface configuration mode.
 - Identify the ACL and the direction to filter.

Create a Standard ACL

RTR(config)# access-list <i>ACL#</i> {permit deny} { test-conditions }
access-list 5 permit 172.34.54.34 0.0.0.0

Apply the ACL to an Interface

RTR(config-if)# {protocol} access-group <i>list-#</i> {in out}
ip access-group 5 out

Implied Deny

When traffic enters the router, the traffic is compared to all ACEs in the order that the entries occur in the ACL. The router continues to process the ACEs until it finds a match. The router will process the packet based on the first match found and no other ACEs will be examined.

If no matches are found when the router reaches the end of the list, the traffic is denied. This is because, by default, there is an implied deny at the end of all ACLs for traffic that was not matched to a configured entry. A single-entry ACL with only one deny entry has the effect of denying all traffic. At least one permit ACE must be configured in an ACL or all traffic is blocked.

Configuring Standard ACLs

To use numbered standard ACLs on a Cisco router, you must first create the standard ACL and then activate the ACL on an interface.

The **access-list** global configuration command defines a standard ACL with a number in the range of 1 through 99. Cisco IOS Software Release 12.0.1 extended these numbers by allowing 1300 to 1999 to be used for standard ACLs. This allows for a maximum of 798 possible standard ACLs. These additional numbers are referred to as expanded IP ACLs.

The full syntax of the standard ACL command is as follows:

```
Router(config)# access-list access-list-number { deny | permit | remark } source [source-wildcard] [log]
```

ACEs can deny or permit an individual host or a range of host addresses. To create a host statement in numbered ACL 10 that permits a specific host with the IP address 192.168.10.0, you would enter:

R1(config)# **access-list 10 permit host 192.168.10.10**

To create a statement that will permit a range of IPv4 addresses in a numbered ACL 10 that permits all IPv4 addresses in the network 192.168.10.0/24, you would enter:

R1(config)# **access-list 10 permit 192.168.10.0 0.0.0.255**

To remove the ACL, the global configuration **no access-list** command is used. Issuing the **show accesslist** command confirms that access list 10 has been removed.

Typically, when an administrator creates an ACL, the purpose of each statement is known and understood. However, to ensure that the administrator and others recall the purpose of a statement, remarks should be included. The **remark** keyword is used for documentation and makes access lists a great deal easier to understand. Each remark is limited to 100 characters.

Internal Logic – Order Matters

Cisco IOS applies an internal logic when accepting and processing standard ACEs. ACEs are processed sequentially. Therefore, the order in which ACEs are entered is important.

105. TOPIC

Configure Standard IPv4 ACLs - 2:

Applying Standard ACLs to Interfaces

Standard ACL Configuration Procedures

After a standard ACL is configured, it is linked to an interface using the **ip access-group** command in interface configuration mode:

Router(config-if)# **ip access-group** { *access-list-number* | *access-list-name* } { **in** | **out** }

To remove an ACL from an interface, first enter the **no ip access-group** command on the interface, and then enter the global **no access-list** command to remove the entire ACL.

Standard ACL Configuration Procedures

After a standard ACL is configured, it is linked to an interface using the **ip access-group** command in interface configuration mode:

Router(config-if)# **ip access-group** { *access-list-number* | *access-list-name* } { **in** | **out** }

To remove an ACL from an interface, first enter the **no ip access-group** command on the interface, and then enter the global **no access-list** command to remove the entire ACL.

Creating a Named ACL

Naming an ACL makes it easier to understand its function. For example, an ACL configured to deny FTP could be called NO_FTP. When you identify your ACL with a name instead of with a number, the configuration mode and command syntax are slightly different.

The steps required to create a standard named ACL.

Step 1. Starting from the global configuration mode, use the **ip access-list** command to create a named ACL. ACL names are alphanumeric, case sensitive, and must be unique. The **ip access-list standard name** is used to create a standard named ACL, whereas the command **ip access-list extended name** is for an extended access list. After entering the command, the router is in named standard ACL configuration mode as indicated by the prompt.

Note: Numbered ACLs use the global configuration command **access-list** whereas named IPv4 ACLs use the **ip access-list** command.

Step 2. From the named ACL configuration mode, use **permit** or **deny** statements to specify one or more conditions for determining whether a packet is forwarded or dropped.

Step 3. Apply the ACL to an interface using the **ip access-group** command. Specify if the ACL should be applied to packets as they enter into the interface (**in**) or applied to packets as they exit the interface (**out**).

Commenting ACLs

You can use the **remark** keyword to include comments (remarks) about entries in any IP standard or extended ACL. The remarks make the ACL easier for you to understand and scan. Each remark line is limited to 100 characters.

The remark can go before or after a **permit** or **deny** statement. You should be consistent about where you put the remark so that it is clear which remark describes which **permit** or **deny** statement. For example, it would be confusing to have some remarks before the associated **permit** or **deny** statements and some remarks after the statements.

To include a comment for IPv4 numbered standard or extended ACLs, use the **access-list accesslist_number remark remark** global configuration command. To remove the remark, use the **no** form of this command.

106. TOPIC

Editing Numbered ACLs

When configuring a standard ACL, the statements are added to the running-config. However, there is no built-in editing feature that allows you to edit a change in an ACL.

There are two ways that a standard numbered ACL can be edited.

Method 1: Using a Text Editor

After someone is familiar with creating and editing ACLs, it may be easier to construct the ACL using a text editor such as Microsoft Notepad. This allows you to create or edit the ACL and then paste it into the router. For an existing ACL, you can use the **show running-config** command to display the ACL, copy and paste it into the text editor, make the necessary changes, and paste it back in.

Configuration: Here are the steps to edit and correct ACL 1:

Step 1. Display the ACL using the **show running-config** command.

Step 2. Highlight the ACL, copy it, and then paste it into Microsoft Notepad. Edit the list as required. After the ACL is correctly displayed in Microsoft Notepad, highlight it and copy it.

Step 3. In global configuration mode, remove the access list using the **no access-list 1** command. Otherwise, the new statements would be appended to the existing ACL. Then paste the new ACL into the configuration of the router.

Step 4. Using the **show running-config** command, verify the changes

Method 2: Using the Sequence Number

The initial configuration of ACL 1 included a host statement for host 192.168.10.99. This was in error. The host should have been configured as 192.168.10.10. To edit the ACL using sequence numbers follow these steps:

Step 1. Display the current ACL using the **show access-lists 1** command. The output from this command will be discussed in more detail later in this section. The sequence number is displayed at the beginning of each statement. The sequence number was automatically assigned when the access list statement was entered. Notice that the misconfigured statement has the sequence number 10.

Step 2. Enter the **ip access-lists standard** command that is used to configure named ACLs. The ACL number, 1, is used as the name. First the misconfigured statement needs to be deleted using the **no 10** command with 10 referring to the sequence number. Next, a new sequence number 10 statement is added using the command, **10 deny host 192.168.10.10**.

Note: Statements cannot be overwritten using the same sequence number as an existing statement. The current statement must be deleted first, and then the new one can be added.

Step 3. Verify the changes using the **show access-lists** command.

Verifying ACLs

The **show ip interface** command is used to verify the ACL on the interface. The output from this command includes the number or name of the access list and the direction in which the ACL was applied.

Viewing ACLs Statistics

Once the ACL has been applied to an interface and some testing has occurred, the **show accesslists** command will show statistics for each statement that has been matched.

During testing of an ACL, the counters can be cleared using the **clear access-list counters** command. This command can be used alone or with the number or name of a specific ACL.

107. TOPIC

Extended IPv4 ACLs:

In this module we are going to talk about extended IPv4 ACLs. Extended IPv4 ACLs are similar to the standard IPv4 ACLs except extended ACLs provide more freedom and more options to filter the packets or provide conditions for the router to either deny, permit or make a remark to the packets. So they can filter on source addresses, destination addresses, protocol and port numbers.

You can find access list by using port numbers and keywords, as given in following examples.

Extended ACL Examples

Using Port Numbers

```
access-list 114 permit tcp 192.168.20.0 0.0.0.255 any eq 23
access-list 114 permit tcp 192.168.20.0 0.0.0.255 any eq 21
access-list 114 permit tcp 192.168.20.0 0.0.0.255 any eq 20
```

Using Keywords

```
access-list 114 permit tcp 192.168.20.0 0.0.0.255 any eq telnet
access-list 114 permit tcp 192.168.20.0 0.0.0.255 any eq ftp
access-list 114 permit tcp 192.168.20.0 0.0.0.255 any eq ftp-data
```



Syntax for extended ACL is:

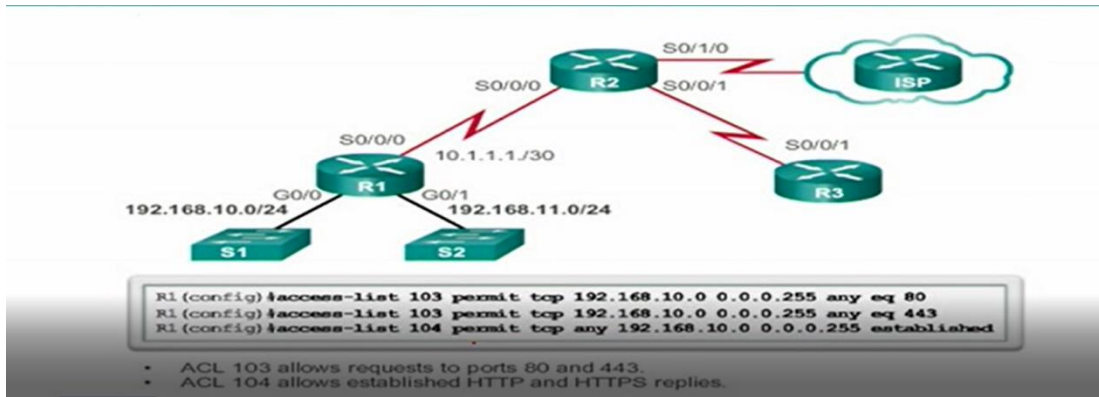
Access-list access-list-number (deny | permit | remark) protocol source [source-wildcard] [operator-operand] [port port-number or name] destination [destination-wildcard][operator-operand][port port-number or name][established]

Parameter	Description
access-list-number	Identifies the access list using a number in the range 100 to 199 (for an extended IP ACL) and 2000 to 2699 (expanded IP ACLs).
deny	Denies access if the conditions are matched.
permit	Permits access if the conditions are matched.
remark	Used to enter a remark or comment.
protocol	Name or number of an Internet protocol. Common keywords include icmp , ip , tcp , or udp . To match any Internet protocol (including ICMP, TCP, and UDP) use the ip keyword.
source	Number of the network or host from which the packet is being sent.
source-wildcard	Wildcard bits to be applied to source.
destination	Number of the network or host to which the packet is being sent.

destination-wildcard	Wildcard bits to be applied to the destination.
operator	(Optional) Compares source or destination ports. Possible operands include lt (less than), gt (greater than), eq (equal), neq (not equal), and range (inclusive range).
port	(Optional) The decimal number or name of a TCP or UDP port.
established	(Optional) For the TCP protocol only: Indicates an established connection.

Configuring Extended ACL:

In this particular example, network administrator configured an ACL in which he is restricting the network access to allow website browsing only from the LAN, which is being attached only to the interface gigabit 0/0. Only ACL 103 is allowing the traffic to go to any destination with that condition that traffic must be using port 80 which is http traffic and port 443 which is HTTP traffic, So it is only allowing HTTP and HTTP traffic. Established keyword is being used there, because its the nature of HTTP and HTTPS traffic to get response back, so the inbound traffic is also allowed.



Applying an access list to an interface using the following command:

(config) ip access-group ACL_NUMBER in | out

Testing Packets with Extended ACLs

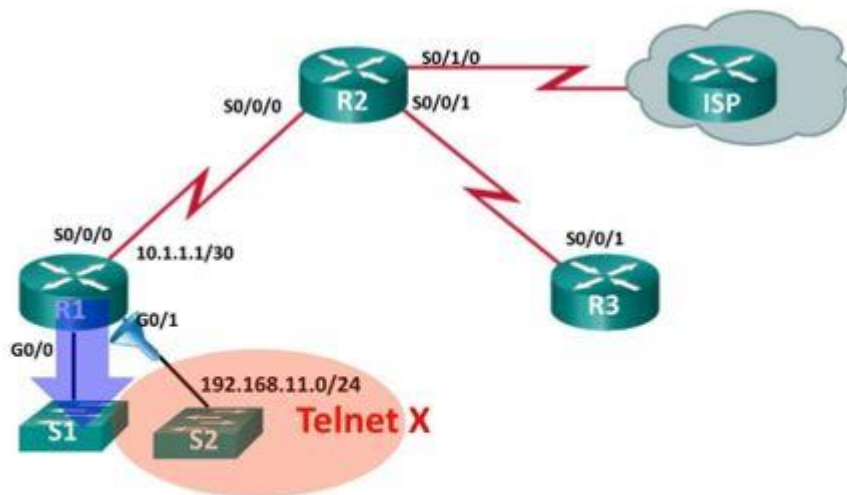
For more precise traffic-filtering control, extended IPv4 ACLs can be created. Extended ACLs are numbered 100 to 199 and 2000 to 2699, providing a total of 799 possible extended numbered ACLs. Extended ACLs can also be named.

Extended ACLs are used more often than standard ACLs because they provide a greater degree of control.

108. TOPIC

Extended IPv4 ACLs - 2:

Deny Telnet and Permit Everything Else



```
R1(config)# access-list 102 deny tcp any 192.168.11.0 0.0.0.255 eq 23
R1(config)# access-list 102 permit ip any any

R1(config)# interface g0/0
R1(config-if)# ip access-group 102 out
```

In this example, the network administrator configured an ACL to allow users from the 192.168.10.0/24 network to browse both insecure and secure websites. Even though it has been configured, the ACL will not filter traffic until it is applied to an interface. To apply an ACL to an interface, first consider whether the traffic to be filtered is going in or out. When a user on the internal LAN accesses a website on the Internet, traffic is going out to the Internet. When an internal user receives an email from the Internet, traffic is coming into the local router. However, when applying an ACL to an interface, in and out take on different meanings. From an ACL consideration, in and out are in reference to the router interface.

In the topology in the figure above, R1 has three interfaces. It has a serial interface, S0/0/0, and two Gigabit Ethernet interfaces, G0/0 and G0/1. Recall that an extended ACL should typically be applied close to the source. In this topology the interface closest to the source of the target traffic is the G0/0 interface.

Web request traffic from users on the 192.168.10.0/24 LAN is inbound to the G0/0 interface. Return traffic from established connections to users on the LAN is outbound from the G0/0 interface. The example applies the ACL to the G0/0 interface in both directions. The inbound ACL, 103, checks for the type of traffic. The outbound ACL, 104, checks for return traffic from established connections. This will restrict 192.168.10.0 Internet access to allow only website browsing.

Creating Named Extended ACLs

Named extended ACLs are created in essentially the same way that named standard ACLs are created. Follow these steps to create an extended ACL, using names:

Step 1. From global configuration mode, use the **ip access-list extended name** command to define a name for the extended ACL.

Step 2. In named ACL configuration mode, specify the conditions to **permit** or **deny**.

Step 3. Return to privileged EXEC mode and verify the ACL with the **show access-lists** *name* command.

Step 4. Save the entries in the configuration file with the **copy running-config startup-config** command.

To remove a named extended ACL, use the **no ip access-list extended** *name* global configuration command.

Verifying Extended ACLs

After an ACL has been configured and applied to an interface, use Cisco IOS **show** commands to verify the configuration.

Unlike standard ACLs, extended ACLs do not implement the same internal logic and hashing function. The output and sequence numbers displayed in the **show access-lists** command output is the order in which the statements were entered. Host entries are not automatically listed prior to range entries.

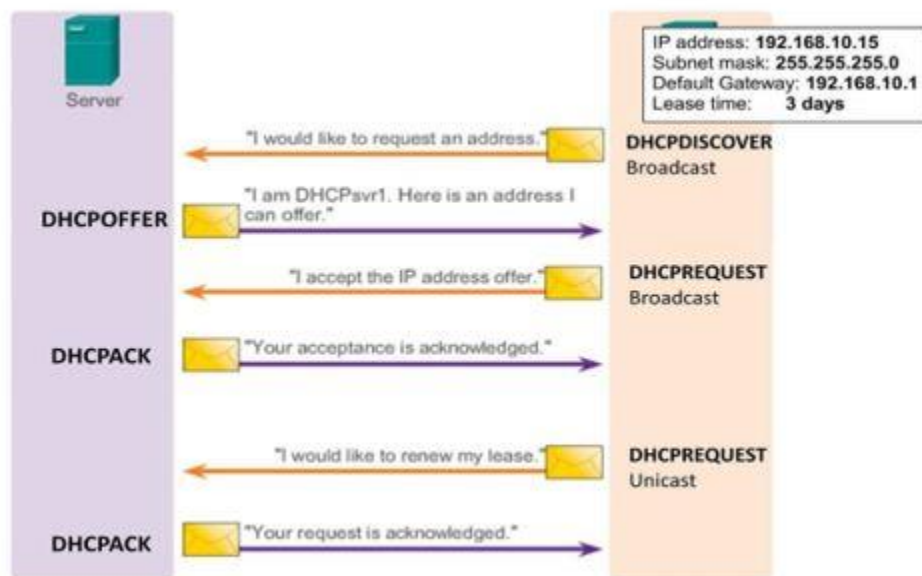
After an ACL configuration has been verified, the next step is to confirm that the ACLs work as planned; blocking and permitting traffic as expected.

109. TOPIC

Dynamic Host Configuration Protocol (DHCPv4):

Dynamic Host Configuration Protocol (DHCP) is a client/server protocol that automatically provides an Internet Protocol (IP) host with its IP address and other related configuration information such as the subnet mask and default gateway. DHCP allows hosts to obtain required TCP/IP configuration information from a DHCP server.

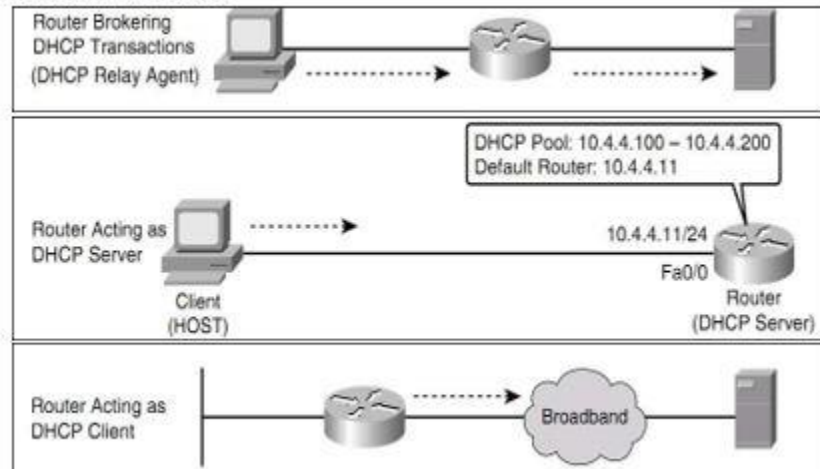
- Hosts were assigned an IP address manually.
- Next came BOOTP.
- The Bootstrap Protocol (BOOTP), defined in RFC 951, is the predecessor of DHCP.
- BOOTP provided a method for diskless workstations to download IP address configurations.
- Diskless workstations are workstation or PCs without disk drives
- E.g., Thin clients, cash register systems.
- Now we have DHCP.
- DHCP emerged as a standard protocol in October 1993 as defined in RFC 1531, succeeding the BOOTP protocol.
- DHCP allows a host to quickly and dynamically obtain configuration parameters from a DHCP Server.
- In an Enterprise, DHCP is almost exclusively always configured on a dedicated server.



110. TOPIC

Configuring DHCPv4:

Router DHCP Roles



IOS DHCP Server

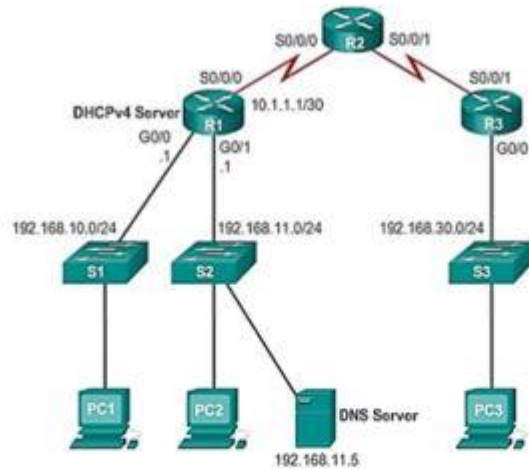
- The DHCP service is enabled by default on newer IOS.
- To disable DHCP, in global config mode:

– **no service dhcp**

- To re-enable:

– **service dhcp global** Steps to Configuring DHCP on a Router.

1. Define the excluded address ranges.



```
R1(config)#ip dhcp excluded-address low-address [high-address]
```

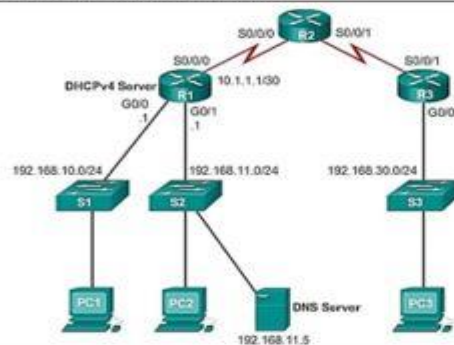
```
R1(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.9
```

```
R1(config)# ip dhcp excluded-address 192.168.10.254
```

2. Create the DHCP address pools.

- Create the DHCP pool using the **ip dhcp pool** command.
- Enters you into DHCP configuration mode

```
R1(config)#ip dhcp pool POOL-NAME
```



```
R1(config)#ip dhcp pool LAN-POOL-1
```

```
R1(dhcp-config)#
```

3. Configure the specifics of the pool.

Enter DHCP configuration mode and configure the specifics

3. Configure the specifics of the pool.

Enter DHCP configuration mode and configure the specifics

Tasks	Command
Define the address pool	network <i>network-number</i> [<i>mask</i> <i>/prefix-length</i>]
Define the default gateway.	default-router <i>address</i> [<i>address2...address8</i>]

Optional Tasks	Command
Define a DNS server.	dns-server <i>address</i> [<i>address2...address8</i>]
Define the domain name.	domain-name <i>domain</i>
Define the duration of the DHCP lease.	lease { <i>days</i> [<i>hours</i>] [<i>minutes</i>] <i>infinite</i> }
Define an option code	option <i>code</i> [<i>instance number</i>] { <i>ascii string</i> <i>hex string</i> <i>ip-address</i> } option 150 ip 192.168.1.254

111. TOPIC

Configuring Standard ACLs:

There are two types of ACLs: standard and extended. Standard ACLs are the oldest, dating back to the early days of Cisco's IOS Software (Release 8.3). Unlike extended ACLs, standard ACLs are limited to controlling traffic based on the source IP address information — as opposed to the source *and destination* IP address information. As you learned above, when a packet tries to enter or leave a router, its IP information is tested against each rule in an ACL. If the packet matches a rule, it's permitted or denied. Right about now, you might be wondering what the packet is permitted or denied to do. That depends on where you apply the ACL — inbound or outbound direction.

Tasks:

- Plan an ACL Implementation
- Configure, Apply and Verify a Standard ACL

This activity focuses on defining filtering criteria, configuring standard ACLs, applying ACLs to router interfaces, and verifying and testing the ACL implementation. The routers are already configured, including IP addresses.

Step 1: Investigate the current network configuration.

Before applying any ACLs to a network, it is important to confirm that you have full connectivity. Verify that the network has full connectivity by choosing a PC and pinging other devices on the network. You should be able to successfully ping every device.

Step 2: Evaluate two network policies and plan ACL implementations.

The following network policies are implemented on R2:

The 192.168.11.0/24 network is not allowed access to the WebServer on the 192.168.20.0/24 network.

All other access is permitted.

To restrict access from the 192.168.11.0/24 network to the WebServer at 192.168.20.254 without interfering with other traffic, an ACL must be created on R2. The access list must be placed on the outbound interface to the WebServer. A second rule must be created on R2 to permit all other traffic.

The following network policies are implemented on R3:

The 192.168.10.0/24 network is not allowed to communicate with the 192.168.30.0/24 network.

All other access is permitted.

To restrict access from the 192.168.10.0/24 network to the 192.168.30.0/24 network without interfering with other traffic, an access list will need to be created on R3. The ACL must be placed on the outbound interface to PC3. A second rule must be created on R3 to permit all other traffic.

Part 2: Configure, Apply, and Verify a Standard ACL

Step 1: Configure and apply a numbered standard ACL on R2.

Create an ACL using the number 1 on R2 with a statement that denies access to the 192.168.20.0/24 network from the 192.168.11.0/24 network.

```
R2(config)# access-list 1 deny 192.168.11.0 0.0.0.255
```

By default, an access list denies all traffic that does not match any rules. To permit all other traffic, configure the following statement:

```
R2(config)# access-list 1 permit any
```

For the ACL to actually filter traffic, it must be applied to some router operation. Apply the ACL by placing it for outbound traffic on the Gigabit Ethernet 0/0 interface.

```
R2(config)# interface GigabitEthernet0/0
```

```
R2(config-if)# ip access-group 1 out
```

Step 2: Configure and apply a numbered standard ACL on R3.

Create an ACL using the number 1 on R3 with a statement that denies access to the 192.168.30.0/24 network from the PC1 (192.168.10.0/24) network.

```
R3(config)# access-list 1 deny 192.168.10.0 0.0.0.255
```

By default, an ACL denies all traffic that does not match any rules. To permit all other traffic, create a second rule for ACL 1.

```
R3(config)# access-list 1 permit any
```

Apply the ACL by placing it for outbound traffic on the Gigabit Ethernet 0/0 interface.

```
R3(config)# interface GigabitEthernet0/0
```

```
R3(config-if)# ip access-group 1 out
```

Step 3: Verify ACL configuration and functionality.

On R2 and R3, enter the show access-list command to verify the ACL configurations. Enter the show run or show ip interface gigabitethernet 0/0 command to verify the ACL placements.

With the two ACLs in place, network traffic is restricted according to the policies detailed in Part 1. Use the following tests to verify the ACL implementations:

A ping from 192.168.10.10 to 192.168.11.10 succeeds.

A ping from 192.168.10.10 to 192.168.20.254 succeeds.

A ping from 192.168.11.10 to 192.168.20.254 fails.

A ping from 192.168.10.10 to 192.168.30.10 fails.

A ping from 192.168.11.10 to 192.168.30.10 succeeds.

A ping from 192.168.30.10 to 192.168.20.254 succeeds.

Week no 11

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Background / Scenario

Two employees need access to services provided by the server. **PC1** only needs FTP access while **PC2** only needs web access. Both computers are able to ping the server, but not each other.

Part 1: Configure, Apply and Verify an Extended Numbered ACL

Step 1: Configure an ACL to permit FTP and ICMP.

a. From global configuration mode on R1, enter the following command to determine the first valid number for an extended access list.

```
R1(config)# access-list ?  
<1-99> IP standard access list  
<100-199> IP extended access list
```

b. Add 100 to the command, followed by a question mark.

```
R1(config)# access-list 100 ?  
deny Specify packets to reject  
permit Specify packets to forward  
remark Access list entry comment
```

c. To permit FTP traffic, enter permit, followed by a question mark.

```
R1(config)# access-list 100 permit ?  
ahp Authentication Header Protocol  
eigrp Cisco's EIGRP routing protocol  
esp Encapsulation Security Payload  
gre Cisco's GRE tunneling  
icmp Internet Control Message Protocol  
ip Any Internet Protocol  
ospf OSPF routing protocol  
tcp Transmission Control Protocol  
udp User Datagram Protocol
```

d. This ACL permits FTP and ICMP. ICMP is listed above, but FTP is not, because FTP uses TCP. So you enter TCP. Enter tcp to further refine the ACL help.

```
R1(config)# access-list 100 permit tcp ?  
A.B.C.D Source address  
any Any source host  
host A single source host
```

e. Enter the network address, followed by a question mark.

```
R1(config)# access-list 100 permit tcp 172.22.34.64 ?  
A.B.C.D Source wildcard bits
```

f. Enter the wildcard mask, followed by a question mark.

```
R1(config)# access-list 100 permit tcp 172.22.34.64 0.0.0.31 ?  
  
A.B.C.D Destination address  
  
any Any destination host
```

eq Match only packets on a given port number

gt Match only packets with a greater port number

host A single destination host

lt Match only packets with a lower port number

neq Match only packets not on a given port number

range Match only packets in the range of port numbers

g. Configure the destination address. In this scenario, we are filtering traffic for a single destination, the server. Enter the host keyword followed by the server's IP address.

```
R1(config)# access-list 100 permit tcp 172.22.34.64 0.0.0.31 host 172.22.34.62 ?
```

dscp Match packets with given dscp value

eq Match only packets on a given port number

established established

gt Match only packets with a greater port number

lt Match only packets with a lower port number

neq Match only packets not on a given port number

precedence Match packets with given precedence value

range Match only packets in the range of port numbers

<cr>

h. Notice that one of the options is <cr> (carriage return). In other words, you can press Enter and the statement would permit all TCP traffic. However, we are only permitting FTP traffic; therefore, enter the eq keyword, followed by a question mark to display the available options. Then, enter ftp and press Enter.

```
R1(config)# access-list 100 permit tcp 172.22.34.64 0.0.0.31 host 172.22.34.62 eq ?
```

<0-65535> Port number

ftp File Transfer Protocol (21)

pop3 Post Office Protocol v3 (110)

smtp Simple Mail Transport Protocol (25)

telnet Telnet (23)

www World Wide Web (HTTP, 80)

```
R1(config)# access-list 100 permit tcp 172.22.34.64 0.0.0.31 host 172.22.34.62 eq ftp
```

Step 2: Apply the ACL on the correct interface to filter traffic.

From **R1**'s perspective, the traffic that ACL 100 applies to is inbound from the network connected to Gigabit Ethernet 0/0 interface. Enter interface configuration mode and apply the ACL.

```
R1(config)# interface gigabitEthernet 0/0
```

```
R1(config-if)# ip access-group 100 in
```

113. TOPIC

Creating Named Extended ACLs :

Named extended ACLs are created in essentially the same way that named standard ACLs are created. Follow these steps to create an extended ACL, using names:

Step 1. From global configuration mode, use the **ip access-list extended** name command to define a name for the extended ACL.

Step 2. In named ACL configuration mode, specify the conditions to permit or deny.

Step 3. Return to privileged EXEC mode and verify the ACL with the show access-lists name command.

Step 4. Save the entries in the configuration file with the copy running-config startup-config command.

To remove a named extended ACL, use the no ip access-list extended name global configuration command.

Verifying Extended ACLs

After an ACL has been configured and applied to an interface, use Cisco IOS show commands to verify the configuration.

Unlike standard ACLs, extended ACLs do not implement the same internal logic and hashing function. The output and sequence numbers displayed in the show access-lists command output is the order in which the statements were entered. Host entries are not automatically listed prior to range entries. After an ACL configuration has been verified, the next step is to confirm that the ACLs work as planned; blocking and permitting traffic as expected.

114. TOPIC

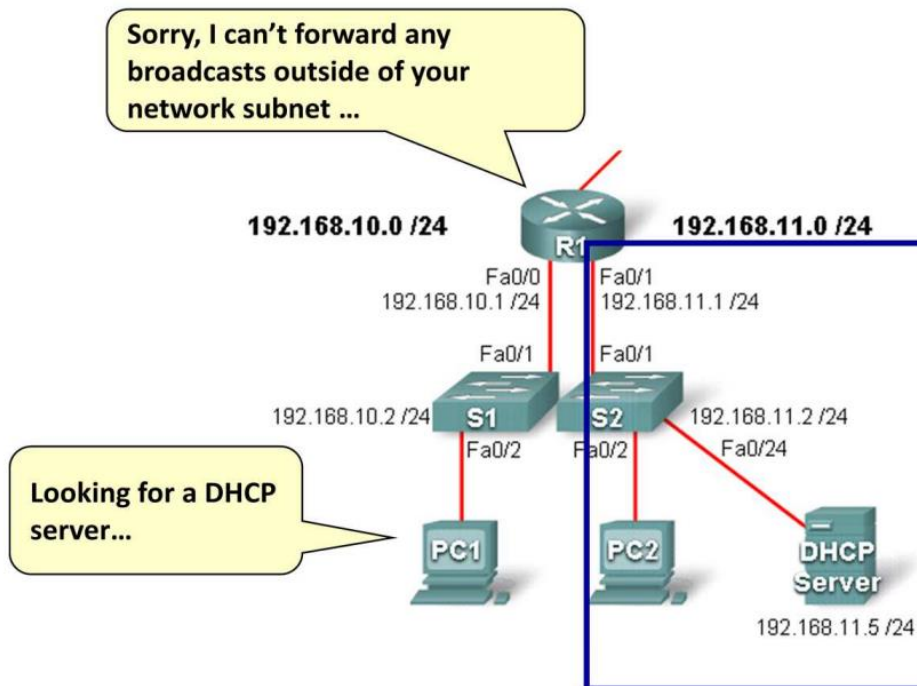
Verify DHCP Configuration

- show ip dhcp server statistics
 - Displays counts for server statistics and messages sent and received for an IOS-based DHCP server.

- show ip dhcp binding
 - Displays DHCP binding information for IP address assignment and subnet allocation.
- show ip dhcp conflict
 - Displays address conflicts found by a Cisco IOS DHCP server when addresses are offered to the client.
- show ip dhcp pool name
 - Displays the subnets pool allocated and the current utilization level for the pool or all the pools if the name argument is not used

115. TOPIC

DHCP Problems



DHCP Relays – Helper Addresses

A host or router can be a DHCP relay agent which forwards the DHCP packets from clients to server or vice versa. It permits the clients to access the IP addresses from the server that are not configured on the same LAN. Remote clients may require DHCP services and send broadcasts to locate these servers. Routers, by default, will not forward client broadcasts beyond their subnet.

Solution:

- Place DHCP servers on all subnets
- Use the Cisco IOS helper address feature.

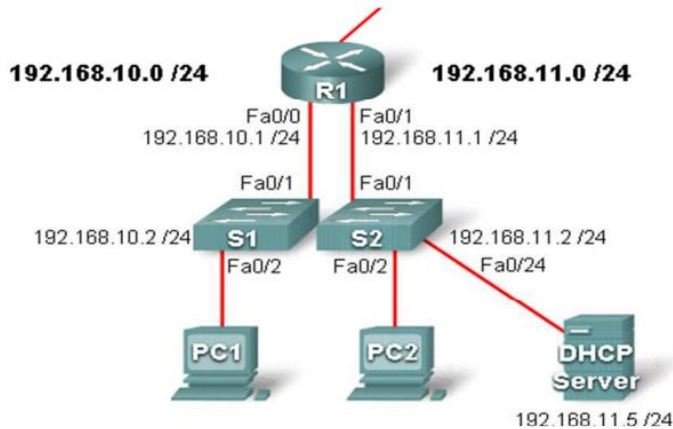
- The IP helper address enables a router forward a UDP broadcast to a specific unicast IP address.
- Configured using the ip helper-address interface configuration command.

Command relays UDP broadcast requests.

Configured on the interface receiving the broadcast.

DHCP Relays

```
R1# config t
R1(config)# interface Fa0/0
R1(config-if)# ip helper-address 192.168.11.5
R1(config-if)# end
```



116. TOPIC

Configure DHCPv4 Client:

Routers can also be DHCP clients.

- Typical in small broadband routers for home use to connect to an ISP using a DSL or cable modem.
- They are set to acquire an IP address automatically from their ISPs.

The following command can be used to enable a Cisco IOS device to obtain an IP address dynamically from DHCP server

```
Router(config-if)#
```

```
ip address dhcp
```

Sometimes, Cisco IOS routers in SOHO (Small Office Home Office) and branch sites have to be configured based on the following manners.

- The method used depends on the ISP.
- However, in its simplest configuration, the Ethernet interface is used to connect to a cable modem.
- To configure an Ethernet interface as a DHCP client, the **ip address dhcp** command must be configured.

117. TOPIC

Internet Concerns

- There are not enough public IPv4 addresses to assign a unique address to each device connected to the Internet.
- Therefore the IETF developed several solutions to help stave off this depletion of global IPv4 addresses:
 - o Subnetting
 - o Variable-length subnet masking (VLSM)
 - o Classless interdomain routing (CIDR)
 - o Route summarization
 - o Private addressing and NAT
 - o Long term solution: IP version 6 (IPv6)

Private Addresses

Class	RFC 1918 Range		CIDR Prefix
A	10.0.0.0	– 10.255.255.255	10.0.0.0/8
B	172.16.0.0	– 172.31.255.255	172.16.0.0/12
C	192.168.0.0	– 192.168.255.255	192.168.0.0/16

Network Address Translation (NAT)

- NAT is a process used to translate network addresses.
- NAT's primary use is to conserve public IPv4 addresses.

- NAT is usually implemented on border network devices (e.g., firewalls or routers).
- NAT allows the networks to use private addresses internally, only translating to public addresses when needed.
- Devices within the organization are assigned private addresses and operate with locally unique addresses.
- When traffic must be sent or received to or from other organizations or the Internet, the border router translates the addresses to a public and globally unique address.
- You need to connect to the Internet and your hosts do not have globally unique IP addresses.
- You change over to a new Internet service provider (ISP) that requires you to renumber your network.
- Two intranets with duplicate addresses merge.
- You want to support basic load sharing.

NAT Terminology

NAT terminology is always applied from the perspective of the device with the translated address:

Inside address: The address of the device which is being translated by NAT.

Outside address: The address of the destination device. NAT also uses the concept of local or global with respect to addresses:

Local address: A local address is any address that appears on the inside portion of the network.

Global address: A global address is any address that appears on the outside portion of the network.

118.TOPIC

NAT Application Types

1. Static address translation (static NAT):
 - One-to-one address mapping between local and global addresses.
2. Dynamic address translation (dynamic NAT):
 - Many-to-many address mapping between local and global addresses.
3. Port Address Translation (PAT):

- Many-to-one address mapping between local and global addresses.
- This method is also known as overloading (NAT overloading).

Static NAT

- Permanently bind an inside local address to an inside global address.
- Mappings are configured by the administrator and remain constant.
- Typically used to configure an internal server that must be accessed from the outside world.

Static NAT Configuration Steps

1. Create a mapping between the inside local address and the inside global addresses.
 - Configured using the ip nat inside source static inside-local inside-global global configuration command.
2. Identify the inside and outside NAT interfaces.
 - Configured using the ip nat inside and ip nat outside interface configuration commands.

119. TOPIC

NAT Application Types

1. Static address translation (static NAT):
 - One-to-one address mapping between local and global addresses.
2. Dynamic address translation (dynamic NAT):
 - Many-to-many address mapping between local and global addresses.
3. Port Address Translation (PAT):
 - Many-to-one address mapping between local and global addresses.
 - This method is also known as overloading (NAT overloading).

Dynamic NAT uses a pool of public addresses and assigns them on a first-come, first-served basis. When an inside device requests access to an outside network, dynamic NAT assigns the inside local address an inside global address from a pool of addresses.

Dynamic NAT Configuration Steps

1. Define the pool of addresses that will be used for translation.

Configured using the `ip nat pool pool-name start-ip end-ip {netmask netmask | prefix length prefix-length}` global configuration command.

2. Configure a standard ACL to identify (permit) only those addresses that are to be translated.
3. Bind the ACL to the pool.

Configured using the `ip nat inside source list acl-# pool pool-name global config` command.

4. Identify the inside and outside NAT interfaces.

Configured using the `ip nat inside` and `ip nat outside` interface configuration commands.

Dynamic NAT Timeout

- Dynamic translations are temporary, and will eventually time out (default 24 hours).

– Timeout can be configured.

- It is important for translation table entries to time out so that addresses in the pool become available for other hosts.

- If translation table entries do not time out fast enough, the entire pool of addresses could be in use.

120. TOPIC

- PAT (also called NAT overload) allows the router to use one inside global address for many inside local addresses.

- With address overloading, many privately addressed nodes can access the Internet using a single global address.

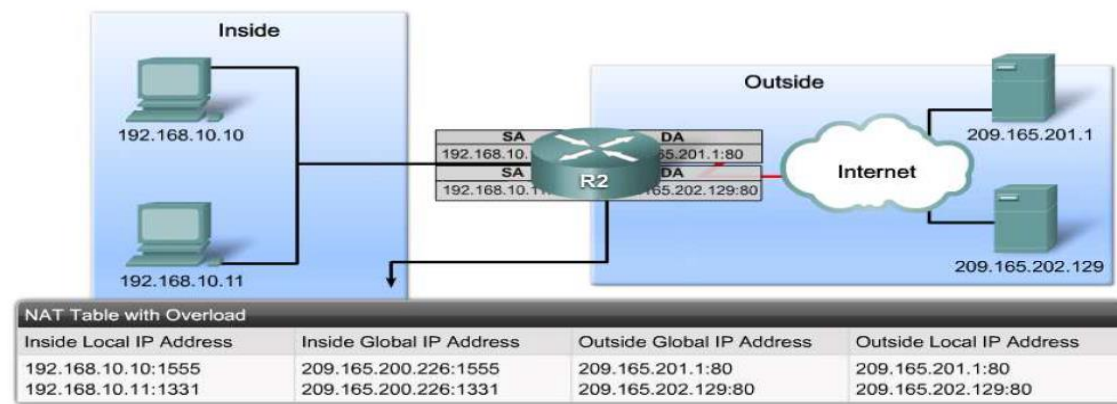
- There are two ways to configure PAT:

- ISP allocates a single public IPv4 address
- ISP allocates more than one public IPv4 address

- **Note:**

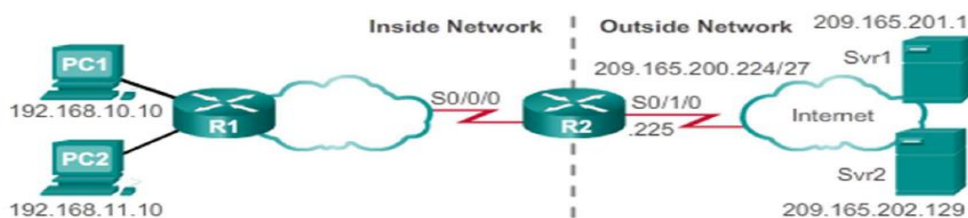
- Over 65,000 inside addresses can theoretically map to a single outside address.
- However, 4000 local addresses per global address is more realistic.
- Each NAT translation consumes about 160 bytes of router DRAM.

NAT Overload (PAT)



- The NAT router keeps track of the different conversations by mapping TCP and UDP port numbers in the translation table.
- Called an extended table entry.

Configuring PAT Using a Pool Example



```

R2(config)# ip nat pool NAT-POOL2 209.165.200.226 209.165.200.240
prefix-length 27
R2(config)# access-list 1 permit 192.168.0.0 0.0.255.255
R2(config)#
R2(config)# ip nat inside source list 1 pool NAT-POOL2 overload
R2(config)# interface Serial0/0/0
R2(config-if)# ip nat inside
R2(config-if)# exit
R2(config)# interface Serial0/1/0
R2(config-if)# ip nat outside
  
```

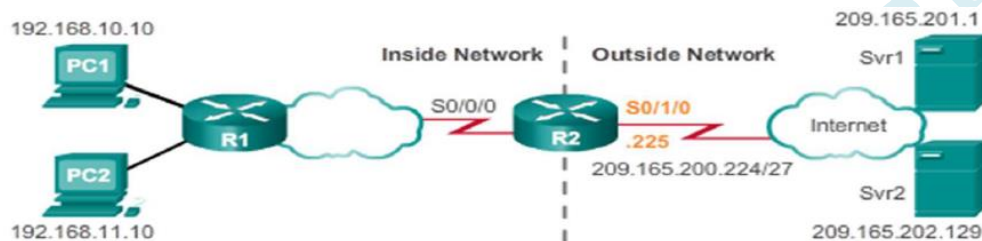
Verifying PAT Using a Pool Example

```
R2# show ip nat translations
```

Pro	Inside global	Inside local	Outside local	Outside global
tcp	209.165.200.226:51839	192.168.10.10:51839	209.165.201.1:80	209.165.201.1:80
tcp	209.165.200.226:42558	192.168.11.10:42558	209.165.202.129:80	209.165.202.129:80

```
R2#
```

Configure PAT Using an Address Example



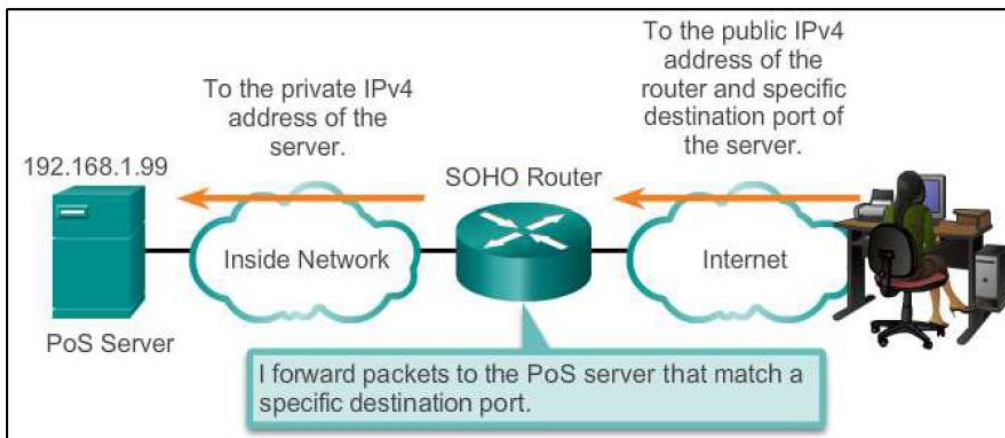
```
R2(config)# access-list 1 permit 192.168.0.0 0.0.255.255
R2(config)#
R2(config)# ip nat source list 1 interface serial 0/1/0 overload
R2(config)#
R2(config)# interface Serial0/0/0
R2(config-if)# ip nat inside
R2(config-if)# exit
R2(config)# interface Serial0/1/0
R2(config-if)# ip nat outside
```

121. TOPIC

• Port forwarding (sometimes referred to as tunneling) is the act of forwarding traffic addressed to a specific network port from one network node to another.

- Helpful in situations where servers have private addresses, not reachable from the outside networks.

- Port forwarding can be enabled for applications by specifying the inside local address that requests should be forwarded to.



Configuring Port Forwarding with IOS

- In IOS, Port forwarding is essentially a static NAT translation with a specified TCP or UDP port number.
- Configured using the **ip nat inside source {static {tcp | udp local-ip local-port global-ip global-port} [extendable]}** global configuration command

Parameter	Description
tcp or udp	Indicates if this is a TCP or UDP port number.
<i>local-ip</i>	This is the IPv4 address assigned to the inside host (typically a private address).
<i>local-port</i>	- Sets the local TCP/UDP port in a range from 1-65535. - This is the port number the server is listening on.
<i>global-ip</i>	- Sets the global TCP/UDP port in a range from 1-65535. - This is the port number the outside client will use to reach the internal server.
extendable	The option is applied by default and allows the router to extend the translation to more than one port if necessary.

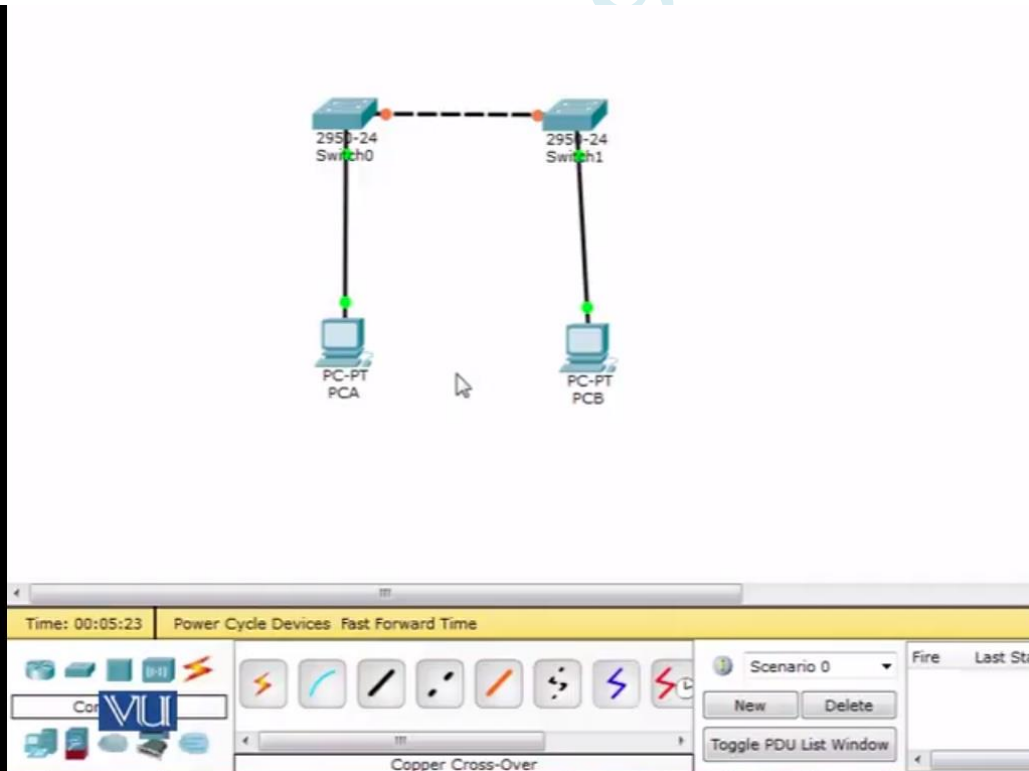
122. TOPIC

We will cover the following concepts in this topic i.e.

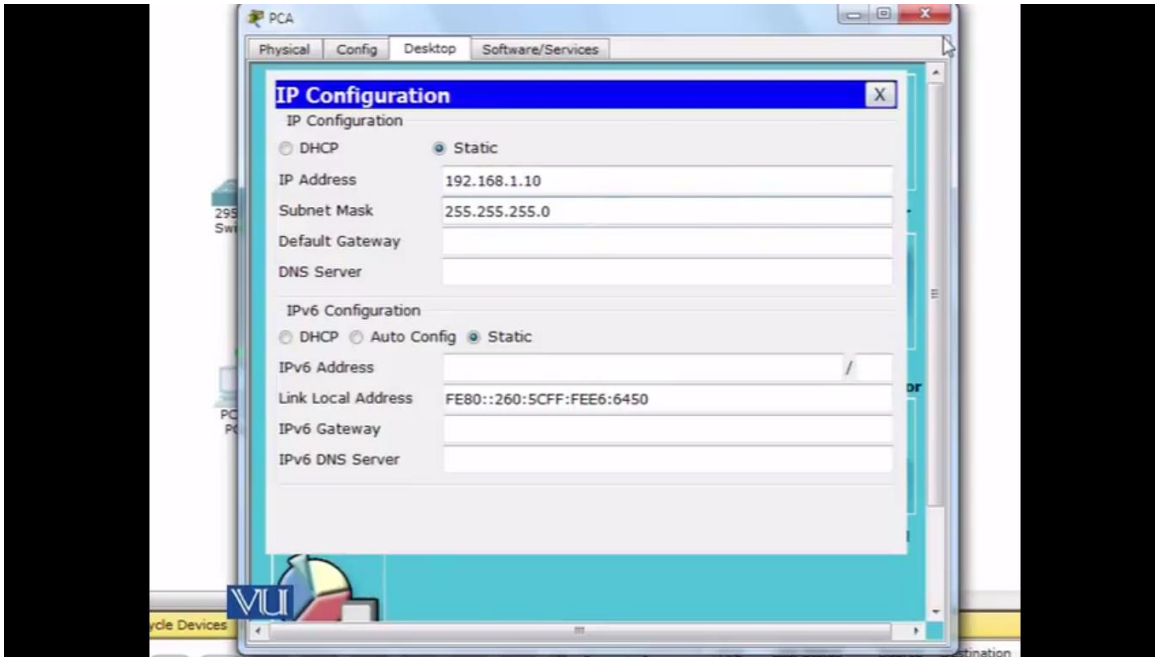
- Setup Network Topology
- Configure PC Hosts
- Configure Switch Settings (Basic)

We will create an Ethernet network topology, for this we need some PC's & switch in Packet Tracer Software.

Here in the screen shot we can see that, 2 pc's and 2 switches (connected via cross copper cable) are placed.



IP configuration of the PC's



IP Config Command:

IP Config/All Command is used to check the complete configuration of the PC in a topology.

Configuration of the Switch (Basic Settings):

In CLI prompt, at first we enter “enable” command to enter privileged execution mode, after that enter “#config T” to enter configuration terminal. After that we assign a name to the switch with “#hostname S1”. Here S1 is the name of switch.

To prevent unauthorized access, we use a local password in the configuration with a command name “#enable secret class”. And for password “#password cisco”, here cisco is the password.

“#line console 0” is used to secure the console option of the switch.

Week no 12

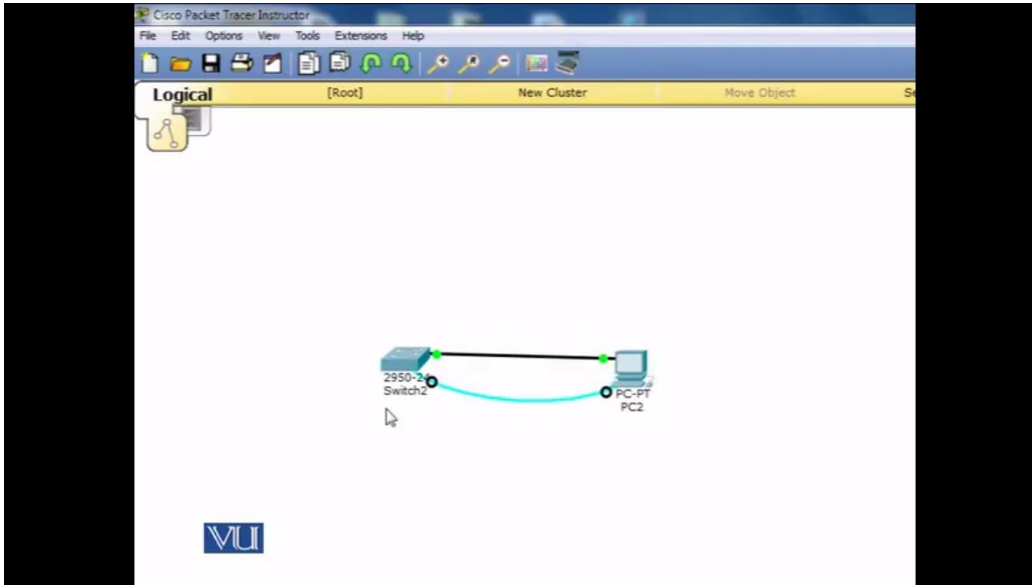
123. TOPIC

We will cover the following concepts in this topic i.e.

- Configure a basic network device (Switch)
- Verify the network connectivity

To configure a switch via a PC device, to Access a switch with console and remote access method.

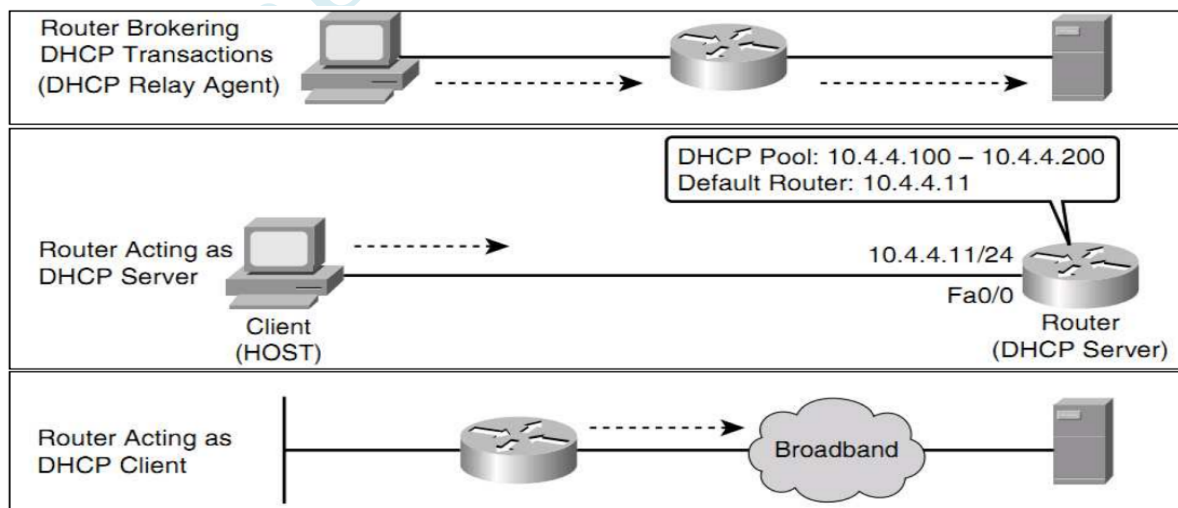
Console Connection: Between PC and Switch via console cable so that we can configure switch via the PC.



We can use the following commands in the screen shot to execute/configure by console.

124. TOPIC

Router DHCP Roles:

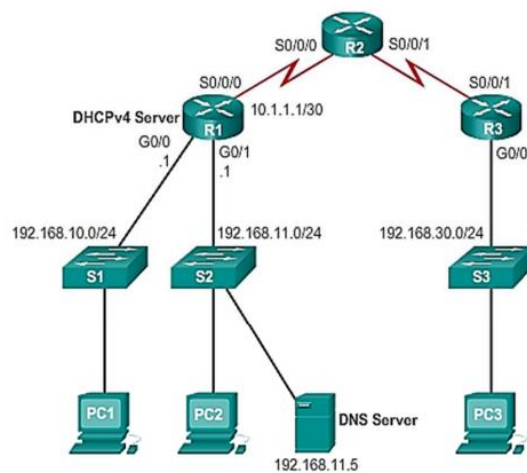


IOS DHCP Server

- The DHCP service is enabled by default on newer IOS.
- To disable DHCP, in global config mode:
 - no service dhcp
- To re-enable:
 - service dhcp global

Steps to Configuring DHCP on a Router

1. Define the excluded address ranges.



```
R1(config)#ip dhcp excluded-address low-address [high-address]
```

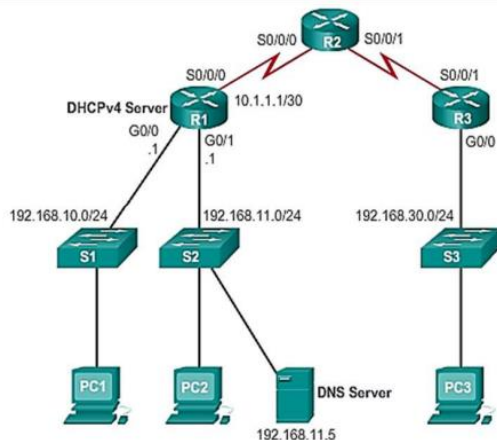
```
R1(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.9
```

```
R1(config)# ip dhcp excluded-address 192.168.10.254
```

2. Create the DHCP address pools.

- Create the DHCP pool using the **ip dhcp pool** command.
 - Enters you into DHCP configuration mode

```
R1(config)#ip dhcp pool POOL-NAME
```



```
R1(config)#ip dhcp pool LAN-POOL-1
```

```
R1(dhcp-config)#
```

3. Configure the specifics of the pool.

- Enter DHCP configuration mode and configure the specifics

Tasks	Command
Define the address pool	network <i>network-number</i> [<i>mask</i> <i>lprefix-length</i>]
Define the default gateway.	default-router <i>address</i> [<i>address2...address8</i>]

Optional Tasks	Command
Define a DNS server.	dns-server <i>address</i> [<i>address2...address8</i>]
Define the domain name.	domain-name <i>domain</i>
Define the duration of the DHCP lease.	lease { <i>days</i> [<i>hours</i>] [<i>minutes</i>] <i>infinite</i> }
Define an option code	option <i>code</i> [<i>instance number</i>] { <i>ascii string</i> <i>hex string</i> <i>ip-address</i> } option 150 ip 192.168.1.254

125. TOPIC

Verify DHCP Configuration

- show ip dhcp server statistics
 - Displays counts for server statistics and messages sent and received for an IOS-based DHCP server.
- show ip dhcp binding
 - Displays DHCP binding information for IP address assignment and subnet allocation.
- show ip dhcp conflict
 - Displays address conflicts found by a Cisco IOS DHCP server when addresses are offered to the client.
- show ip dhcp pool name
 - Displays the subnets pool allocated and the current utilization level for the pool or all the pools if the name argument is not used.

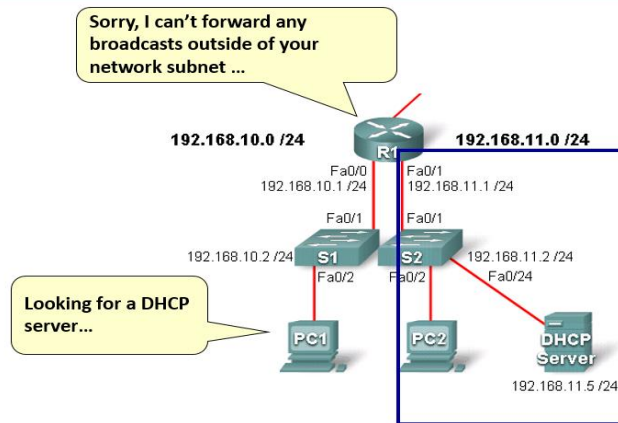
126. TOPIC

The main problem with the routers is that they cannot broadcast traffic from their network to another network.

- Router divides the broadcast domain and limits them.
- Switches divides the collision domain and limits them.

As shown in the screenshot, we observe that when a PC tried to access the DHCP, it gets error because of the router limitation.

DHCP Problems



Remote clients may require DHCP services and send broadcasts to locate these servers. Routers, by default, will not forward client broadcasts beyond their subnet.

Solution:

- Place DHCP servers on all subnets
- Use the Cisco IOS helper address feature.

DHCP RELAYS:

HELPER ADDRESSES:

- Port 37: Time
- Port 49: TACACS
- Port 53: DNS
- Port 67: DHCP/BOOTP client
- Port 68: DHCP/BOOTP server
- Port 69: TFTP
- Port 137: NetBIOS name service
- Port 138: NetBIOS datagram service

HELPER ADDRESS EXAMPLE:

CS407

Routing and switching
Reading Material

Made by Mah jabeen
[mahjabeen97869@gmail.com]

```
Router(config-if)#
```

```
ip helper-address address
```

- Enables forwarding and specifies destination address for main UDP broadcast packets
- Changes destination address from broadcast to unicast or directed broadcast address

```
Router(config)#
```

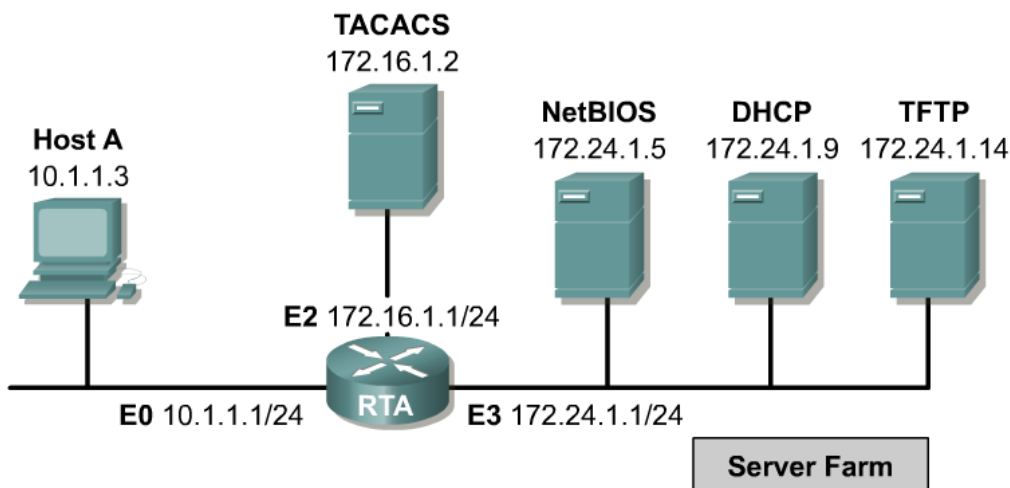
```
ip forward-protocol { udp [ port ] }
```

- Specifies which protocols will be forwarded

HELPER ADDRESSES:

- Port 37: Time
- Port 49: TACACS
- Port 53: DNS
- Port 67: DHCP/BOOTP client
- Port 68: DHCP/BOOTP server
- Port 69: TFTP
- Port 137: NetBIOS name service
- Port 138: NetBIOS datagram service

HELPER ADDRESS EXAMPLE:



```
RTA(config)# interface e0
```

```
RTA(config-if)# ip helper-address 172.16.1.2
```

```
RTA(config-if)# ip helper-address 172.24.1.9
```

```
RTA(config-if)# ip helper-address 172.24.1.255
```

127. TOPIC

Lecture-23.127 Configuring DHCPv4 Client (2)

Reading Tab:

- Routers can also be DHCP clients.
 - Typical in small broadband routers for home use to connect to an ISP using a DSL or cable modem.
 - They are set to acquire an IP address automatically from their ISPs.

The following command can be used to enable a Cisco IOS device to obtain an IP address dynamically from DHCP server

```
Router(config-if)#
```

```
ip address dhcp
```

Sometimes, Cisco IOS routers in SOHO (Small Office Home Office) and branch sites have to be configured based on the following manners.

- The method used depends on the ISP.
- However, in its simplest configuration, the Ethernet interface is used to connect to a cable modem.
- To configure an Ethernet interface as a DHCP client, the **ip address dhcp** command must be configured.

128. TOPIC

Lecture-23.128 Packet Tracer - Configuring Basic DHCPv4 on a Router , Part-1

129. TOPIC

130. TOPIC

131. TOPIC

Internet Concerns

- There are not enough public IPv4 addresses to assign a unique address to each device connected to the Internet.
- Therefore the IETF developed several solutions to help stave off this depletion of global IPv4 addresses:
 - o Subnetting
 - o Variable-length subnet masking (VLSM)
 - o Classless interdomain routing (CIDR)
 - o Route summarization
 - o Private addressing and NAT
 - o Long term solution: IP version 6 (IPv6)

Private Addresses

Class	RFC 1918 Range		CIDR Prefix
A	10.0.0.0	- 10.255.255.255	10.0.0.0/8
B	172.16.0.0	- 172.31.255.255	172.16.0.0/12
C	192.168.0.0	- 192.168.255.255	192.168.0.0/16

Network Address Translation (NAT)

- NAT is a process used to translate network addresses.

- NAT's primary use is to conserve public IPv4 addresses.
- NAT is usually implemented on border network devices (e.g., firewalls or routers).
- NAT allows the networks to use private addresses internally, only translating to public addresses when needed.
- Devices within the organization are assigned private addresses and operate with locally unique addresses.
- When traffic must be sent or received to or from other organizations or the Internet, the border router translates the addresses to a public and globally unique address.
- You need to connect to the Internet and your hosts do not have globally unique IP addresses.
- You change over to a new Internet service provider (ISP) that requires you to renumber your network.
- Two intranets with duplicate addresses merge.
- You want to support basic load sharing.

NAT Terminology

NAT terminology is always applied from the perspective of the device with the translated address:

Inside address: The address of the device which is being translated by NAT.

Outside address: The address of the destination device. NAT also uses the concept of local or global with respect to addresses:

Local address: A local address is any address that appears on the inside portion of the network.

Global address: A global address is any address that appears on the outside portion of the network.

132. TOPIC

Lecture-24.132 Types of NAT Applications - Static NAT

Reading Tab:

NAT Application Types

- Static address translation (static NAT):

- One-to-one address mapping between local and global addresses.
- Dynamic address translation (dynamic NAT):
 - Many-to-many address mapping between local and global addresses.
- Port Address Translation (PAT):
 - Many-to-one address mapping between local and global addresses.
 - This method is also known as overloading (NAT overloading).

Static NAT

- Permanently bind an inside local address to an inside global address.
- Mappings are configured by the administrator and remain constant.
- Typically used to configure an internal server that must be accessed from the outside world. **Static NAT**

Configuration Steps

1. Create a mapping between the inside local address and the inside global addresses.
 - Configured using the `ip nat inside source static inside-local inside-global` global configuration command.
2. Identify the inside and outside NAT interfaces.
 - Configured using the `ip nat inside` and `ip nat outside` interface configuration commands.

Verifying Static NAT Example

To verify the static NAT, following command will be executed.

```
R2# show ip nat translations
Pro Inside global    Inside local    Outside local    Outside global
--- 209.165.201.5    192.168.10.254    ---             ---
R2#
```

133. TOPIC

Reading Tab:

Dynamic NAT:

Dynamic NAT uses a pool of public addresses and assigns them on a first-come, first-served basis. When an inside device requests access to an outside network, dynamic NAT assigns the inside local address an inside global address from a pool of addresses.

Dynamic NAT Configuration Steps

1. Define the pool of addresses that will be used for translation

Configured using the **ip nat pool** pool-name start-ip end-ip {**netmask** netmask | **prefix-length** prefix-length} global configuration command.

2. Configure a standard ACL to identify (permit) only those addresses that are to be translated.
3. Bind the ACL to the pool.

- Configured using the **ip nat inside source list** acl-# **pool** pool-name global config command.

4. Identify the inside and outside NAT interfaces.

- Configured using the **ip nat inside** and **ip nat outside** interface configuration commands.

Dynamic NAT Timeout

- Dynamic translations are temporary, and will eventually time out (default 24 hours).

- Timeout can be configured.

- It is important for translation table entries to time out so that addresses in the pool become available for other hosts.

- If translation table entries do not time out fast enough, the entire pool of addresses could be in use.

```
Router(config)# ip nat translation timeout sec
```

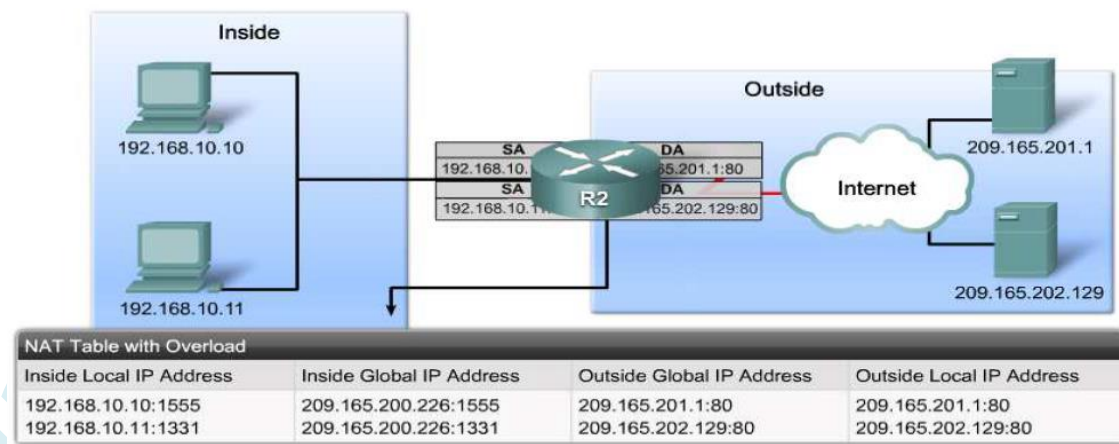
```
Router(config)# ip nat translation timeout 120
```

Lecture-24.134 Port Address Translation (PAT)

Reading Tab:

- PAT (also called NAT overload) allows the router to use one inside global address for many inside local addresses.
 - With address overloading, many privately addressed nodes can access the Internet using a single global address.
- There are two ways to configure PAT:
 - ISP allocates a single public IPv4 address
 - ISP allocates more than one public IPv4 address
- Note:
 - Over 65,000 inside addresses can theoretically map to a single outside address.
 - However, 4000 local addresses per global address is more realistic.
 - Each NAT translation consumes about 160 bytes of router DRAM.

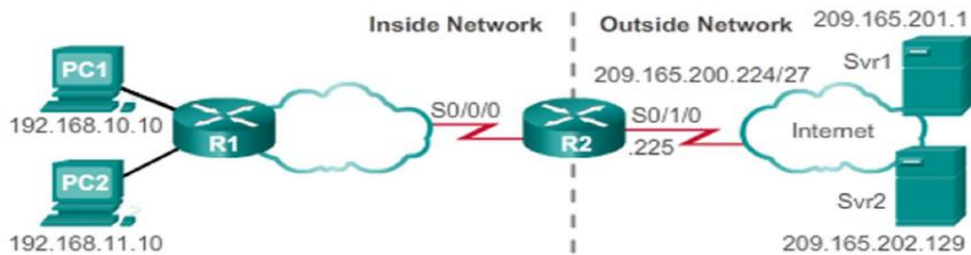
NAT Overload (PAT)



The NAT router keeps track of the different conversations by mapping TCP and UDP port numbers in the translation table.

- Called an extended table entry.

Configuring PAT Using a Pool Example

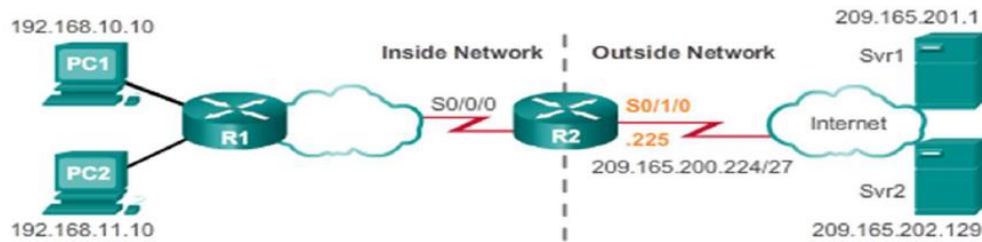


```
R2(config)# ip nat pool NAT-POOL2 209.165.200.226 209.165.200.240
prefix-length 27
R2(config)# access-list 1 permit 192.168.0.0 0.0.255.255
R2(config)#
R2(config)# ip nat inside source list 1 pool NAT-POOL2 overload
R2(config)# interface Serial0/0/0
R2(config-if)# ip nat inside
R2(config-if)# exit
R2(config)# interface Serial0/1/0
R2(config-if)# ip nat outside
```

Verifying PAT Using a Pool Example

```
R2# show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
tcp 209.165.200.226:51839 192.168.10.10:51839 209.165.201.1:80 209.165.201.1:80
tcp 209.165.200.226:42558 192.168.11.10:42558 209.165.202.129:80 209.165.202.129:80
R2#
```

Configure PAT Using an Address Example



```

R2(config)# access-list 1 permit 192.168.0.0 0.0.255.255
R2(config)#
R2(config)# ip nat source list 1 interface serial 0/1/0 overload
R2(config)#
R2(config)# interface Serial0/0/0
R2(config-if)# ip nat inside
R2(config-if)# exit
R2(config)# interface Serial0/1/0
R2(config-if)# ip nat outside

```

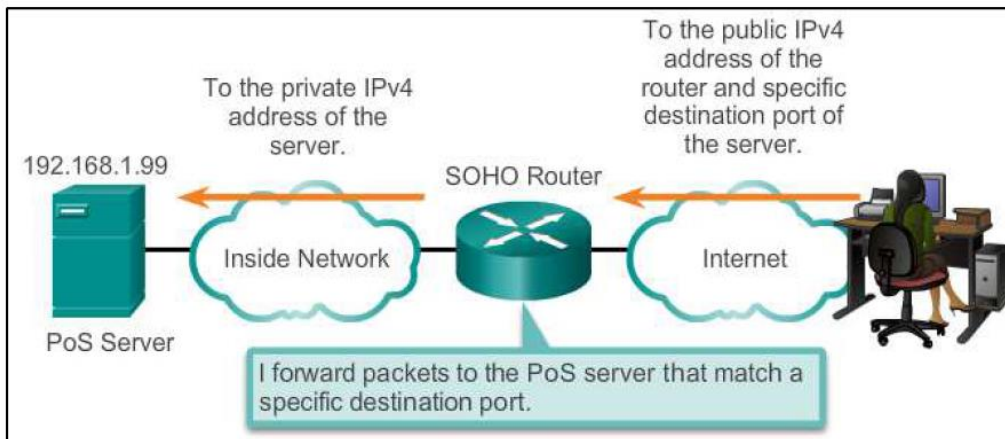
Week no 13

135. TOPIC

Lecture-25.135 Port Forwarding

Reading Tab:

- Port forwarding (sometimes referred to as tunneling) is the act of forwarding traffic addressed to a specific network port from one network node to another.
 - Helpful in situations where servers have private addresses, not reachable from the outside networks.
 - Port forwarding can be enabled for applications by specifying the inside local address that requests should be forwarded to.



Configuring Port Forwarding with IOS

- In IOS, Port forwarding is essentially a static NAT translation with a specified TCP or UDP port number.
- Configured using the **ip nat inside source {static {tcp | udp local-ip local-port global-ip global-port} [extendable]}** global configuration command

Parameter	Description
tcp or udp	Indicates if this is a TCP or UDP port number.
<i>local-ip</i>	This is the IPv4 address assigned to the inside host (typically a private address).
<i>local-port</i>	- Sets the local TCP/UDP port in a range from 1-65535. - This is the port number the server is listening on.
<i>global-ip</i>	- Sets the global TCP/UDP port in a range from 1-65535. - This is the port number the outside client will use to reach the internal server.
extendable	The option is applied by default and allows the router to extend the translation to more than one port if necessary.

136. TOPIC

137. TOPIC

Lecture-25.137 Packet Tracer - Configuring NAT Port Overload Translation (PAT) , Part-2

138. TOPIC

Lecture-25.138 Packet Tracer - Configuring NAT Port Overload Translation (PAT) , Part-3

139. TOPIC

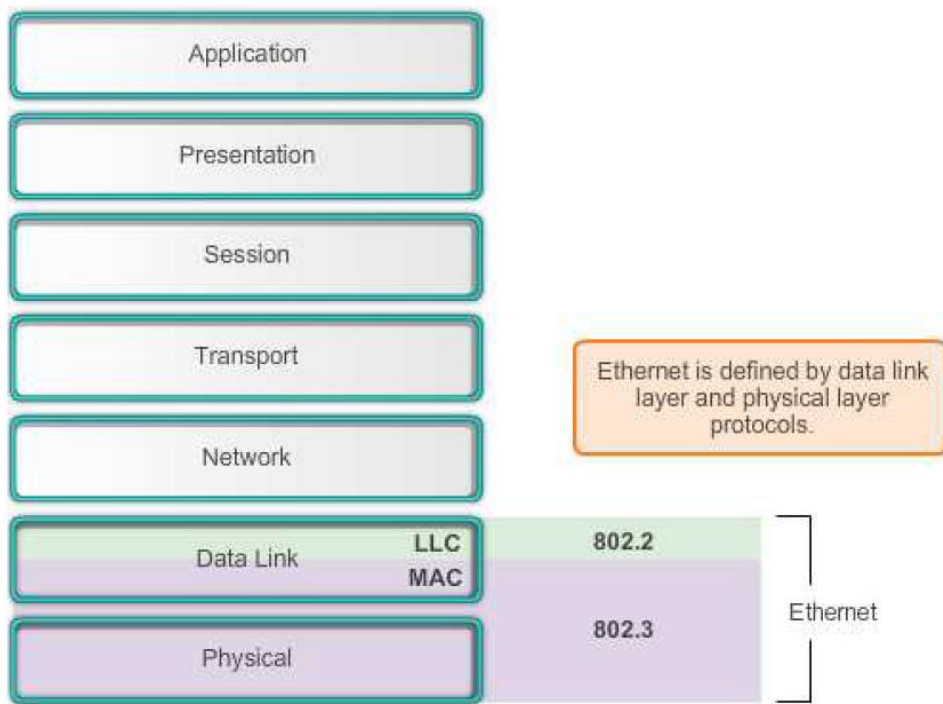
Lecture-25.139 Ethernet Protocols

Reading Tab:

Ethernet is the most widely used physical layer LAN technology used today. The standard Ethernet based LANs transfer data at speed up to 10 Mbps. Ethernet operates in the data link layer and the physical layer. It is a family of networking technologies that are defined in the IEEE 802.2 and 802.3 standards. Ethernet supports data bandwidths of:

- 10 Mb/s
- 100 Mb/s
- 1000 Mb/s (1 Gb/s)
- 10,000 Mb/s (10 Gb/s)
- 40,000 Mb/s (40 Gb/s)
- 100,000 Mb/s (100 Gb/s)

As shown in Figure below, Ethernet standards define both the Layer 2 protocols and the Layer 1 technologies. For the Layer 2 protocols, as with all 802 IEEE standards, Ethernet relies on the two separate sub layers of the data link layer to operate, the Logical Link Control and the MAC sub layers.



LLC is the abbreviation of Logic Link Control Layer and MAC is the abbreviation of Medium Access Control layer. These layers are the sub layers of the Data link layer in the OSI reference model.

LLC sublayer

The Ethernet LLC sub layer handles the communication between the upper layers and the lower layers. The LLC sub layer is typically between the networking software and the device hardware. The LLC sub layer takes the network protocol data, which is typically an IPv4 packet, and adds control information to help deliver the packet to the destination node. The LLC is used to communicate with the upper layers of the application, and transition the packet to the lower layers for delivery. LLC is implemented in software, and its implementation is independent of the hardware. In a computer, the LLC can be considered the driver software for the NIC. The NIC driver is a program that interacts directly with the hardware on the NIC to pass the data between the MAC sub layer and the physical media.

MAC sub layer

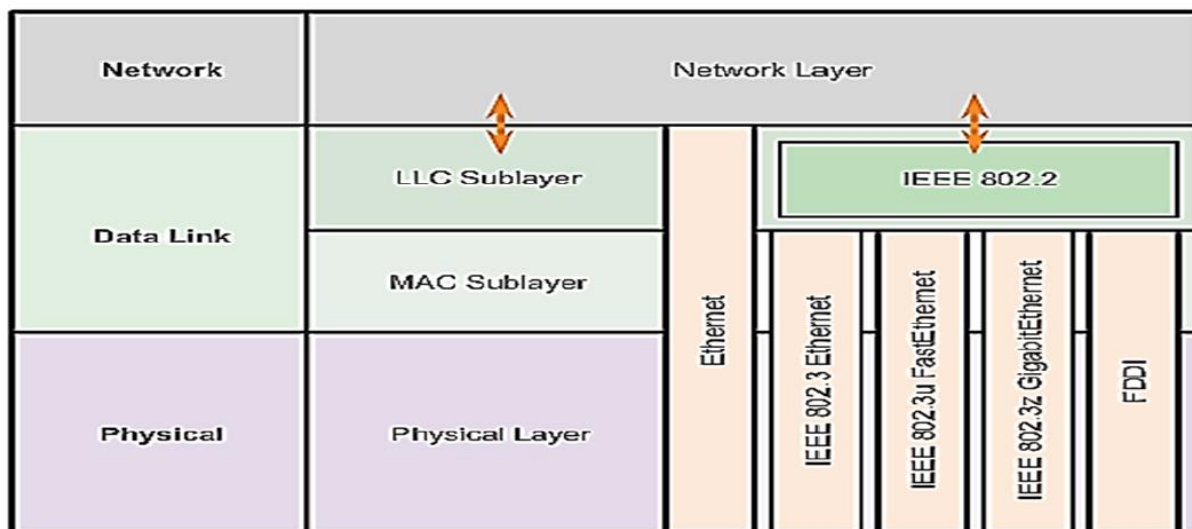
MAC constitutes the lower sub layer of the data link layer. MAC is implemented by hardware, typically in the computer NIC. The specifics are specified in the IEEE 802.3 standards. Figure below lists common IEEE Ethernet standards.

Ethernet MAC sub layer has two primary responsibilities:

- Data encapsulation
- Media access control

Data encapsulation

The data encapsulation process includes frame assembly before transmission, and frame disassembly upon reception of a frame. In forming the frame, the MAC layer adds a header and trailer to the network layer PDU.



Data encapsulation provides three primary functions:

Frame delimiting: The framing process provides important delimiters that are used to identify a group of bits that make up a frame. This process provides synchronization between the transmitting and receiving nodes.

Addressing: The encapsulation process also provides for data link layer addressing. Each Ethernet header added in the frame contains the physical address (MAC address) that enables a frame to be delivered to a destination node.

Error detection: Each Ethernet frame contains a trailer with a cyclic redundancy check (CRC) of the frame contents. After reception of a frame, the receiving node creates a CRC to compare to the one in the frame. If these two CRC calculations match, the frame can be trusted to have been received without error.

The use of frames aids in the transmission of bits as they are placed on the media and in the grouping of bits at the receiving node.

Media Access Control

The second responsibility of the MAC sub layer is media access control. Media access control is responsible for the placement of frames on the media and the removal of frames from the media. As its name implies, it controls access to the media. This sub layer communicates directly with the physical layer.

The underlying logical topology of Ethernet is a multi-access bus; therefore, all nodes (devices) on a single network segment share the medium. Ethernet is a contention-based method of networking. Recall that a contention-based method, or non-deterministic method, means that any device can try to transmit data across the shared medium whenever it has data to send. However, much like if two people try to talk simultaneously, if multiple devices on a single medium attempt to forward data simultaneously, the data will collide resulting in corrupted, unusable data. For this reason, Ethernet provides a method for controlling how the nodes share access through the use of a Carrier Sense Multiple Access (CSMA) technology.

Media Access Control

The CSMA process is used to first detect if the media is carrying a signal. If a carrier signal on the media from another node is detected, it means that another device is transmitting. When the device attempting to transmit sees that the media is busy, it will wait and try again after a short time period. If no carrier signal is detected, the device transmits its data. It is possible that the CSMA process will fail and two devices will transmit at the same time. This is called a data collision. If this occurs, the data sent by both devices will be corrupted and will need to be resent.

Contention-based media access control methods do not require mechanisms for tracking whose turn it is to access the media; therefore, they do not have the overhead of controlled access methods. However, the contention-based systems do not scale well under heavy media use. As use and the number of nodes increases, the probability of successful media access without a collision decreases. Additionally, the recovery mechanisms required to correct errors due to these collisions further diminishes the throughput.

CSMA is usually implemented in conjunction with a method for resolving media contention. The two commonly used methods are:

CSMA/Collision Detection

In CSMA/Collision Detection (CSMA/CD), the device monitors the media for the presence of a data signal. If a data signal is absent, indicating that the media is free, the device transmits the data. If signals are then detected that show another device was transmitting at the same time, all devices stop sending and try again later. Traditional forms of Ethernet were developed to use this method.

The widespread incorporation of switched technologies in modern networks has largely displaced the original need for CSMA/CD in local-area networks. Almost all wired connections between devices in a LAN today are full-duplex connections - a device is able to send and receive simultaneously. This means, that while Ethernet networks are designed with CSMA/CD technology, with today's intermediate devices, collisions do not occur and the processes utilized by CSMA/CD are really unnecessary.

However, wireless connections in a LAN environment still have to take collisions into account. Wireless LAN devices utilize the CSMA/Collision Avoidance (CSMA/CA) media access method.

CSMA/Collision Avoidance

In CSMA/CA, the device examines the media for the presence of a data signal. If the media is free, the device sends a notification across the media of its intent to use it. The device then sends the data. This method is used by 802.11 wireless networking technologies.

MAC Address: Ethernet Identity

The underlying logical topology of Ethernet is a multi-access bus. Every network device is connected to the same, shared media, and all the nodes receive all frames transmitted. The issue is if all devices are receiving every frame, how can each individual device identify if it is the intended receiver without the overhead of having to process and de-encapsulate the frame to get to the IP address? The issue becomes even more problematic in large, high traffic volume networks where lots of frames are forwarded.

To prevent the excessive overhead involved in the processing of every frame, a unique identifier called a MAC address was created to identify the actual source and destination nodes within an Ethernet network. Regardless of which variety of Ethernet is used, MAC addressing provided a method for device identification at the lower level

of the OSI model. As you may recall, MAC addressing is added as part of a Layer 2 PDU. An Ethernet MAC address is a 48-bit binary value expressed as 12 hexadecimal digits (4 bits per hexadecimal digit).

MAC Address Structure

MAC addresses must be globally unique. The MAC address value is a direct result of IEEE-enforced rules for vendors to ensure globally unique addresses for each Ethernet device. The rules established by IEEE require any vendor that sells Ethernet devices to register with IEEE. The IEEE assigns the vendor a 3-byte (24-bit) code, called the Organizationally Unique Identifier (OUI).

IEEE requires a vendor to follow two simple rules:

- All MAC addresses assigned to a NIC or other Ethernet device must use that vendor's assigned OUI as the first 3 bytes.
- All MAC addresses with the same OUI must be assigned a unique value (vendor code or serial number) in the last 3 bytes.

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Lecture-26.140 Ethernet MAC

Reading Tab:

MAC Address

- Layer 2 Ethernet MAC address is a 48-bit binary value expressed as 12 hexadecimal digits

OUI unique

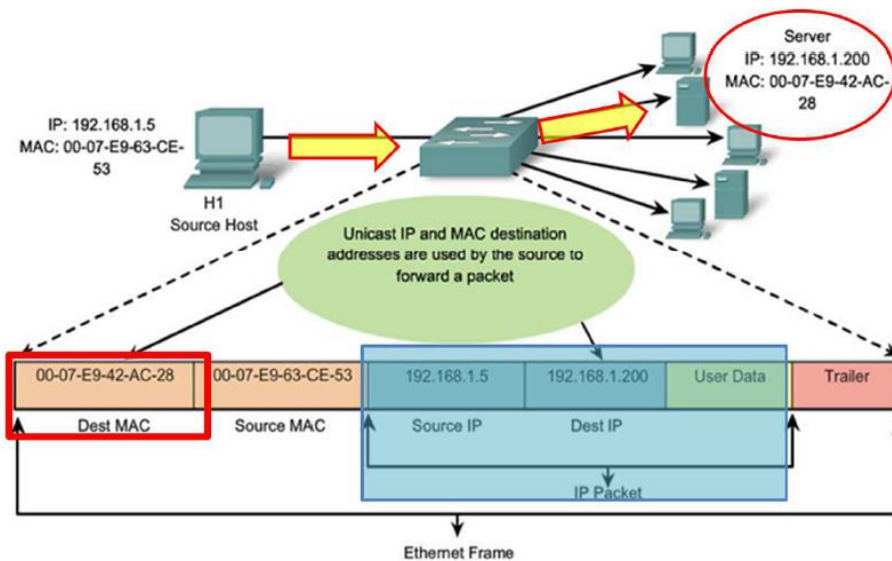
An Intel MAC address: **00-21-CC-BA-44-C4**

Unicast MAC Address

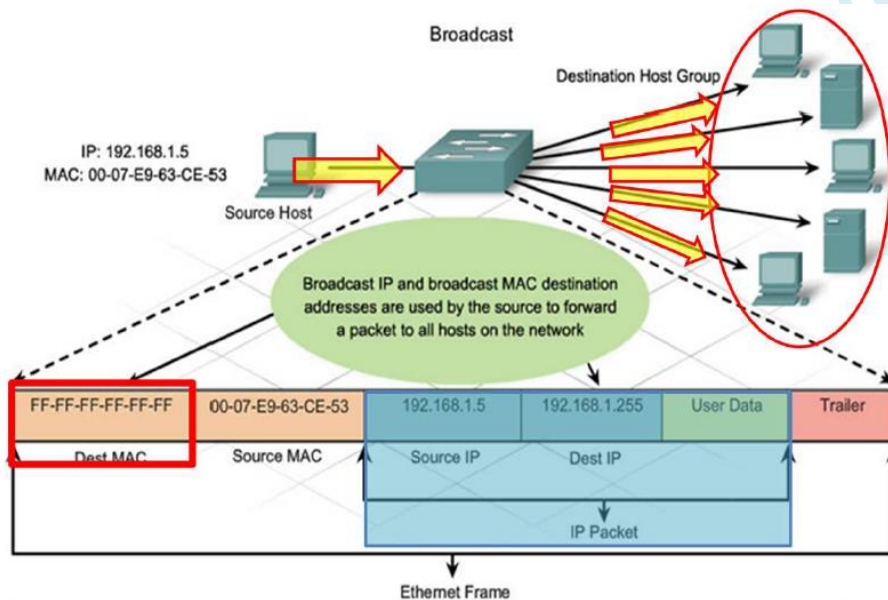
CS407

Routing and switching
Reading Material

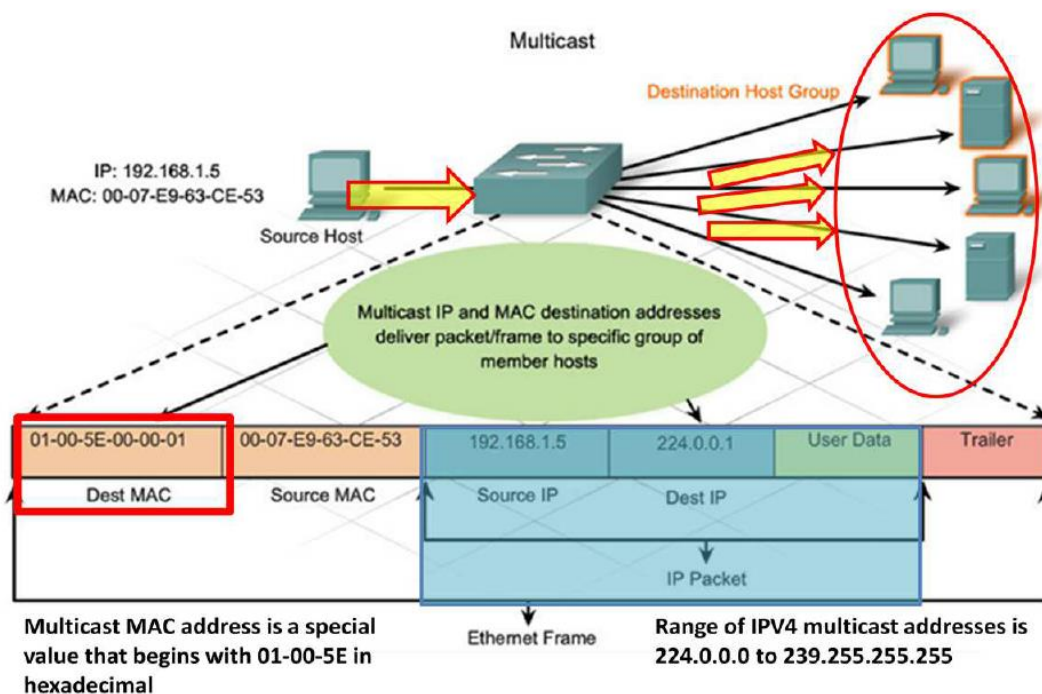
Made by Mah jabeen
[mahjabeen97869@gmail.com]



Broadcast MAC Address



Multicast MAC Address



MAC and IP

There are two primary addresses assigned to a host device:

- Physical address (the MAC address)
- Logical address (the IP address)

Both the MAC address and IP address work together to identify a device on the network. The process of using the MAC address and the IP address to find a computer is similar to the process of using a name and address of an individual to send a letter.

A person's name usually does not change. A person's address on the other hand, relates to where they live and can change.

Similar to the name of a person, the MAC address on a host does not change; it is physically assigned to the host NIC and is known as the physical address. The physical address remains the same regardless of where the host is placed.

The IP address is similar to the address of a person. This address is based on where the host is actually located. Using this address, it is possible for a frame to determine the location of where a frame should be sent. The IP address, or network address, is known as a logical address because it is assigned logically.

It is assigned to each host by a network administrator based on the local network that the host is connected to. Both the physical MAC and logical IP addresses are required for a computer to communicate on a hierarchical network, just like both the name and address of a person are required to send a letter.

End-to-End Connectivity: MAC and IP

A source device will send a packet based on an IP address. One of the most common ways a source device determines the IP address of a destination device is through Domain Name Service (DNS), in which an IP address is associated to a domain name. For example, www.cisco.com is equal to 209.165.200.225. This IP address will get the packet to the network location of the destination device. It is this IP address that routers will use to determine the best path to reach a destination. So, in short, IP addressing determines the end-to-end behavior of an IP packet.

However, along each link in a path, an IP packet is encapsulated in a frame specific to the particular data link technology associated with that link, such as Ethernet. End devices on an Ethernet network do not accept and process frames based on IP addresses, rather, a frame is accepted and processed based on MAC addresses.

On Ethernet networks, MAC addresses are used to identify, at a lower level, the source and destination hosts. When a host on an Ethernet network communicates, it sends frames containing its own MAC address as the source and the MAC address of the intended recipient as the destination. All hosts that receive the frame will read the destination MAC address. If the destination MAC address matches the MAC address configured on the host NIC, only then will the host process the message.

How are the IP addresses of the IP packets in a data flow associated with the MAC addresses on each link along the path to the destination? This is done through a process called Address Resolution Protocol (ARP).

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Lecture-26.141 Address Resolution Protocol (ARP)

Reading Tab:

Recall that each node on an IP network has both a MAC address and an IP address. In order to send data, the node must use both of these addresses. The node must use its own MAC and IP addresses in the source fields and must provide both a MAC address and an IP address for the destination. While the IP address of the destination will be provided by a higher OSI layer, the sending node needs a way to find the MAC address of the destination for a given Ethernet link. This is the purpose of ARP. ARP relies on certain types of Ethernet broadcast messages and Ethernet unicast messages, called ARP requests and ARP replies.

The ARP protocol provides two basic functions:

- Resolving IPv4 addresses to MAC addresses
- Maintaining a table of mappings

ARP Table –

- Used to find data link layer address that is mapped to destination IPv4 address
- As a node receives frames from the media, it records the source IP and MAC address as a

mapping in the ARP table

ARP request –

- Layer 2 broadcast to all devices on the Ethernet LAN
- The node that matches the IP address in the broadcast will reply
- If no device responds, the packet is dropped because a frame cannot be created

ARP Process – Communicating Locally

Creating the Frame

When ARP receives a request to map an IPv4 address to a MAC address, it looks for the cached map in its ARP table. If an entry is not found, the encapsulation of the IPv4 packet fails and the Layer 2 processes notify ARP that it needs a map. The ARP processes then send out an ARP request packet to discover the MAC address of the destination device on the local network. If a device receiving the request has the destination IP address, it responds with an ARP reply. A map is created in the ARP table. Packets for that IPv4 address can now be encapsulated in frames. If no device responds to the ARP request, the packet is dropped because a frame cannot be created. This encapsulation failure is reported to the upper layers of the device. If the device is an intermediary device, like a router, the upper layers may choose to respond to the source host with an error in an ICMPv4 packet

ARP Process – Communicating Remotely

All frames must be delivered to a node on the local network segment. If the destination IPv4 host is on the local network, the frame will use the MAC address of this device as the destination MAC address. If the destination IPv4 host is not on the local network, the source node needs to deliver the frame to the router interface that is the gateway or next hop used to reach that destination. The source node will use the MAC address of the gateway as the destination address for frames containing an IPv4 packet addressed to hosts on other networks.

The gateway address of the router interface is stored in the IPv4 configuration of the hosts. When a host creates a packet for a destination, it compares the destination IP address and its own IP address to determine if the two IP addresses are located on the same Layer 3 network. If the receiving host is not on the same network, the source uses the ARP process to determine a MAC address for the router interface serving as the gateway.

In the event that the gateway entry is not in the table, the normal ARP process will send an ARP request to retrieve the MAC address associated with the IP address of the router interface.

Removing Entries from ARP Table

For each device, an ARP cache timer removes ARP entries that have not been used for a specified period of time. The times differ depending on the device and its operating system. For example, some Windows operating systems store ARP cache entries for 2 minutes. If the entry is used again during that time, the ARP timer for that entry is extended to 10 minutes.

Reading Tab:

The role of switched networks has evolved dramatically in the last two decades. It was not long ago that flat Layer 2 switched networks were the norm. Flat Layer 2 data networks relied on the basic properties of Ethernet and the widespread use of hub repeaters to propagate LAN traffic throughout an organization. Networks have fundamentally changed to switched LANs in a hierarchical network. A switched LAN allows more flexibility, traffic management, and additional features, such as:

- Quality of service
- Additional security
- Support for wireless networking and connectivity
- Support for new technologies, such as IP telephony and mobility services
- Layer 3 functionality

Switch Form Factor

There are various types of switches used in business networks. It is important to deploy the appropriate types of switches based on network requirements. When selecting the type of switch, the network designer must choose between a fixed or a modular configuration, and stackable or non-stackable. Another consideration is the thickness of the switch, which is expressed in number of rack units. This is important for switches that are mounted in a rack.

There following are the major types of switches:

- **Fixed Configuration Switches**
- **Modular Configuration Switches**
- **Stackable Configuration Switches**

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Reading Tab:

Port Density

The port density of a switch refers to the number of ports available on a single switch.

Fixed configuration switches typically support up to 48 ports on a single device. They have options for up to four additional ports for small form-factor pluggable (SFP) devices. High-port densities allow for better use of limited space and power. If there are two switches that each contain 24 ports, they would be able to support up to 46 devices, because at least one port per switch is lost with the connection of each switch to the rest of the network. In addition, two power outlets are required. Alternatively, if there is a single 48-port switch, 47 devices can be supported, with only one port used to connect the switch to the rest of the network, and only one power outlet needed to accommodate the single switch. Modular switches can support very high-port densities through the addition of multiple switch port line cards. For example, some Catalyst 6500 switches can support in excess of 1,000 switch ports.

Forwarding Rates

Forwarding rates define the processing capabilities of a switch by rating how much data the switch can process per second. Switch product lines are classified by forwarding rates. Entry-level switches have lower forwarding rates than enterprise-level switches. Forwarding rates are important to consider when selecting a switch. If the switch forwarding rate is too low, it cannot accommodate full wire-speed communication across all of its switch ports. Wire speed is the data rate that each Ethernet port on the switch is capable of attaining. Data rates can be 100 Mb/s, 1 Gb/s, 10 Gb/s, or 100 Gb/s.

Power over Ethernet (PoE)

PoE allows the switch to deliver power to a device over the existing Ethernet cabling. This feature can be used by IP phones and some wireless access points. PoE allows more flexibility when installing wireless access points and IP phones, allowing them to be installed anywhere that there is an Ethernet cable. A network administrator should ensure that the PoE features are required, because switches that support PoE are expensive. The relatively new Cisco Catalyst 2960-C and 3560-C Series compact switches support PoE pass-through.

PoE pass-through allows a network administrator to power PoE devices connected to the switch, as well as the switch itself, by drawing power from certain upstream switches.

144. TOPIC

Lecture-26.144 Frame Forwarding

Reading Tab:

The concept of switching and forwarding frames is universal in networking and telecommunications. Various types of switches are used in LANs, WANs, and the public switched telephone network (PSTN). The fundamental concept of switching refers to a device making a decision based on two criteria:

- Ingress port
- Destination address

The decision on how a switch forwards traffic is made in relation to the flow of that traffic. The term ingress is used to describe where a frame enters the device on a port. The term egress is used to describe frames leaving the device from a particular port. When a switch makes a decision, it is based on the ingress port and the destination address of the message.

A LAN switch maintains a table that it uses to determine how to forward traffic through the switch. The only intelligence of the LAN switch is its ability to use its table to forward traffic based on the ingress port and the destination address of a message. With a LAN switch, there is only one master switching table that describes a strict association between addresses and ports; therefore, a message with a given destination address always exits the same egress port, regardless of the ingress port it enters. Cisco LAN switches forward Ethernet frames based on the destination MAC address of the frames.

Dynamically Populating a MAC Address Table

Switches use MAC addresses to direct network communications through the switch to the appropriate port toward the destination. A switch is made up of integrated circuits and the accompanying software that controls the data paths through the switch. For a switch to know which port to use to transmit a frame, it must first learn which devices exist on each port. As the switch learns the relationship of ports to devices, it builds a table called a MAC address, or content addressable memory (CAM) table. CAM is a special type of memory used in high-speed searching applications.

Switch Forwarding Methods

As networks grew and enterprises began to experience slower network performance, Ethernet bridges (an early version of a switch) were added to networks to limit the size of the collision domains. In the 1990s, advancements in integrated circuit technologies allowed for LAN switches to replace Ethernet bridges. These LAN switches were able to move the Layer 2 forwarding decisions from software to application-specific-integrated circuits (ASICs). ASICs reduce the packet-handling time within the device, and allow the device to handle an increased number of ports without degrading performance. This method of forwarding data frames at Layer 2 was referred to as store-and-forward switching. This term distinguished it from cut-through switching.

145. TOPIC

Lecture-26.145 Switching Domains

[Reading Tab:](#)

Collision Domains

In hub-based Ethernet segments, network devices compete for the medium, because devices must take turns when transmitting. The network segments that share the same bandwidth between devices are known as collision domains, because when two or more devices within that segment try to communicate at the same time, collisions may occur.

It is possible, however, to use a switch device, operating the OSI data link layer, to divide a network into segments and reduce the number of devices that compete for bandwidth. When a switch is used, each port represents a new segment. Each new segment is a new collision domain. More bandwidth is available to the devices on the segment, and collisions in one collision domain do not interfere with the other segments. This is also known as micro segmentation.

Broadcast Domains

Although switches filter most frames based on MAC addresses, they do not filter broadcast frames. For other devices on the LAN to receive broadcast frames, switches must flood these frames out all ports except the one on which the broadcast was received. A collection of interconnected switches forms a single broadcast domain. Only a network layer device, such as a router, can divide a Layer 2 broadcast domain. Routers are used to segment both collision and broadcast domains.

When a device sends a Layer 2 broadcast, the destination MAC address in the frame is set to all binary ones. A frame with a destination MAC address of all binary ones is received by all devices in the broadcast domain.

The Layer 2 broadcast domain is referred to as the MAC broadcast domain. The MAC broadcast domain consists of all devices on the LAN that receive broadcast frames from a host. When a switch receives a broadcast frame, it forwards the frame out each of its ports, except the ingress port where the broadcast frame was received. Each device connected to the switch receives a copy of the broadcast frame and processes it. Broadcasts are sometimes necessary for initially locating other devices and network services, but they also reduce network efficiency. Network bandwidth is used to propagate the broadcast traffic. Too many broadcasts and a heavy traffic load on a network can result in congestion: a slow-down in the network performance

Week no 14

146.TOPIC

Lecture-27.146 Basic Switch Configuration

Reading Tab:

Bootup Process

After a Cisco switch is powered on, it goes through the following boot sequence:

1. First, the switch loads a power-on self-test (POST) program stored in ROM. POST checks the CPU subsystem. It tests the CPU, DRAM, and the portion of the flash device that makes up the flash file system.
2. Next, the switch loads the boot loader software. The boot loader is a small program stored in ROM and is run immediately after POST successfully completes.
3. The boot loader performs low-level CPU initialization. It initializes the CPU registers, which control where physical memory is mapped, the quantity of memory, and its speed.
4. The boot loader initializes the flash file system on the system board.
5. Finally, the boot loader locates and loads a default IOS operating system software image into memory and hands control of the switch over to the IOS.

The boot loader finds the Cisco IOS image and attempts to automatically boot by using information in the BOOT environment variable. If this variable is not set, the switch attempts to load and execute the first executable file it can by performing a recursive, depth-first search throughout the flash file system.

In a depth-first search of the file system, the search begins at the first top-level directory. The search proceeds through the directory from the lowest level subdirectory, up the tree. If the search is unsuccessful, the next top-level directory is located and the bottom up search pattern is repeated. On Catalyst 2960 Series switches, the image file is normally contained in a directory that has the same name as the image file (excluding the .bin file extension). The IOS operating system then initializes the interfaces using the Cisco IOS commands found in the configuration file, startup-config, which is stored in NVRAM.

The BOOT environment variable is set using the **boot system** global configuration mode command. Notice that the IOS is located in a distinct folder and the folder path is specified. Use the **show bootvar** command (**show boot** in older IOS versions) to see what the current IOS boot file is set to.

147. TOPIC

Lecture-27.147 Configure Switch Ports

Reading Tab:

Duplex Communication

Full-duplex communication improves the performance of a switched LAN. Full-duplex communication increases effective bandwidth by allowing both ends of a connection to transmit and receive data simultaneously. This is also known as bidirectional. This method of optimizing network performance requires micro-segmentation. A micro-segmented LAN is created when a switch port has only one device connected and is operating at full-duplex. This results in a micro size collision domain of a single device. However, because there is only one device connected, a micro-segmented LAN is collision free.

Duplex and Speed

Switch ports can be manually configured with specific duplex and speed settings. Use the **duplex** interface configuration mode command to manually specify the duplex mode for a switch port. Use the **speed** interface configuration mode command to manually specify the speed for a switch port. The default setting for both duplex and speed for switch ports on Cisco Catalyst 2960 and 3560 switches is auto. The 10/100/1000 ports operate in either half- or full-duplex mode when they are set to 10 or 100 Mb/s, but when they are set to 1000 Mb/s (1 Gb/s), they operate only in full-duplex mode. Auto negotiation is useful when the speed and duplex settings of the device connecting to the port are unknown or may change. When connecting to known devices, such as servers, dedicated workstations, or network devices, best practice is to manually set the speed and duplex settings.

When troubleshooting switch port issues, the duplex and speed settings should be checked.

All fiber optic ports, such as 100BASE-FX ports, operate only at one preset speed and are always full duplex.

148. TOPIC

Creation of topology in packet tracer:

- select switch and an end device
- connect them with a straight through cable as shown in the snapshot below:

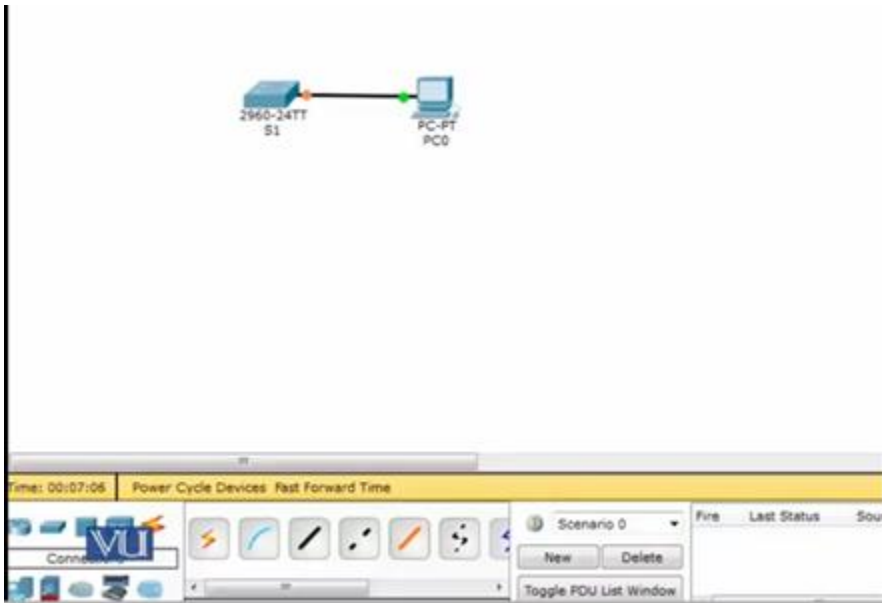
CS407

Routing and switching

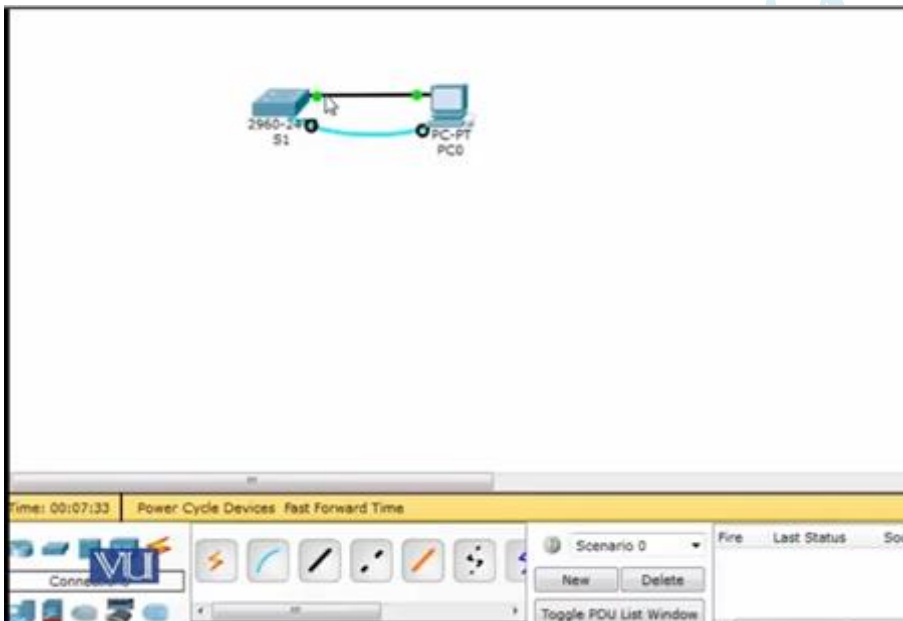
Reading Material

Made by Mah jabeen

[mahjabeen97869@gmail.com]



Connect console cable:



For configuration, go to CLI of that switch:

```
IOS Command Line Interface

MotherBoard serial number : FOC1032489J
Power supply serial number : DCA102133JA
Model revision number : B0
Motherboard revision number : C0
Model number : WS-C2960-24TT
System serial number : FOC103321EY
Top Assembly Part Number : 800-26671-02
Top Assembly Revision Number : B0
Version ID : V02
CLEI Code Number : COM3K00BBA
Hardware Board Revision Number : 0x01

Switch  Ports  Model          SW Version      SW Image
-----  -
* 1    24    WS-C2960-24TT  12.2            C2960-LANBASE-M

Cisco IOS Software, C2960 Software (C2960-LANBASE-M), Version 12.2(25)FX, RELEASE
SOFTWARE (fc1)
Copyright (c) 1986-2006 by Cisco Systems, Inc.
Compiled Wed 12-Oct-05 22:05 by pt_team

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to
up
```

Write down the following commands as shown in the screenshot:

```
IOS Command Line Interface

System serial number : FOC103321EY
Top Assembly Part Number : 800-26671-02
Top Assembly Revision Number : B0
Version ID : V02
CLEI Code Number : COM3K00BBA
Hardware Board Revision Number : 0x01

Switch  Ports  Model          SW Version      SW Image
-----  -
* 1    24    WS-C2960-24TT  12.2            C2960-LANBASE-M

Cisco IOS Software, C2960 Software (C2960-LANBASE-M), Version 12.2(25)FX, RELEASE
SOFTWARE (fc1)
Copyright (c) 1986-2006 by Cisco Systems, Inc.
Compiled Wed 12-Oct-05 22:05 by pt_team

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to
up

Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
```

then write show interface VLAN1

```
IOS Command Line Interface

Switch(config)#show interface vianl
^
% Invalid input detected at '^' marker.

Switch(config)#exit
Switch#
*SYS-5-CONFIG_I: Configured from console by console

Switch#show interface vianl
Vianl is administratively down, line protocol is down
Hardware is CPU Interface, address is 00d0.bcb3.92c6 (bia 00d0.bcb3.92c6)
MTU 1500 bytes, BW 1000000 Kbit, DLY 1000000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
ARP type: ARPA, ARP Timeout 04:00:00
Last input 21:40:21, output never, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
1682 packets input, 530955 bytes, 0 no buffer
Received 0 broadcasts (0 IP multicast)
0 runs, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
56355 packets output, 0 bytes, 0 underruns
0 output errors, 23 interface resets
0 output buffer failures, 0 output buffers swapped out
Switch#
```

it shows down as it is not configured yet.

Write config as shown in the snapshot below:

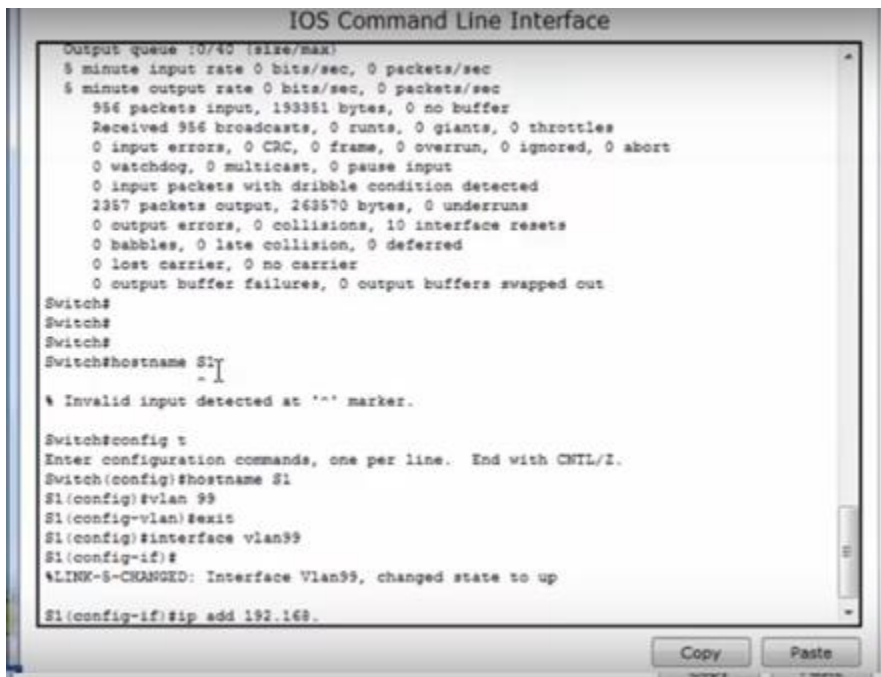
```
IOS Command Line Interface

input flow-control is off, output flow-control is off
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:08, output 00:00:06, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue :0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
956 packets input, 193351 bytes, 0 no buffer
Received 956 broadcasts, 0 runs, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
0 watchdog, 0 multicast, 0 pause input
0 input packets with dribble condition detected
2357 packets output, 263570 bytes, 0 underruns
0 output errors, 0 collisions, 10 interface resets
0 babble, 0 late collision, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out

Switch#
Switch#
Switch#
Switch#hostname S1
^
% Invalid input detected at '^' marker.

Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#
```

In global configuration mode, type the following commands:



```
IOS Command Line Interface

Output queue 10/40 (size/max)
 5 minute input rate 0 bits/sec, 0 packets/sec
 5 minute output rate 0 bits/sec, 0 packets/sec
 956 packets input, 193351 bytes, 0 no buffer
Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
 0 watchdog, 0 multicast, 0 pause input
 0 input packets with dribble condition detected
2357 packets output, 263570 bytes, 0 underruns
 0 output errors, 0 collisions, 10 interface resets
 0 babble, 0 late collision, 0 deferred
 0 lost carrier, 0 no carrier
 0 output buffer failures, 0 output buffers swapped out

Switch#
Switch#
Switch#
Switch#hostname S1
~
% Invalid input detected at '^' marker.

Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#vlan 99
S1(config-vlan)#exit
S1(config)#interface vlan99
S1(config-if)#
%LINK-5-CHANGED: Interface Vlan99, changed state to up
S1(config-if)#ip add 192.168.
```

Follow the lecture video for complete steps.

149. TOPIC

Lecture-27.149 Secure Remote Access – SSH

Reading Content:

Secure Shell (SSH) is a protocol that provides a secure (encrypted) management connection to a remote device. SSH should replace Telnet for management connections. Telnet is an older protocol that uses unsecure plaintext transmission of both the login authentication (username and password) and the data transmitted between the communicating devices. SSH provides security for remote connections by providing strong encryption when a device is authenticated (username and password) and also for the transmitted data between the communicating devices. SSH is assigned to TCP port 22. Telnet is assigned to TCP port 23.

To enable SSH on a Catalyst 2960 switch, the switch must be using a version of the IOS software including cryptographic (encrypted) features and capabilities. Use the **show version** command on the switch to see which IOS the switch is currently running.

Configuring SSH

Before configuring SSH, the switch must be minimally configured with a unique hostname and the correct network connectivity settings.

150. TOPIC

Lecture-27.150 VLAN

Reading Content:

Within a switched internetwork, VLANs provide segmentation and organizational flexibility. VLANs provide a way to group devices within a LAN. A group of devices within a VLAN communicate as if they were attached to the same wire. VLANs are based on logical connections, instead of physical connections.

VLANs allow an administrator to segment networks based on factors such as function, project team, or application, without regard for the physical location of the user or device. Devices within a VLAN act as if they are in their own independent network, even if they share a common infrastructure with other VLANs. Any switch port can belong to a VLAN, and unicast, broadcast, and multicast packets are forwarded and flooded only to end stations within the VLAN where the packets are sourced. Each VLAN is considered a separate logical network, and packets destined for stations that do not belong to the VLAN must be forwarded through a device that supports routing. A VLAN creates a logical broadcast domain that can span multiple physical LAN segments. VLANs improve network performance by separating large broadcast domains into smaller ones. If a device in one VLAN sends a broadcast Ethernet frame, all devices in the VLAN receive the frame, but devices in other VLANs do not.

VLANs enable the implementation of access and security policies according to specific groupings of users. Each switch port can be assigned to only one VLAN (with the exception of a port connected to an IP phone or to another switch).

151. TOPIC

Lecture-28.151 VLANs in Multi-Switched Environment

Reading Content:

VLAN Trunks

A trunk is a point-to-point link between two network devices that carries more than one VLAN. A VLAN trunk extends VLANs across an entire network. Cisco supports IEEE 802.1Q for coordinating trunks on Fast Ethernet, Gigabit Ethernet, and 10-Gigabit Ethernet interfaces.

VLANs would not be very useful without VLAN trunks. VLAN trunks allow all VLAN traffic to propagate between switches, so that devices which are in the same VLAN, but connected to different switches, can communicate without the intervention of a router.

A VLAN trunk does not belong to a specific VLAN; rather, it is a conduit for multiple VLANs between switches and routers. A trunk could also be used between a network device and server or other device that is equipped with an appropriate 802.1Q-capable NIC. By default, on a Cisco Catalyst switch, all VLANs are supported on a trunk port.

Network without VLANs

In normal operation, when a switch receives a broadcast frame on one of its ports, it forwards the frame out all other ports except the port where the broadcast was received.

Network with VLANs

When VLANs are implemented on a switch, the transmission of unicast, multicast, and broadcast traffic from a host in a particular VLAN are restricted to the devices that are in that VLAN.

152. TOPIC

Lecture-28.152 VLAN Implementation

Reading Content:

Normal Range VLANs

Different Cisco Catalyst switches support various numbers of VLANs. The number of supported VLANs is large enough to accommodate the needs of most organizations. For example, the Catalyst 2960 and 3560 Series switches support over 4,000 VLANs. Normal range VLANs on these switches are numbered 1 to 1,005 and extended range VLANs are numbered 1,006 to 4,094.

Normal Range VLANs

- Used in small- and medium-sized business and enterprise networks.
- Identified by a VLAN ID between 1 and 1005.
- IDs 1002 through 1005 are reserved for Token Ring and FDDI VLANs.
- IDs 1 and 1002 to 1005 are automatically created and cannot be removed.
- Configurations are stored within a VLAN database file, called vlan.dat. The vlan.dat file is located in the flash memory of the switch.

- The VLAN Trunking Protocol (VTP), which helps manage VLAN configurations between switches, can only learn and store normal range VLANs.

Extended Range VLANs

- Enable service providers to extend their infrastructure to a greater number of customers. Some global enterprises could be large enough to need extended range VLAN IDs.
- Are identified by a VLAN ID between 1006 and 4094.
- Configurations are not written to the vlan.dat file.
- Support fewer VLAN features than normal range VLANs.
- Are, by default, saved in the running configuration file.
- VTP does not learn extended range VLANs.

153. TOPIC

Lecture-28.153 VLAN Trunks

Reading Content:

Native VLANs

A VLAN trunk is an OSI Layer 2 link between two switches that carries traffic for all VLANs (unless the allowed VLAN list is restricted manually or dynamically). To enable trunk links, configure the ports on either end of the physical link with parallel sets of commands.

To configure a switch port on one end of a trunk link, use the **switchport mode trunk** command. With this command, the interface changes to permanent trunking mode. The port enters into a Dynamic Trunking Protocol (DTP) negotiation to convert the link into a trunk link even if the interface connecting to it does not agree to the change. DTP is described in the next topic. In this course, the **switchport mode trunk** command is the only method implemented for trunk configuration.

Use the Cisco IOS **switchport trunk allowed vlan** *vlan-list* command to specify the list of VLANs to be allowed on the trunk link.

154. TOPIC

Lecture-28.154 Dynamic Trunking Protocol (DTP)

Reading Content:

Ethernet trunk interfaces support different trunking modes. An interface can be set to trunking or nontrunking, or to negotiate trunking with the neighbor interface. Trunk negotiation is managed by the Dynamic Trunking Protocol (DTP), which operates on a point-to-point basis only, between network devices.

DTP is a Cisco proprietary protocol that is automatically enabled on Catalyst 2960 and Catalyst 3560 Series switches. Switches from other vendors do not support DTP. DTP manages trunk negotiation only if the port on the neighbor switch is configured in a trunk mode that supports DTP.

To enable trunking from a Cisco switch to a device that does not support DTP, use the **switchport mode trunk** and **switchport nonegotiate** interface configuration mode commands. This causes the interface to become a trunk, but not generate DTP frames.

To Clear A Switch

S1# **delete vlan.dat**

Delete filename [vlan.dat]?

Delete flash:/vlan.dat? [confirm]

S1# **erase startup-config**

Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]

[OK]

Erase of nvram: complete

%SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram

S1# **reload**

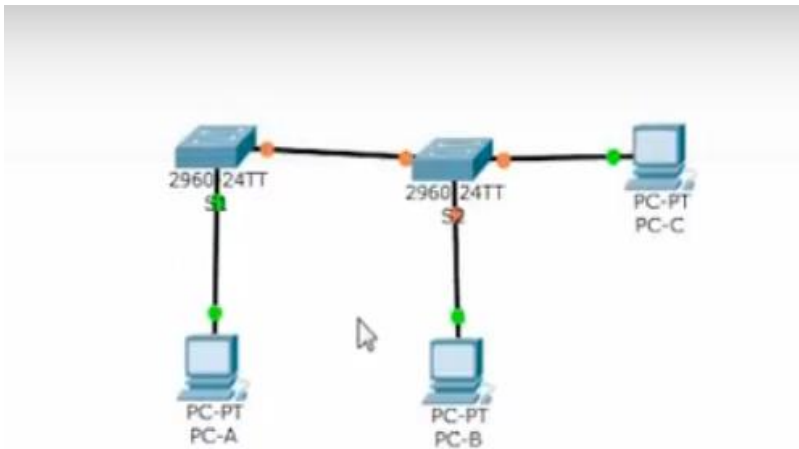
Proceed with reload? [confirm]

155. TOPIC

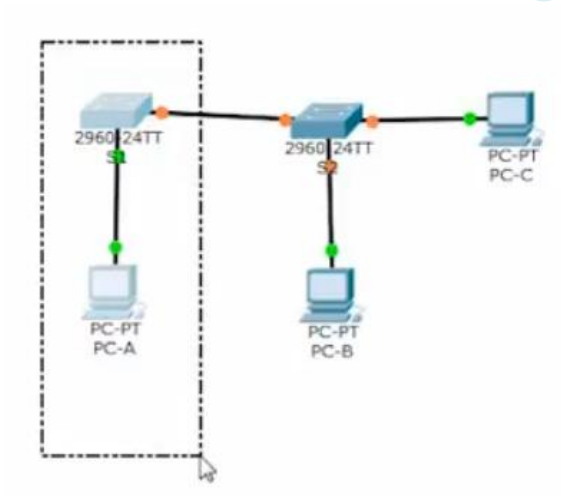
Lecture-28.155 Packet Tracer - Configuring VLANs and Trunking , Part -1

- Build the Network
- Create VLANs and Assign Switch Ports
- Configure an 802.1Q Trunk between Switches

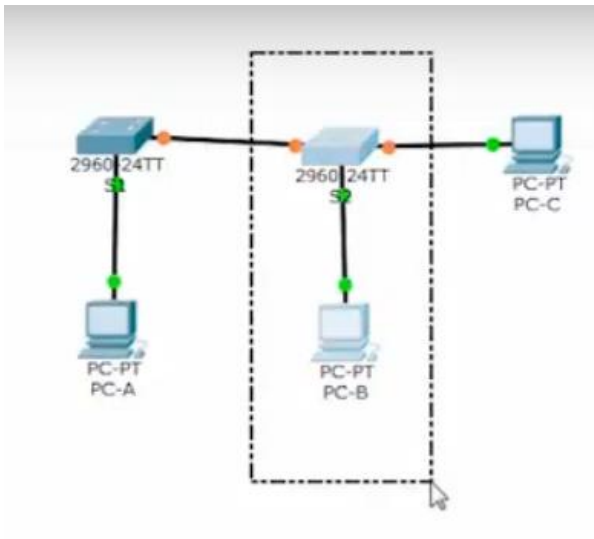
Create a topology of switches and PCs in packet tracer as shown in the screenshot:



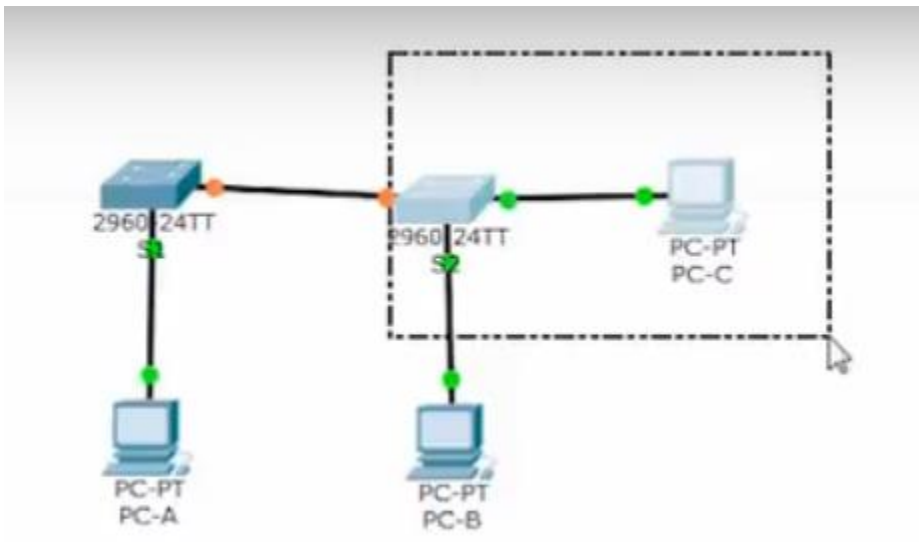
First VLAN:



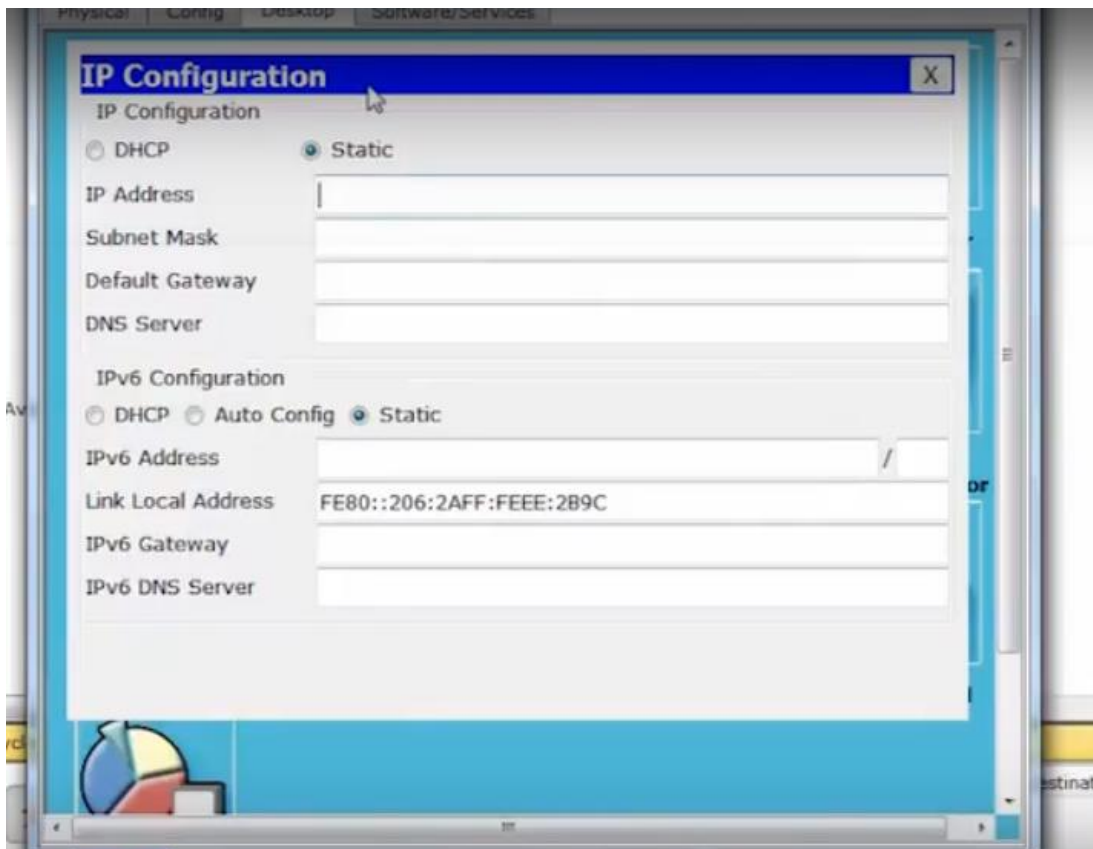
Second VLAN:



Third VLAN:



Now assigns IP addresses to all devices by this IP configuration window:



156. TOPIC

Lecture-28.156 Packet Tracer - Configuring VLANs and Trunking , Part -2

Reading Content:

This activity in this video is a continuation of previous activity in which we were configuring VLAN. And now we have to configure trunks.

Show vlan brief command usage as shown in the screenshot:

IOS Command Line Interface

```
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int f0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

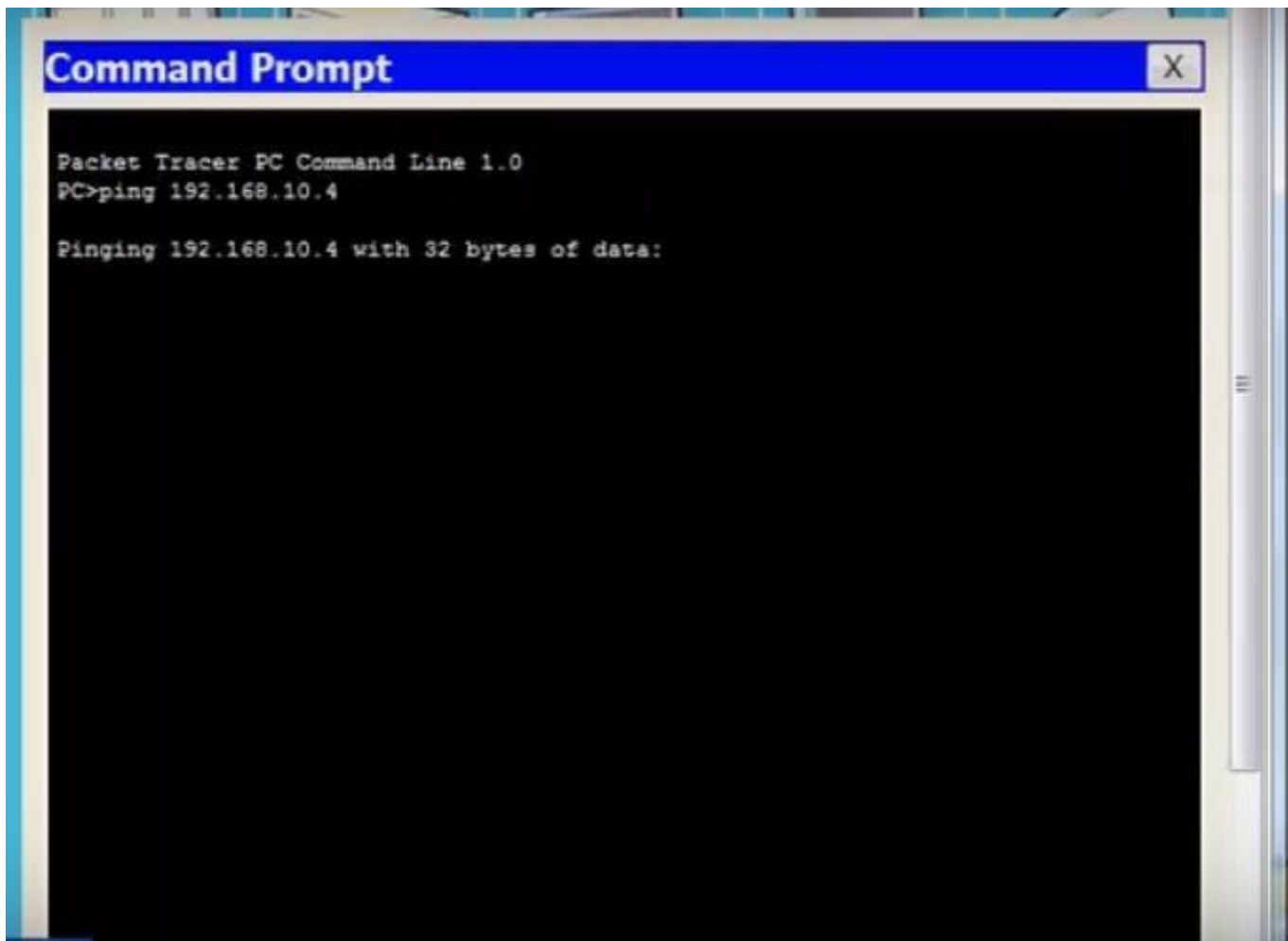
Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10 Student	active	Fa0/6
20 Faculty	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch#
```

Ping command usage but its not pinging as we have not created trunk yet.

For trunk creation, please refer to the lecture video.



Week no 15

157. TOPIC

Lecture-29.157 Purpose of Spanning Tree Protocol

Reading Content:

Redundancy in a Hierarchical Network

- Network redundancy is a key to maintaining network reliability. Multiple physical links between devices provide redundant paths. The network can then continue to operate when a single link or port has failed. Redundant links can also share the traffic load and increase capacity.

- Multiple paths must be managed so that Layer 2 loops are not created, the best paths are

chosen for use, and an alternate path is immediately available should a primary path fail. The Spanning Tree Protocols are used to manage Layer 2 redundancy.

- Redundant devices, such as multilayer switches or routers, provide the capability for a client to use an alternate default gateway should the primary default gateway fail. A client may now have multiple paths to more than one possible default gateway. First Hop Redundancy Protocols are used to manage how a client is assigned a default gateway, and to be able to use an alternate default gateway should the primary default gateway fail.

Issues with Layer 1 Redundancy–MAC DB Instability

MAC Database Instability

Ethernet frames do not have a time to live (TTL) attribute, like IP packets. As a result, if there is no mechanism enabled to block continued propagation of these frames on a switched network, they continue to propagate between switches endlessly, or until a link is disrupted and breaks the loop. This continued propagation between switches can result in MAC database instability. This can occur due to broadcast frames forwarding.

Broadcast frames are forwarded out all switch ports, except the original ingress port. This ensures that all devices in a broadcast domain are able to receive the frame. If there is more than one path for the frame to be forwarded out, an endless loop can result. When a loop occurs, it is possible for the MAC address table on a switch to constantly change with the updates from the broadcast frames, resulting in MAC database instability.

158. TOPIC

Lecture-29.158 Spanning Tree Algorithm - Introduction

[Reading Content:](#)

Redundancy increases the availability of the network topology by protecting the network from a single point of failure, such as a failed network cable or switch. When physical redundancy is introduced into a design, loops and duplicate frames occur. Loops and duplicate frames have severe consequences for a switched network. The STP was developed to address these issues.

STP ensures that there is only one logical path between all destinations on the network by intentionally blocking redundant paths that could cause a loop. A port is considered blocked when user data is prevented from entering or leaving that port. This does not include bridge protocol data unit (BPDU) frames that are used by STP to prevent loops. Blocking the

redundant paths is critical to preventing loops on the network. The physical paths still exist to provide redundancy, but these paths are disabled to prevent the loops from occurring. If the path is ever needed to compensate for a network cable or switch failure, STP recalculates the paths and unblocks the necessary ports to allow the redundant path

to become active.

159. TOPIC

Lecture-29.159 Spanning Tree Protocol - Part 2

Reading Content:

Spanning Tree Algorithm – Port Rules

The Spanning Tree Algorithm (STA) is used to determine which switch ports on a network must be put in blocking state to prevent loops from occurring. The STA designates a single switch as the root bridge and uses it as the reference point for all path calculations. In the figure below, the root bridge (switch S1) is chosen through an election process. All switches participating in spanning tree exchange BPDU frames to determine which switch has the lowest bridge ID (BID) on the network. The switch with the lowest BID automatically becomes the root bridge for the STA calculations.

A BPDU is a messaging frame exchanged by switches for STP. Each BPDU contains a BID that identifies the switch that sent the BPDU. The BID contains a priority value, the MAC address of the sending switch, and an optional extended system ID. The lowest BID value is determined by the combination of these three fields.

After the root bridge has been determined, the STA calculates the shortest path to it. Each switch uses the STA to determine which ports to block. While the STA determines the best paths to the root bridge for all switch ports in the broadcast domain, traffic is prevented from being forwarded through the network. The STA considers both path and port costs when determining which ports to block. The path costs are calculated using port cost values associated with port speeds for each switch port along a given path. The sum of the port cost values determines the overall path cost to the root bridge. If there is more than one path to choose from, STA chooses the path with the lowest path cost.

160. TOPIC

Lecture-29.160 Spanning Tree Protocol - Part 3

Reading Content:

802.1D BPDUs Frame Format

The spanning tree algorithm depends on the exchange of BPDUs to determine a root bridge. A BPDU frame contains 12 distinct fields that convey path and priority information used to determine the root bridge and paths to the root bridge.

- The first four fields identify the protocol, version, message type, and status flags.
- The next four fields are used to identify the root bridge and the cost of the path to the root bridge.

- The last four fields are all timer fields that determine how frequently BPDU messages are sent and how long the information received through the BPDU process (next topic) is retained.

The Protocol ID field indicates the type of protocol being used.

- The Version field indicates the version of the protocol.
- The Message type field indicates the type of message.
- The Flags field includes one of the following:

- o Topology change (TC) bit, which signals a topology change in the event a path to the root bridge has been disrupted.

- o Topology change acknowledgment (TCA) bit, which is set to acknowledge receipt of a configuration message with the TC bit set.

- The Root ID field indicates the root bridge by listing its 2-byte priority followed by its 6-byte MAC address ID. When a switch first boots, the root ID is the same as the bridge ID. However, as the election process occurs, the lowest bridge ID replaces the local root ID to identify the root bridge switch.

- The Cost of path field indicates the cost of the path from the bridge sending the configuration message to the root bridge.

161. TOPIC

Lecture-29.161 Spanning Tree Protocol - Part 4

Reading Content:

BPDU Propagation and Process

Each switch in the broadcast domain initially assumes that it is the root bridge for a spanning tree instance, so the BPDU frames sent contain the BID of the local switch as the root ID. By default, BPDU frames are sent every two seconds after a switch is booted; that is, the default value of the Hello timer specified in the BPDU frame is two seconds. Each switch maintains local information about its own BID, the root ID, and the path cost to the root.

When adjacent switches receive a BPDU frame, they compare the root ID from the BPDU frame with the local root ID. If the root ID in the BPDU is lower than the local root ID, the switch updates the local root ID and the ID in its BPDU messages. These messages indicate the new root bridge on the network. The distance to the root bridge is also indicated by the path cost update. For example, if the BPDU was received on a Fast Ethernet switch port, the path cost would increment by 19. If the local root ID is lower than the root ID received in the BPDU frame, the BPDU frame is discarded.

After a root ID has been updated to identify a new root bridge, all subsequent BPDU frames sent from that switch contain the new root ID and updated path cost. That way, all other adjacent switches are able to see the lowest root ID identified at all times. As the BPDU frames pass between other adjacent switches, the path cost is continually updated to indicate the total path cost to the root bridge. Each switch in the spanning tree uses its path costs to identify the best possible path to the root bridge.

162. TOPIC

Lecture-29.162 Varieties of the Spanning Tree Protocol

[Reading content:](#)

Several varieties of spanning tree protocols have emerged after the original IEEE 802.1D.

The varieties of spanning tree protocols include:

STP - This is the original IEEE 802.1D version (802.1D-1998 and earlier) that provides a loop-free topology in a network with redundant links. Common Spanning Tree (CST) assumes one spanning tree instance for the entire bridged network, regardless of the number of VLANs.

PVST+ - This is a Cisco enhancement of the original 802.1D standard that provides a separate 802.1D spanning tree instance for each VLAN configured in the network. The separate instance supports PortFast, UplinkFast, BackboneFast, BPDU guard, BPDU filter, root guard, and loop guard.

802.1D-2004 - This is an updated version of the STP standard, incorporating IEEE 802.1w.

Rapid Spanning Tree Protocol (RSTP) or IEEE 802.1w - This is an evolution of STP that provides faster convergence than STP.

Rapid PVST+ - This is a Cisco enhancement of RSTP that uses PVST+. Rapid PVST+ provides a separate instance of 802.1w per VLAN. The separate instance supports PortFast, BPDU guard, BPDU filter, root guard, and loop guard.

Multiple Spanning Tree Protocol(MSTP) - This is an IEEE standard inspired by the earlier Cisco proprietary Multiple Instance STP (MISTP) implementation. MSTP maps multiple VLANs into the same spanning tree instance. The Cisco implementation of MSTP is MST, which provides up to 16 instances of RSTP and combines many VLANs with the same physical and logical topology into a common RSTP instance. Each instance supports PortFast, BPDU guard, BPDU filter, root guard, and loop guard.

A network professional, whose duties include switch administration, may be required to decide which type of spanning tree protocol to implement.

Week no 16

163. TOPIC

Lecture-30.163 Link Aggregation

Reading Content:

In the figure below, traffic coming from several links (usually 100 or 1000 Mb/s) aggregates on the access switch and must be sent to distribution switches. Because of the traffic aggregation, links with higher bandwidth must be available between the access and distribution switches.

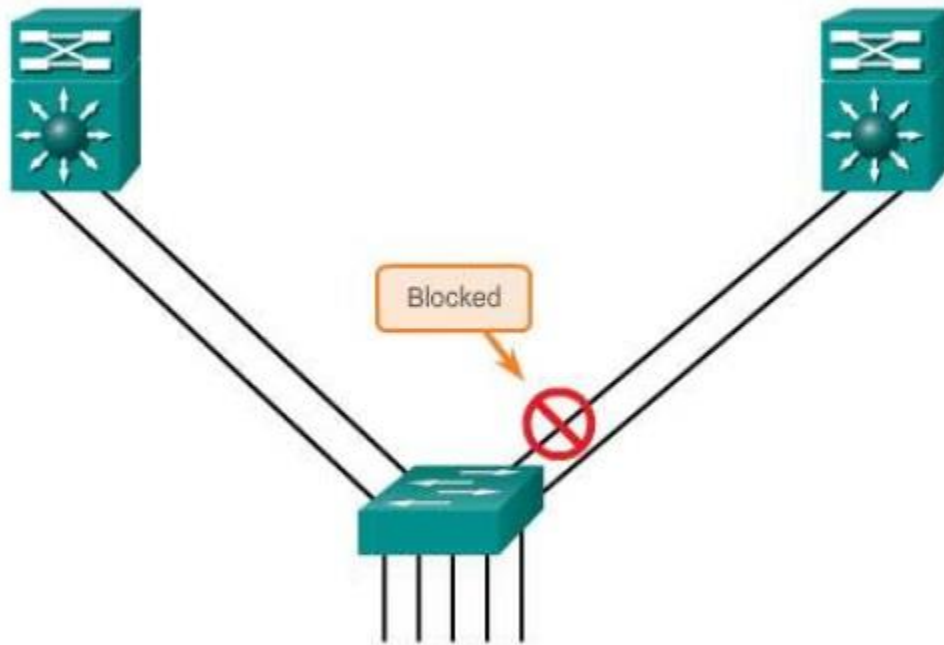
It may be possible to use faster links, such as 10 Gb/s, on the aggregated link between the access and distribution layer switches. However, adding faster links is expensive. Additionally, as the speed increases on the access links, even the fastest possible port on the aggregated link is no longer fast enough to aggregate the traffic coming from all access links.

It is also possible to multiply the number of physical links between the switches to increase the overall speed of switch-to-switch communication. However, by default, STP is enabled on switch devices. STP will block redundant links to prevent routing loops.

For these reasons, the best solution is to implement an EtherChannel configuration.

CS407
Routing and switching
Reading Material
Made by Mah jabeen
[mahjabeen97869@gmail.com]

Redundant Links with STP



By default, STP will block redundant links.

164. TOPIC

Lecture-30.164 Packet Tracer - Switched Network with Redundant Links

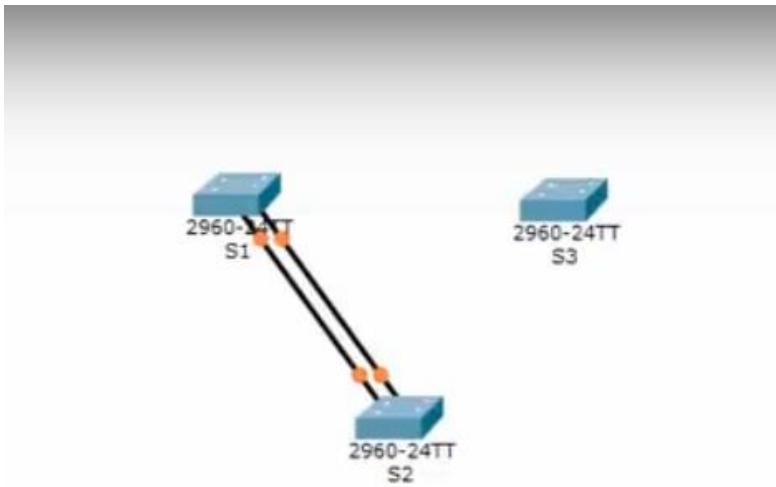
Reading Content:

In this topic, we will see how can we build a network with redundant links.

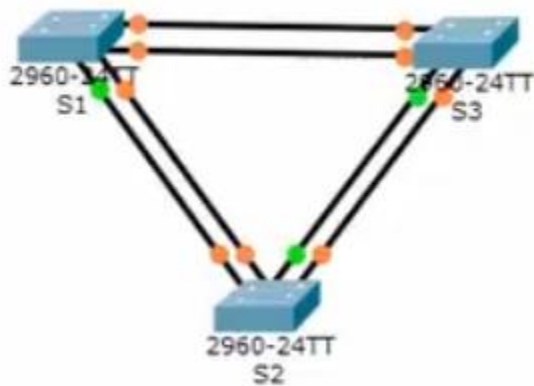
In a packet tracer create a topology and then connect them with multiple redundant links.

Then we will check the working of Spanning Tree Algorithm.

Create the topology and redundant links as shown in the figure below:



The final topology will look like this:



Now we will configure VLANs as shown in the lecture video.

Use ping command to check the connectivity:

```
Switch#ping 192.168.1.1
```

I

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
```

```
..!!!
```

```
Success rate is 60 percent (3/5), round-trip min/avg/max = 0/0/0 ms
```

The next step is to check the root bridge

for that, deactivate all ports by the command “shut down”

configure trunk in ports by following the commands as shown in the figure:

```
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface f0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#interface f0/2
Switch(config-if)#switchport mode trunk
Switch(config-if)#interface f0/3
Switch(config-if)#switchport mode trunk
Switch(config-if)#interface f0/4
Switch(config-if)#switchport mode trunk
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

now we will enable only two ports:

Spanning tree takes at least 30 secs to converge.

```
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int f0/2
Switch(config-if)#no shut

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to down
Switch(config-if)#no shut
Switch(config-if)#int f0/4
Switch(config-if)#no shut

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to down
Switch(config-if)#
```

Use the command show spanning-tree as shown in the figure below:

```
Switch#show spanning-tree
VLAN0001
Spanning tree enabled protocol ieee
Root ID    Priority    32769
           Address    0006.2A85.5D6A
           Cost        19
           Port        4(FastEthernet0/4)
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
           Address    000D.BD78.CE44
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/4	Root	FWD	19	128.4	P2p
Fa0/2	Desg	FWD	19	128.2	P2p

165. TOPIC

Lecture-30.165 Inter Vlan Routing

Reading Content:

VLANs are used to segment switched networks. Layer 2 switches, such as the Catalyst 2960 Series, can be configured by a network professional with over 4,000 VLANs. However, Layer 2 switches have very limited IPv4 and IPv6 functionality and cannot perform the routing function of routers. While Layer 2 switches are gaining more IP functionality, such as the ability to perform static routing, these switches do not support dynamic routing. With the large number of VLANs possible on these switches, static routing is insufficient.

A VLAN is a broadcast domain, so computers on separate VLANs are unable to communicate without the intervention of a routing device. Any device that supports Layer 3 routing, such as a router or a multilayer switch, can be used to perform the necessary routing functionality. Regardless of the device used, the process of forwarding network traffic from one VLAN to another VLAN using routing is known as inter-VLAN routing.

Legacy Inter-VLAN Routing

Historically, the first solution for inter-VLAN routing relied on routers with multiple physical interfaces. Each interface had to be connected to a separate network and configured with a distinct subnet. In this legacy approach, inter-VLAN routing is performed by connecting different physical router interfaces to different physical switch ports. The switch ports connected to the router are placed in access mode and each physical interface is assigned to a different VLAN. Each router interface can then accept traffic from the VLAN associated with the switch interface that it is connected to, and traffic can be routed to the other VLANs connected to the other interfaces.

Router-on-a-Stick Inter-VLAN Routing

While legacy inter-VLAN routing requires multiple physical interfaces on both the router and the switch, a more common, present-day implementation of inter-VLAN routing does not. Instead, some router software permits configuring a router interface as a trunk link, meaning only one physical interface is required on the router and the switch to route packets between multiple VLANs.

166. TOPIC

Lecture-30.166 Configuring Legacy Inter-VLAN Routing

Reading Content:

Legacy inter-VLAN routing requires routers to have multiple physical interfaces. The router accomplishes the routing by having each of its physical interfaces connected to a unique VLAN. Each interface is also configured with an IP address for the subnet associated with the particular VLAN to which it is connected. By configuring the IP addresses on the physical interfaces, network devices connected to each of the VLANs can communicate with the router using the physical interface connected to the same VLAN. In this configuration, network devices can use the router as a gateway to access the devices connected to the other VLANs.

The routing process requires the source device to determine if the destination device is local or remote to the local subnet. The source device accomplishes this by comparing the source and destination IP addresses against the subnet mask. When the destination IP address has been determined to be on a remote network, the source device must identify where it needs to forward the packet to reach the destination device.

The source device examines the local routing table to determine where it needs to send the data. Devices use their default gateway as the Layer 2 destination for all traffic that must leave the local subnet. The default gateway is the route that the device uses when it has no other explicitly defined route to the destination network. The IP address of the router interface on the local subnet acts as the default gateway for the sending device.

167. TOPIC

Lecture-30.167 Configure Router on Stick Intervlan Routing

Reading Content:

Legacy inter-VLAN routing using physical interfaces has a significant limitation. Routers have a limited number of physical interfaces to connect to different VLANs. As the number of VLANs increases on a network, having one physical router interface per VLAN quickly exhausts the physical interface capacity of a router. An alternative in larger networks is to use VLAN trunking and subinterfaces.

VLAN trunking allows a single physical router interface to route traffic for multiple VLANs. This technique is termed router-on-a-stick and uses virtual subinterfaces on the router to overcome the hardware limitations based on physical router interfaces.

Subinterfaces are software-based virtual interfaces that are assigned to physical interfaces. Each subinterface is configured independently with its own IP address and subnet mask. This allows a single physical interface to simultaneously be part of multiple logical networks.

When configuring inter-VLAN routing using the router-on-a-stick model, the physical interface of the router must be connected to a trunk link on the adjacent switch. On the router, subinterfaces are created for each unique VLAN on the network. Each subinterface is assigned an IP address specific to its subnet/VLAN and is also configured to tag frames for that VLAN. This way, the router can keep the traffic from each subinterface separated as it traverses the trunk link back to the switch.

Functionally, the router-on-a-stick model is the same as using the legacy inter-VLAN routing model, but instead of using the physical interfaces to perform the routing, subinterfaces of a single physical interface are used.

168. TOPIC

Lecture-30.168 Layer3 Switching

Reading content:

Router-on-a-stick is simple to implement because routers are usually available in every network. Most enterprise networks use multilayer switches to achieve high-packet processing rates using hardware-based switching. Layer 3 switches usually have packet-switching throughputs in the millions of packets per second (pps), whereas traditional routers provide packet switching in the range of 100,000 pps to more than 1 million pps.

All Catalyst multilayer switches support the following types of Layer 3 interfaces:

Routed port - A pure Layer 3 interface similar to a physical interface on a Cisco IOS router.

Switch virtual interface (SVI) - A virtual VLAN interface for inter-VLAN routing. In other words, SVIs are the virtual-routed VLAN interfaces.

High-performance switches, such as the Catalyst 6500 and Catalyst 4500, perform almost every function involving OSI Layer 3 and higher using hardware-based switching that is based on Cisco Express Forwarding.

All Layer 3 Cisco Catalyst switches support routing protocols, but several models of Catalyst switches require enhanced software for specific routing protocol features. Catalyst 2960 Series switches running IOS Release 12.2(55) or later, support static routing.

Catalyst switches use different default settings for interfaces. All members of the Catalyst 3560 and 4500 families of switches use Layer 2 interfaces by default. Members of the Catalyst 6500 family of switches running Cisco IOS use Layer 3 interfaces by default. Default interface configurations do not appear in the running or startup configuration. Depending on which Catalyst family of switches is used, the **switchport** or **no switchport** interface configuration mode commands might be present in the running config or startup configuration files.

Best of luck for your studies

Remember me and my family in your prayers



JAZAK ALLAH

